Topology Assignment Solutions

Topology: MAT307/M405/MAT803

Problem Sets 1–7

Submitted by **B23**

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Harisankar B	PS2: Q1–5; PS4: Q9–11
Kanishk	PS2: Q6–8
Karmukilan	PS2: Q9–11
Srijit Bhadra	PS2: Q12–15
Arunodaya	PS3: Q1–5
Pula Karan	PS3: Q6–10
Dhrubojoti	PS4: Q1–4
Aaditya	PS4: Q5–8
Mayuri	PS5: Q1–4
Prabhu	PS5: Q5–6
Jai	PS5: Q7–10
Yashpal Prasad	PS6: Q1–3
Indrajeet Kumawat	PS6: Q4–6
Himanshu Sharma	PS6: Q7–9
Karan Gangadharan	PS7: Q1–4
Alokjeet Sahoo	PS7: Q5–6

Coverage

Solved problems are shown in **green**; pending ones in grey.

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PS3	**1 2 3 4 5 6 7 8 9 10**	10/10
PS4	**1 2 3 4 5 6 7 8 9 10 11**	11/11
PS5	**1 2 3 4 5 6 7 8 9 10**	10/10
PS6	**1 2 3 4 5 6 7 8 9**	9/9
PS7	**1 2 3 4 5 6**	6/6
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Problem Set 1

Problem 1.1

Let A be a set and $\mathcal{P}(A)$ be its power set. Prove that there is no injective map $f\colon \mathcal{P}(A) \to A$ and no surjective map $g\colon A \to \mathcal{P}(A)$.

Solution *submitted by* **Ansh**

Proof. We establish both claims rigorously using Cantor's diagonalization argument, and we also show how the non-existence of an injection follows directly from the non-existence of a surjection.

Part I: Non-existence of a surjective map $g\colon A \to \mathcal{P}(A)$. Suppose, for the sake of contradiction, that there exists a surjective function $g\colon A \to \mathcal{P}(A)$. Define the diagonal subset $B \subseteq A$ by

$$B := \{x \in A \mid x \notin g(x)\}.$$

Note that this construction is well-defined for every set A (including $A = \emptyset$, in which case $\mathcal{P}(\emptyset) = \{\emptyset\} \neq \emptyset$, so no map $g\colon \emptyset \to \mathcal{P}(\emptyset)$ exists at all).

Since $B \subseteq A$, we have $B \in \mathcal{P}(A)$. Because g is assumed to be surjective, there exists at least one element $x_0 \in A$ such that

$$g(x_0) = B.$$

By the law of excluded middle, either $x_0 \in B$ or $x_0 \notin B$. We examine both cases:

(a) *Case 1 ($x_0 \in B$):* By the defining property of B, an element $x \in A$ belongs to B if and only if $x \notin g(x)$. Consequently, $x_0 \in B \implies x_0 \notin g(x_0)$. However, since $g(x_0) = B$, this implies $x_0 \notin B$, contradicting our assumption that $x_0 \in B$.

(b) *Case 2 ($x_0 \notin B$):* Since $g(x_0) = B$, the assumption $x_0 \notin B$ means $x_0 \notin g(x_0)$. By the definition of B, any element of A satisfying $x_0 \notin g(x_0)$ must belong to B; hence $x_0 \in B$, contradicting our assumption that $x_0 \notin B$.

In both cases we arrive at the logical contradiction $x_0 \in B \iff x_0 \notin B$. Therefore, our initial assumption is false, and no surjective map $g\colon A \to \mathcal{P}(A)$ exists.

Part II: Non-existence of an injective map $f\colon \mathcal{P}(A) \to A$. We provide two independent proofs of this result.

Method 1 (Direct Diagonal Proof). Suppose, for the sake of contradiction, that there exists an injective map $f\colon \mathcal{P}(A) \to A$. Consider the subset $C \subseteq A$ defined by

$$C := \{f(S) \in A \mid S \in \mathcal{P}(A) \text{ and } f(S) \notin S\}.$$

Since $C \subseteq A$, we have $C \in \mathcal{P}(A)$. Let $c := f(C) \in A$. We ask whether $c \in C$ or $c \notin C$:

(a) *Case 1 ($c \in C$):* By definition of C, there exists some subset $S \in \mathcal{P}(A)$ such that $c = f(S)$ and $f(S) \notin S$. Since $f(C) = c = f(S)$ and f is injective, it follows that $C = S$. Substituting $S = C$ into $f(S) \notin S$ yields $c = f(C) \notin C$, a contradiction.

(b) *Case 2 ($c \notin C$):* Since $c = f(C)$, the statement $c \notin C$ is equivalent to $f(C) \notin C$. Taking $S = C \in \mathcal{P}(A)$ in the defining condition for C, we see that $f(C)$ satisfies the membership criterion for C, which implies $c = f(C) \in C$, a contradiction.

Thus, in both cases we obtain a contradiction. Hence, no injective map $f\colon \mathcal{P}(A) \to A$ can exist.

Method 2 (Reduction to Part I). Suppose $f\colon \mathcal{P}(A) \to A$ is injective. Since $\mathcal{P}(A)$ is always non-empty (as $\emptyset \in \mathcal{P}(A)$), its image $f(\mathcal{P}(A)) \subseteq A$ is non-empty, which implies $A \neq \emptyset$. Define a map $g\colon A \to \mathcal{P}(A)$ by

$$g(a) := \begin{cases} S & \text{if } a = f(S) \text{ for some } S \in \mathcal{P}(A), \\ \emptyset & \text{if } a \notin f(\mathcal{P}(A)). \end{cases}$$

Because f is injective, for each $a \in f(\mathcal{P}(A))$ the preimage $S \in \mathcal{P}(A)$ with $f(S) = a$ is unique; thus g is a well-defined function. Moreover, for every $S \in \mathcal{P}(A)$, we have $g(f(S)) = S$, showing that $g\colon A \to \mathcal{P}(A)$ is surjective. This directly contradicts Part I. Therefore, no injective map $f\colon \mathcal{P}(A) \to A$ exists. □

Problem 1.2

Let X be an infinite set. Is the collection $\mathcal{T}_\infty := \{U \subseteq X \mid X \setminus U \text{ is infinite or } U = \emptyset\}$ a topology on X? Justify your answer.

Solution

submitted by **Ansh**

Answer. No, τ_∞ is not a topology on X. It violates both the first axiom ($X \in \tau_\infty$) and the second axiom (closure under arbitrary unions; in fact, it even fails closure under finite unions).

Justification. For τ_∞ to be a topology on X, it must satisfy the three standard topological axioms:

(T1) $\emptyset \in \tau_\infty$ and $X \in \tau_\infty$;

(T2) τ_∞ is closed under arbitrary unions;

(T3) τ_∞ is closed under finite intersections.

We examine each axiom in turn:

1. Violation of Axiom (T1) ($X \notin \tau_\infty$). By explicit definition, $\emptyset \in \tau_\infty$. However, consider the whole space $U = X$. Since X is an infinite set, $X \neq \emptyset$. Furthermore, the complement of X in X is

$$X \setminus X = \emptyset,$$

which has cardinality 0 and is therefore a finite set, not an infinite set. Because $U = X$ neither equals $\emptyset$ nor has an infinite complement in X, we have

$$X \notin \tau_\infty.$$

Thus, Axiom (T1) is violated.

2. Violation of Axiom (T2) (Failure of closure under unions). Even if one were to adjoin X to the collection, τ_∞ fails to be closed under arbitrary unions—and even fails under the union of two sets:

- **Counterexample via arbitrary unions of singletons:** For any element $x \in X$, consider the singleton $U_x = \{x\}$. Since X is infinite, removing a single element leaves an infinite set, so $X \setminus \{x\}$ is infinite and $\{x\} \in \tau_\infty$ for every $x \in X$. Fix a point $x_0 \in X$ and consider the union

$$U := \bigcup_{x \in X \setminus \{x_0\}} \{x\} = X \setminus \{x_0\}.$$

Since X is infinite, $U = X \setminus \{x_0\}$ is non-empty (and $U \neq X$). Yet its complement is $X \setminus U = \{x_0\}$, which is a finite set of cardinality 1. Therefore, $U \notin \tau_\infty$.

- **Counterexample via a finite union of two sets:** Since X is infinite, X contains a countably infinite subset $C = \{x_1, x_2, x_3, \dots\}$ with $x_i \neq x_j$ for $i \neq j$. Partition C into the set of even-indexed elements $E = \{x_2, x_4, x_6, \dots\}$ and odd-indexed elements $O = \{x_1, x_3, x_5, \dots\}$, both of which are infinite. Define

$$U_1 := X \setminus (E \cup \{x_1\}) \quad \text{and} \quad U_2 := X \setminus O.$$

Then $X \setminus U_1 = E \cup \{x_1\}$ is infinite, so $U_1 \in \tau_\infty$; similarly, $X \setminus U_2 = O$ is infinite, so $U_2 \in \tau_\infty$. However, by De Morgan's laws,

$$U_1 \cup U_2 = X \setminus \big((E \cup \{x_1\}) \cap O\big) = X \setminus \{x_1\}.$$

Since $U_1 \cup U_2 = X \setminus \{x_1\} \neq \emptyset$ and its complement $X \setminus (U_1 \cup U_2) = \{x_1\}$ is finite (of size 1), we have $U_1 \cup U_2 \notin \tau_\infty$. (Moreover, taking $V_1 = X \setminus E \in \tau_\infty$ and $V_2 = X \setminus O \in \tau_\infty$ gives $V_1 \cup V_2 = X \notin \tau_\infty$.)

3. Remark on Axiom (T3) (Closure under finite intersections holds). For completeness, note that if $U_1, U_2 \in \tau_\infty$, then either one of them is $\emptyset$ (so $U_1 \cap U_2 = \emptyset \in \tau_\infty$) or both $X \setminus U_1$ and $X \setminus U_2$ are infinite, in which case $X \setminus (U_1 \cap U_2) = (X \setminus U_1) \cup (X \setminus U_2)$ contains $X \setminus U_1$ and is therefore infinite, so $U_1 \cap U_2 \in \tau_\infty$. Nevertheless, because Axioms (T1) and (T2) fail, τ_∞ is not a topology on X.

Problem 1.3

Let $X = \{a, b, c\}$. Determine which of the following collections of subsets form a valid topology on X. For those that fail, explicitly state which axiom or axioms are violated:

(i) $\mathcal{T}_1 = \{\emptyset, \{a\}, \{b\}, X\}$;

(ii) $\mathcal{T}_2 = \{\emptyset, \{a\}, \{a, b\}, X\}$;

(iii) $\mathcal{T}_3 = \{\emptyset, \{a, b\}, \{b, c\}, X\}$.

Solution

submitted by **Ansh**

Recall that a collection $\mathcal{T} \subseteq \mathcal{P}(X)$ is a topology on X if and only if: (T1) $\emptyset, X \in \mathcal{T}$; (T2) $\mathcal{T}$ is closed under arbitrary unions; and (T3) $\mathcal{T}$ is closed under finite intersections.

Part (i): $\mathcal{T}_1 = \{\emptyset, \{a\}, \{b\}, X\}$.

- **Axiom (T1):** Satisfied, since $\emptyset \in \mathcal{T}_1$ and $X \in \mathcal{T}_1$.
- **Axiom (T2) (Arbitrary Unions): Violated.** Taking the union of $\{a\} \in \mathcal{T}_1$ and $\{b\} \in \mathcal{T}_1$ yields

$$\{a\} \cup \{b\} = \{a, b\} \notin \mathcal{T}_1.$$

- **Axiom (T3) (Finite Intersections):** Satisfied, since $\{a\} \cap \{b\} = \emptyset \in \mathcal{T}_1$, and the intersection of any element of $\mathcal{T}_1$ with $\emptyset$, X, or itself remains in $\mathcal{T}_1$.
- **Conclusion:** $\mathcal{T}_1$ is **not a valid topology** on X because it violates the **closure under unions axiom (T2)**.

Part (ii): $\mathcal{T}_2 = \{\emptyset, \{a\}, \{a,b\}, X\}$. Observe that the sets in $\mathcal{T}_2$ form a strictly nested chain under set inclusion:

$$\emptyset \subsetneq \{a\} \subsetneq \{a,b\} \subsetneq \{a,b,c\} = X.$$

- **Axiom (T1):** Satisfied, since $\emptyset \in \mathcal{T}_2$ and $X \in \mathcal{T}_2$.
- **Axiom (T2) (Arbitrary Unions):** Satisfied. Because $\mathcal{T}_2$ is finite and totally ordered by inclusion ($U \subseteq V$ or $V \subseteq U$ for all $U, V \in \mathcal{T}_2$), the union of any non-empty subcollection of $\mathcal{T}_2$ equals the largest element of that subcollection, which belongs to $\mathcal{T}_2$ (and the empty union is $\emptyset \in \mathcal{T}_2$). Explicitly, the only non-trivial pairwise union is $\{a\} \cup \{a,b\} = \{a,b\} \in \mathcal{T}_2$.
- **Axiom (T3) (Finite Intersections):** Satisfied. For any non-empty finite subcollection of $\mathcal{T}_2$, its intersection equals the smallest element in that subcollection, which belongs to $\mathcal{T}_2$. Explicitly, $\{a\} \cap \{a,b\} = \{a\} \in \mathcal{T}_2$.
- **Conclusion: $\mathcal{T}_2$ is a valid topology** on X.

Part (iii): $\mathcal{T}_3 = \{\emptyset, \{a,b\}, \{b,c\}, X\}$.

- **Axiom (T1):** Satisfied, since $\emptyset \in \mathcal{T}_3$ and $X \in \mathcal{T}_3$.
- **Axiom (T2) (Arbitrary Unions):** Satisfied, because $\{a,b\} \cup \{b,c\} = \{a,b,c\} = X \in \mathcal{T}_3$, and any union involving $\emptyset$, X, or a single set remains in $\mathcal{T}_3$.
- **Axiom (T3) (Finite Intersections): Violated.** Taking the intersection of $\{a,b\} \in \mathcal{T}_3$ and $\{b,c\} \in \mathcal{T}_3$ yields

$$\{a,b\} \cap \{b,c\} = \{b\} \notin \mathcal{T}_3.$$

- **Conclusion:** $\mathcal{T}_3$ is **not a valid topology** on X because it violates the **closure under finite intersections axiom (T3)**.

Collection	(T1)	(T2) Unions	(T3) Intersections	Topology?
(i) $\mathcal{T}_1$	Holds	Fails ($\{a\} \cup \{b\} = \{a,b\} \notin \mathcal{T}_1$)	Holds	**No**
(ii) $\mathcal{T}_2$	Holds	Holds	Holds	**Yes**
(iii) $\mathcal{T}_3$	Holds	Holds	Fails ($\{a,b\} \cap \{b,c\} = \{b\} \notin \mathcal{T}_3$)	**No**

Problem 1.4

Let X be a topological space and $A \subset X$. Prove that A is open in X if and only if for each $x \in A$, $\exists$ an open set U in X such that $x \in U \subseteq A$.

Solution

submitted by **Ansh**

Proof. Let (X, τ) be a topological space and let $A \subseteq X$. We prove both directions of the biconditional statement.

Forward Direction ($\Longrightarrow$). Suppose that A is open in X, so $A \in \tau$. Let $x \in A$ be arbitrary. Choose $U := A$. Since A is open in X, U is an open set in X, and by reflexivity of set inclusion ($A \subseteq A$), we immediately have

$$x \in U \subseteq A.$$

Thus, for each $x \in A$, there exists an open set U in X such that $x \in U \subseteq A$.

Reverse Direction ($\Longleftarrow$). Conversely, suppose that for every $x \in A$, there exists an open set $U_x \in \tau$ such that $x \in U_x \subseteq A$. We claim that

$$A = \bigcup_{x \in A} U_x.$$

We establish this set equality by proving mutual inclusion:

1. *Proof that* $A \subseteq \bigcup_{x \in A} U_x$*:* Let $y \in A$ be arbitrary. By hypothesis, the corresponding open set U_y satisfies $y \in U_y$. Since U_y is one of the sets in the indexed family $\{U_x\}_{x \in A}$, it follows from the definition of the union that $y \in \bigcup_{x \in A} U_x$. Hence, $A \subseteq \bigcup_{x \in A} U_x$. (Note that if $A = \emptyset$, the empty union is $\emptyset$, so the inclusion holds vacuously.)
2. *Proof that* $\bigcup_{x \in A} U_x \subseteq A$*:* Let $y \in \bigcup_{x \in A} U_x$ be arbitrary. By definition of the union, there exists some index $x_0 \in A$ such that $y \in U_{x_0}$. By hypothesis, $U_{x_0} \subseteq A$, which implies $y \in A$. Hence, $\bigcup_{x \in A} U_x \subseteq A$.

Combining both inclusions yields $A = \bigcup_{x \in A} U_x$. Because each U_x is an open set in X and a topology τ is closed under arbitrary unions, we conclude that $A = \bigcup_{x \in A} U_x \in \tau$; that is, A is open in X. □

Problem 1.5

Let U be an open set and V be a closed set in X. Prove that $U \setminus V$ is open in X and $V \setminus U$ is closed in X.

Solution

submitted by **Ansh**

Proof. Let (X, τ) be a topological space, let $U \subseteq X$ be an open set in X, and let $V \subseteq X$ be a closed set in X.

First, recall that for any two subsets $A, B \subseteq X$, the set difference $A \setminus B$ is given by the identity

$$A \setminus B = \{x \in X \mid x \in A \text{ and } x \notin B\} = A \cap (X \setminus B).$$

Furthermore, by the definition of a closed set in a topological space, a subset $F \subseteq X$ is closed in X if and only if its complement $X \setminus F$ is open in X.

Part I: Proof that $U \setminus V$ is open in X. Applying the set-difference identity to $U \setminus V$, we have

$$U \setminus V = U \cap (X \setminus V).$$

Since V is closed in X, its complement $X \setminus V$ is open in X by definition. By hypothesis, U is also open in X. By the finite intersection axiom of a topology, the intersection of two open sets in X is open in X. Therefore, $U \setminus V = U \cap (X \setminus V)$ is open in X.

Part II: Proof that $V \setminus U$ is closed in X. We provide both a direct proof from the open-set axioms via De Morgan's laws and a dual proof via closed-set intersections:

Method 1 (Direct proof via De Morgan's Laws). To show that $V \setminus U$ is closed in X, it suffices by definition to prove that its complement $X \setminus (V \setminus U)$ is open in X. Using $V \setminus U = V \cap (X \setminus U)$

and De Morgan's laws, we compute:

$$\begin{aligned} X \setminus (V \setminus U) &= X \setminus \big(V \cap (X \setminus U)\big) \\ &= (X \setminus V) \cup \big(X \setminus (X \setminus U)\big) \\ &= (X \setminus V) \cup U. \end{aligned}$$

Since V is closed in X, its complement $X \setminus V$ is open in X. By hypothesis, U is also open in X. Because a topology is closed under unions, $(X \setminus V) \cup U$ is open in X. Consequently, $X \setminus (V \setminus U)$ is open in X, which by definition means that $V \setminus U$ is closed in X.

Method 2 (Via intersections of closed sets). Using the set-difference identity, we write

$$V \setminus U = V \cap (X \setminus U).$$

Since U is open in X and $X \setminus (X \setminus U) = U$, the set $X \setminus U$ is closed in X. We are also given that V is closed in X. Since the intersection of any two closed sets in X is closed in X, it follows immediately that $V \setminus U = V \cap (X \setminus U)$ is closed in X.

This completes the proof of both statements. □

Problem 1.6

Let $X = \mathbb{N}$ and define $\mathcal{T}_1, \mathcal{T}_2 \subseteq \mathcal{P}(\mathbb{N})$ as follows:

(i) $\mathcal{T}_1 := \{\emptyset, \mathbb{N}, \{1\}, \{1,2\}, \{1,2,3\}, \{1,2,3,4\}, \dots\}$.

(ii) $\mathcal{T}_2 := \{\emptyset, \mathbb{N}, \mathbb{N} \setminus \{1\}, \mathbb{N} \setminus \{1,2\}, \mathbb{N} \setminus \{1,2,3\}, \mathbb{N} \setminus \{1,2,3,4\}, \dots\}$.

Determine whether $\mathcal{T}_1$ and $\mathcal{T}_2$ define topologies on $\mathbb{N}$.

Solution

submitted by **Sahil**

Throughout, a topology on a set X is a collection $\mathcal{T} \subseteq \mathcal{P}(X)$ such that (1) $\varnothing, X \in \mathcal{T}$, (2) arbitrary unions of members of $\mathcal{T}$ belong to $\mathcal{T}$, and (3) finite intersections of members of $\mathcal{T}$ belong to $\mathcal{T}$.

Let

$$A_n = \{1, 2, \dots, n\}, \qquad B_n = \mathbb{N} \setminus A_n \quad (n \geq 1).$$

Then the two collections in the question can be written as

$$\mathcal{T}_1 = \{\varnothing, \mathbb{N}\} \cup \{A_n : n \geq 1\}, \qquad \mathcal{T}_2 = \{\varnothing, \mathbb{N}\} \cup \{B_n : n \geq 1\}.$$

We verify the topology axioms separately.

(a) The collection $\mathcal{T}_1$ is a topology. Clearly, $\varnothing, \mathbb{N} \in \mathcal{T}_1$. For finite intersections, observe that

$$A_m \cap A_n = A_{\min\{m,n\}}.$$

Intersections with $\varnothing$ or $\mathbb{N}$ also belong to $\mathcal{T}_1$. Thus $\mathcal{T}_1$ is closed under finite intersections.

Now let $\{U_i\}_{i \in I}$ be any family of members of $\mathcal{T}_1$. If every U_i is empty, then $\bigcup_{i \in I} U_i = \varnothing$. If at least one $U_i = \mathbb{N}$, the union is $\mathbb{N}$. In the remaining case, the nonempty sets in the family are of the form A_n. If their indices are unbounded in $\mathbb{N}$, then every positive integer belongs to some A_n, so their union is $\mathbb{N}$. If their indices are bounded, they have a largest value m (a nonempty bounded subset of $\mathbb{N}$ has a maximum), and the union is A_m. In every case the union belongs to $\mathcal{T}_1$. Hence $\mathcal{T}_1$ is a topology on $\mathbb{N}$.

(b) The collection $\mathcal{T}_2$ is a topology. Clearly, $\varnothing, \mathbb{N} \in \mathcal{T}_2$. For finite intersections, De Morgan's law gives

$$B_m \cap B_n = (\mathbb{N} \setminus A_m) \cap (\mathbb{N} \setminus A_n) = \mathbb{N} \setminus (A_m \cup A_n) = B_{\max\{m,n\}}.$$

Intersections involving $\varnothing$ or $\mathbb{N}$ also remain in $\mathcal{T}_2$.

Consider an arbitrary family $\{U_i\}_{i \in I} \subseteq \mathcal{T}_2$. If one member is $\mathbb{N}$, its union is $\mathbb{N}$. If every member is empty, its union is empty. Otherwise, discard the empty members and write the remaining members as B_{n_i}. The set of indices $\{n_i : i \in I,\ U_i \neq \varnothing\}$ is a nonempty subset of $\mathbb{N}$, so it has a least element, say m. Since the sets B_n increase as n decreases, we have

$$\bigcup_{i \in I} B_{n_i} = B_m.$$

Indeed, B_m is one of the sets in the union and contains every B_{n_i} because $m \leq n_i$. Thus the union belongs to $\mathcal{T}_2$. Therefore $\mathcal{T}_2$ is also a topology on $\mathbb{N}$.

Conclusion: Both $\mathcal{T}_1$ and $\mathcal{T}_2$ define topologies on $\mathbb{N}$.

Problem 1.7

Show that if $\mathscr{B}$ is a basis for a topology on X, then the topology generated by $\mathscr{B}$ equals the intersection of all topologies on X that contain $\mathscr{B}$.

Solution

submitted by **Sahil**

Let $\mathcal{B}$ be a basis for a topology on X, and let $\mathcal{T}_{\mathcal{B}}$ denote the topology generated by $\mathcal{B}$. By the definition of the topology generated by a basis,

$$\mathcal{T}_{\mathcal{B}} = \Big\{ \bigcup_{\alpha \in A} B_\alpha : A \text{ is an index set and } B_\alpha \in \mathcal{B} \Big\}.$$

Let $\mathfrak{F}$ be the collection of all topologies on X that contain $\mathcal{B}$. We prove

$$\boxed{\mathcal{T}_{\mathcal{B}} = \bigcap_{\mathcal{T} \in \mathfrak{F}} \mathcal{T}.}$$

First, let $U \in \mathcal{T}_{\mathcal{B}}$. Then U is a union of elements of $\mathcal{B}$. If $\mathcal{T} \in \mathfrak{F}$, then $\mathcal{B} \subseteq \mathcal{T}$. Since every topology is closed under arbitrary unions, it follows that $U \in \mathcal{T}$. This holds for every $\mathcal{T} \in \mathfrak{F}$, and hence

$$\mathcal{T}_{\mathcal{B}} \subseteq \bigcap_{\mathcal{T} \in \mathfrak{F}} \mathcal{T}.$$

For the reverse inclusion, note that $\mathcal{T}_{\mathcal{B}}$ is itself a topology on X containing $\mathcal{B}$. Consequently, $\mathcal{T}_{\mathcal{B}} \in \mathfrak{F}$. The intersection of all members of $\mathfrak{F}$ is therefore contained in this particular member:

$$\bigcap_{\mathcal{T} \in \mathfrak{F}} \mathcal{T} \subseteq \mathcal{T}_{\mathcal{B}}.$$

Combining the two inclusions proves the required equality.

Interpretation: The topology generated by a basis is the smallest topology on X that contains every basis element.

Problem 1.8

Let X be an infinite set and $\mathcal{T}_f$ be the finite complement topology on X. If U and V are non-empty open sets in $(X, \mathcal{T}_f)$, prove that $U \cap V \neq \emptyset$.

Solution *submitted by* **Sahil**

Let X be infinite and let

$$\mathcal{T}_f = \{\varnothing\} \cup \{U \subseteq X : X \setminus U \text{ is finite}\}$$

be the finite-complement topology. Suppose $U, V \in \mathcal{T}_f$ are nonempty. By the definition of $\mathcal{T}_f$, both complements $X \setminus U$ and $X \setminus V$ are finite. Using De Morgan's law,

$$X \setminus (U \cap V) = (X \setminus U) \cup (X \setminus V).$$

The right-hand side is a union of two finite sets, and is therefore finite. If $U \cap V = \varnothing$, then the left-hand side equals X, which would imply that X is finite. This contradicts the assumption that X is infinite. Hence

$$\boxed{U \cap V \neq \varnothing.}$$

Problem 1.9

Let $X \neq \emptyset$ be a set and let $a \in X$ be a fixed point. Prove that the *particular point* collection

$$\mathcal{T}_a = \{V \subseteq X \mid V = \emptyset \text{ or } a \in V\}$$

defines a valid topology on X.

Solution *submitted by* **Sahil**

Let $X \neq \varnothing$ and fix $a \in X$. Define

$$\mathcal{T}_a = \{V \subseteq X : V = \varnothing \text{ or } a \in V\}.$$

We check the topology axioms.

(i) Empty set and whole space. By definition, $\varnothing \in \mathcal{T}_a$. Also $a \in X$, so $X \in \mathcal{T}_a$.

(ii) Arbitrary unions. Let $\{V_i\}_{i \in I}$ be any family in $\mathcal{T}_a$. If every V_i is empty, then $\bigcup_{i \in I} V_i = \varnothing$. Otherwise, at least one member, say V_j, is nonempty. Since $V_j \in \mathcal{T}_a$, it contains a. Therefore $a \in \bigcup_{i \in I} V_i$, so the union belongs to $\mathcal{T}_a$.

(iii) Finite intersections. It is enough to check intersections of two members, since repeated application gives the result for every finite intersection. Let $U, V \in \mathcal{T}_a$. If either set is empty, then $U \cap V = \varnothing \in \mathcal{T}_a$. If neither is empty, both contain a, and hence $a \in U \cap V$. Thus $U \cap V \in \mathcal{T}_a$.

All three axioms hold. Therefore $\boxed{\mathcal{T}_a \text{ is a topology on } X.}$

Problem 1.10

Let $X \neq \emptyset$ be a set and let $a \in X$ be a fixed point. Prove that the *excluded point* collection

$$\mathcal{T}^a := \{V \subseteq X \mid V = X \text{ or } a \notin V\}$$

defines a valid topology on X.

Solution

submitted by **Sahil**

Let $X \neq \varnothing$ and fix $a \in X$. Define

$$\mathcal{T}_a^e = \{V \subseteq X : V = X \text{ or } a \notin V\}.$$

Thus the open sets are X itself and all subsets of X that do not contain a. We verify the topology axioms.

(i) Empty set and whole space. Since $a \notin \varnothing$, we have $\varnothing \in \mathcal{T}_a^e$. Also $X \in \mathcal{T}_a^e$ by definition.

(ii) Arbitrary unions. Let $\{V_i\}_{i \in I}$ be any family in $\mathcal{T}_a^e$. If some $V_j = X$, then $\bigcup_{i \in I} V_i = X$. If no member equals X, then every V_i omits a. Consequently, their union also omits a:

$$a \notin V_i \text{ for all } i \in I \quad \Longrightarrow \quad a \notin \bigcup_{i \in I} V_i.$$

In either case the union belongs to $\mathcal{T}_a^e$.

(iii) Finite intersections. Again it is enough to consider two sets. Let $U, V \in \mathcal{T}_a^e$. If both equal X, then $U \cap V = X$. Otherwise, at least one of them, say U, is not X. By the definition of the collection, $a \notin U$. Since $U \cap V \subseteq U$, the intersection also omits a. Hence $U \cap V \in \mathcal{T}_a^e$.

All three topology axioms are satisfied. Therefore

$$\boxed{\mathcal{T}_a^e \text{ is a topology on } X.}$$

Problem 1.11

Let X be an infinite set and let $a \in X$ be a fixed point. Define a collection of subsets $\mathcal{T}_a \subseteq \mathcal{P}(X)$ by:

$$\mathcal{T}_a = \mathcal{P}(X \setminus \{a\}) \cup \{V \subseteq X \mid a \in V \text{ and } X \setminus V \text{ is finite}\}.$$

Prove that $\mathcal{T}_a$ defines a valid topology on X. What happens when X is a finite set?

Solution

submitted by **Satyajit**

Call the first family $\mathcal{A} = \mathcal{P}(X \setminus \{a\})$ (sets *not* containing a; every subset of $X \setminus \{a\}$ is in it) and the second family $\mathcal{C} = \{V \subseteq X : a \in V \text{ and } X \setminus V \text{ is finite}\}$ (sets containing a with finite complement). Note that every member of $\mathcal{T}_a$ either does not contain a (and is in $\mathcal{A}$) or contains a (and then is in $\mathcal{C}$).

Claim: $\mathcal{T}_a$ is a topology.

(T1) $\emptyset, X \in \mathcal{T}_a$. $\emptyset \subseteq X \setminus \{a\}$, so $\emptyset \in \mathcal{A}$. Also $a \in X$ and $X \setminus X = \emptyset$ is finite, so $X \in \mathcal{C}$.

(T2) Arbitrary unions. Let $\{V_\lambda\}_{\lambda \in \Lambda} \subseteq \mathcal{T}_a$ and $V = \bigcup_\lambda V_\lambda$.

- If no V_λ contains a (this includes $\Lambda = \emptyset$), then every $V_\lambda \subseteq X \setminus \{a\}$, so $V \subseteq X \setminus \{a\}$ and $V \in \mathcal{A}$.
- If some $V_{\lambda_0} \ni a$, then $V_{\lambda_0} \in \mathcal{C}$, so $X \setminus V_{\lambda_0}$ is finite. Since $V_{\lambda_0} \subseteq V$ we get $X \setminus V \subseteq X \setminus V_{\lambda_0}$, hence $X \setminus V$ is finite; and $a \in V$. So $V \in \mathcal{C}$.

(T3) Finite intersections. By induction it suffices to treat $U, V \in \mathcal{T}_a$.

- If $a \notin U$ or $a \notin V$, then $a \notin U \cap V$, i.e. $U \cap V \subseteq X \setminus \{a\}$, so $U \cap V \in \mathcal{A}$.
- If $a \in U$ and $a \in V$, then $U, V \in \mathcal{C}$, so $X \setminus U$ and $X \setminus V$ are finite. Then $X \setminus (U \cap V) = (X \setminus U) \cup (X \setminus V)$ is finite, and $a \in U \cap V$. So $U \cap V \in \mathcal{C}$.

Hence $\mathcal{T}_a$ is a topology. ■

What happens if X is finite? Every $V \subseteq X$ has finite complement (a subset of the finite set X). So any $V \ni a$ lies in $\mathcal{C}$, and any $V \not\ni a$ lies in $\mathcal{A}$. Therefore $\mathcal{T}_a = \mathcal{P}(X)$, the **discrete topology**.

(For infinite X it is *not* discrete: $\{a\}$ is not open, because $a \in \{a\}$ but $X \setminus \{a\}$ is infinite.)

Problem 1.12

Let X be an infinite set and let $a, b \in X$ be two distinct points. Define a collection of subsets $\mathcal{T} \subseteq \mathcal{P}(X)$ by:

$$\mathcal{T} = \mathcal{P}(X \setminus \{a, b\}) \cup \{V \subseteq X \mid a \in V \text{ or } b \in V \text{ and } X \setminus V \text{ is finite}\}.$$

Is $\mathcal{T}$ a topology on X?

Solution

submitted by **Satyajit**

We use the usual convention for logical connectives: "and" has precedence over "or". Thus, the condition defining the second family is interpreted as

$$a \in V \quad \text{or} \quad (b \in V \text{ and } X \setminus V \text{ is finite}).$$

Answer. The collection $\mathcal{T}$ is **not a topology** on X. We show that it is not closed under finite intersections.

Counterexample. Consider

$$U = \{a, b\}, \qquad V = X \setminus \{a\}.$$

We first verify that both sets belong to $\mathcal{T}$.

Since $a \in U$, the set U belongs to the second family, and hence $U \in \mathcal{T}$.

Next, $b \in V$ and

$$X \setminus V = \{a\},$$

which is finite. Therefore $V \in \mathcal{T}$ as well.

However,

$$U \cap V = \{a, b\} \cap (X \setminus \{a\}) = \{b\}.$$

We now show that $\{b\} \notin \mathcal{T}$.

First, $\{b\} \notin \mathcal{P}(X \setminus \{a, b\})$, because $b \notin X \setminus \{a, b\}$. Also, $a \notin \{b\}$, while $X \setminus \{b\}$ is infinite, since X is infinite. Thus $\{b\}$ does not satisfy

$$a \in W \quad \text{or} \quad (b \in W \text{ and } X \setminus W \text{ is finite})$$

either. Consequently, $\{b\} \notin \mathcal{T}$.

We have therefore found $U, V \in \mathcal{T}$ such that

$$U \cap V = \{b\} \notin \mathcal{T}.$$

Hence $\mathcal{T}$ is not closed under finite intersections, and so

$$\boxed{\mathcal{T} \text{ is not a topology on } X.}$$

□

Remark (On the logical grouping). If the intended condition had instead been

$$(a \in V \text{ or } b \in V) \text{ and } X \setminus V \text{ is finite,}$$

then it would define a different collection. The counterexample above addresses the expression as printed, using the standard precedence of "and" over "or".

Problem 1.13

Let $\mathscr{B}_1$, $\mathscr{B}_2$, and $\mathscr{B}_3$ be respectively the collections of all open balls, open rectangles, and open triangles in the plane $\mathbb{R}^2$.

(i). Prove that all three collections satisfy the basis axioms on $\mathbb{R}^2$.

(ii). Let $\mathcal{T}_i$ be the topology generated by the basis $\mathscr{B}_i$ for $i = 1, 2, 3$. Let $f\colon (\mathbb{R}^2, \mathcal{T}_i) \to (\mathbb{R}^2, \mathcal{T}_j)$ be the identity map. Prove that for any choice of $i, j \in \{1, 2, 3\}$, the map f is continuous.

(iii). Conclude that $\mathcal{T}_1 = \mathcal{T}_2 = \mathcal{T}_3$.

Solution

submitted by **Satyajit**

Let $\mathscr{B}_1, \mathscr{B}_2, \mathscr{B}_3$ be the open balls, open rectangles $(\alpha, \beta) \times (\gamma, \delta)$, and open triangles (interiors of non-degenerate triangles) in $\mathbb{R}^2$. Let $\mathcal{T}_E$ denote the Euclidean topology and $B(p, \varepsilon)$ the open ball of radius ε about p.

Recall: $\mathscr{B}$ is a **basis** on Y if (B1) every point of Y lies in some member of $\mathscr{B}$, and (B2) if $y \in B \cap B'$ with $B, B' \in \mathscr{B}$, there is $B'' \in \mathscr{B}$ with $y \in B'' \subseteq B \cap B'$.

Two lemmas used for all three collections. Lemma A. Every member of $\mathscr{B}_1, \mathscr{B}_2, \mathscr{B}_3$ is Euclidean-open.
Proof. Balls are open. A rectangle is a product of open intervals, hence open. An open triangle is an intersection of three open half-planes, hence open. □

Lemma B. For every $p \in \mathbb{R}^2$ and $\varepsilon > 0$, each $\mathscr{B}_i$ has a member B with $p \in B \subseteq B(p, \varepsilon)$.
Proof. Balls: $B(p, \varepsilon)$ itself. *Rectangles:* for $p = (p_1, p_2)$ take $R = (p_1 - \frac{\varepsilon}{2}, p_1 + \frac{\varepsilon}{2}) \times (p_2 - \frac{\varepsilon}{2}, p_2 + \frac{\varepsilon}{2})$. A point of R is at distance $< \sqrt{(\varepsilon/2)^2 + (\varepsilon/2)^2} = \frac{\varepsilon}{\sqrt{2}} < \varepsilon$ from p. *Triangles:* take the open equilateral triangle with centroid p and circumradius $\varepsilon/2$. It lies in the closed disc of radius $\varepsilon/2$ about p, hence in $B(p, \varepsilon)$, and contains its centroid p. □

(i) Basis axioms. (B1) Given p, Lemma B with $\varepsilon = 1$ gives a member of $\mathscr{B}_i$ containing p.

(B2) Let $B, B' \in \mathscr{B}_i$ and $p \in B \cap B'$. By Lemma A, $B \cap B'$ is Euclidean-open, so there is $\varepsilon > 0$ with $B(p, \varepsilon) \subseteq B \cap B'$. By Lemma B there is $B'' \in \mathscr{B}_i$ with $p \in B'' \subseteq B(p, \varepsilon) \subseteq B \cap B'$.

So $\mathscr{B}_1, \mathscr{B}_2, \mathscr{B}_3$ are bases. ■

(ii) The identity maps are continuous. Claim: $\mathcal{T}_i = \mathcal{T}_E$ for $i = 1, 2, 3$.

- $\mathcal{T}_i \subseteq \mathcal{T}_E$: members of $\mathcal{T}_i$ are unions of members of $\mathscr{B}_i$, which are Euclidean-open (Lemma A).
- $\mathcal{T}_E \subseteq \mathcal{T}_i$: let G be Euclidean-open and $p \in G$. Pick ε with $B(p, \varepsilon) \subseteq G$ and, by Lemma B, $B_p \in \mathscr{B}_i$ with $p \in B_p \subseteq B(p, \varepsilon) \subseteq G$. Then $G = \bigcup_{p \in G} B_p \in \mathcal{T}_i$.

Now let $f\colon (\mathbb{R}^2, \mathcal{T}_i) \to (\mathbb{R}^2, \mathcal{T}_j)$, $f(x) = x$. If $G \in \mathcal{T}_j$, then $f^{-1}(G) = G \in \mathcal{T}_j = \mathcal{T}_E = \mathcal{T}_i$. So f is continuous for all $i, j \in \{1, 2, 3\}$. ■

(iii) Conclusion. The identity $(\mathbb{R}^2, \mathcal{T}_i) \to (\mathbb{R}^2, \mathcal{T}_j)$ is continuous exactly when every $G \in \mathcal{T}_j$ is in $\mathcal{T}_i$, i.e. $\mathcal{T}_j \subseteq \mathcal{T}_i$. By (ii) this holds for all pairs (i, j), in both directions, so $\mathcal{T}_i \subseteq \mathcal{T}_j$ and $\mathcal{T}_j \subseteq \mathcal{T}_i$. Hence

$$\mathcal{T}_1 = \mathcal{T}_2 = \mathcal{T}_3 \ (= \mathcal{T}_E). \qquad \blacksquare$$

Problem 1.14

Along with the finite complement topology ($\mathcal{T}_f$) and the countable complement topology ($\mathcal{T}_c$), we consider the topologies generated by the following six bases:

(i) **Standard Topology ($\mathcal{T}_{\text{std}}$):** Generated by the basis of all open intervals:

$$\mathscr{B}_1 = \{(a, b) \mid a < b \text{ and } a, b \in \mathbb{R}\}.$$

(ii) **K-Topology ($\mathcal{T}_K$):** Generated by the basis:

$$\mathscr{B}_2 = \{(a, b) \mid a < b\} \cup \{(a, b) \setminus K \mid a < b\}, \quad \text{where } K = \{1, \tfrac{1}{2}, \tfrac{1}{3}, \dots\}.$$

(iii) **Lower Limit Topology ($\mathcal{T}_\ell$):** Generated by the basis of all half-open intervals closed on the left:

$$\mathscr{B}_3 = \{[a, b) \mid a < b \text{ and } a, b \in \mathbb{R}\}.$$

(iv) **Upper Limit Topology ($\mathcal{T}_u$):** Generated by the basis of all half-open intervals closed on the right:

$$\mathscr{B}_4 = \{(a, b] \mid a < b \text{ and } a, b \in \mathbb{R}\}.$$

(v) **Left-Ray Topology ($\mathcal{T}_{\text{left}}$):** Generated by the basis of open left rays:

$$\mathscr{B}_5 = \{(-\infty, b) \mid b \in \mathbb{R}\}.$$

(vi) **Right-Ray Topology ($\mathcal{T}_{\text{right}}$):** Generated by the basis of open right rays:

$$\mathscr{B}_6 = \{(a, \infty) \mid a \in \mathbb{R}\}.$$

For each pair of these topologies, determine whether one contains the other (i.e., whether one is finer, coarser, or if they are completely incomparable). Summarize your results by drawing a partial order hierarchy diagram representing the containment relations among all eight topologies.

Solution

submitted by **Satyajit**

Topologies on $\mathbb{R}$: $\mathcal{T}_f$ (finite complement), $\mathcal{T}_c$ (countable complement), $\mathcal{T}_{\text{std}}$ (basis (a,b)), $\mathcal{T}_K$ (basis (a,b) and $(a,b)\setminus K$, $K=\{1,\frac{1}{2},\frac{1}{3},\dots\}$), $\mathcal{T}_\ell$ (basis $[a,b)$), $\mathcal{T}_u$ (basis $(a,b]$), $\mathcal{T}_{\text{left}}$ (basis $(-\infty,b)$), $\mathcal{T}_{\text{right}}$ (basis (a,∞)).

Here $\mathcal{T}_f=\{\emptyset\}\cup\{U:\mathbb{R}\setminus U \text{ finite}\}$ and $\mathcal{T}_c=\{\emptyset\}\cup\{U:\mathbb{R}\setminus U \text{ countable}\}$. Also $\mathcal{T}_{\text{left}}=\{\emptyset,\mathbb{R}\}\cup\{(-\infty,b)\}$ (a union of left rays is a left ray or $\mathbb{R}$), and similarly $\mathcal{T}_{\text{right}}=\{\emptyset,\mathbb{R}\}\cup\{(a,\infty)\}$.

Convention. $\mathcal{T}\subseteq\mathcal{T}'$ means $\mathcal{T}'$ is *finer*. To show a set is not open in a topology generated by a basis, show that some point of it lies in no basis element contained in the set.

Part 1: Inclusions. **1.** $\mathcal{T}_f\subsetneq\mathcal{T}_c$**.** A finite set is countable, so every nonempty $\mathcal{T}_f$-open set is in $\mathcal{T}_c$. Strict: $\mathbb{R}\setminus\mathbb{Q}\in\mathcal{T}_c$ ($\mathbb{Q}$ countable) but $\notin\mathcal{T}_f$ ($\mathbb{Q}$ infinite).

2. $\mathcal{T}_f\subsetneq\mathcal{T}_{\text{std}}$**.** If $U=\mathbb{R}\setminus\{x_1<\cdots<x_k\}$, then $U=(-\infty,x_1)\cup(x_1,x_2)\cup\cdots\cup(x_k,\infty)$, a union of open intervals (a ray is a union of bounded intervals, e.g. $(-\infty,x_1)=\bigcup_n(x_1-n,x_1)$). Strict: $(0,1)\in\mathcal{T}_{\text{std}}$ is nonempty with infinite complement, so $\notin\mathcal{T}_f$.

3. $\mathcal{T}_{\text{left}},\mathcal{T}_{\text{right}}\subsetneq\mathcal{T}_{\text{std}}$**.** $(-\infty,b)=\bigcup_n(b-n,b)$ and $(a,\infty)=\bigcup_n(a,a+n)$ are standard-open. Strict: $(0,1)$ is neither $\emptyset$, $\mathbb{R}$, a left ray, nor a right ray, so it is open in neither ray topology.

4. $\mathcal{T}_{\text{std}}\subsetneq\mathcal{T}_K$**.** Every (a,b) is a basis element of $\mathcal{T}_K$. Strict: $(-1,1)\setminus K\in\mathcal{T}_K$ contains 0, but no interval $(-\delta,\delta)$ lies in it (it contains $1/n$ for large n), so it is not standard-open.

5. $\mathcal{T}_{\text{std}}\subsetneq\mathcal{T}_\ell$ **and** $\mathcal{T}_{\text{std}}\subsetneq\mathcal{T}_u$**.**

$$(a,b)=\bigcup_{x\in(a,b)}[x,b)=\bigcup_{x\in(a,b)}(a,x].$$

Strict: $[0,1)\in\mathcal{T}_\ell$ is not standard-open (no interval around 0 fits inside), and $(0,1]\in\mathcal{T}_u$ is not standard-open (no interval around 1 fits inside).

6. $\mathcal{T}_K\subsetneq\mathcal{T}_u$**.** Every (a,b) is standard-open, hence $\mathcal{T}_u$-open. Take $x\in(a,b)\setminus K$; we find a basis set $(c,x]$ with $x\in(c,x]\subseteq(a,b)\setminus K$:

- if $x\le 0$: take $c=a<x$; then $(a,x]\subseteq(a,b)$ contains no positive numbers, hence no point of K;
- if $x>0$: let $k=\max\{1/n:1/n<x\}=1/n_0$ with n_0 the least integer $>1/x$, and $c=\max(a,k)<x$. Every element of K that is $<x$ is $\le k\le c$, and $x\notin K$, so $(c,x]\cap K=\emptyset$ and $(c,x]\subseteq(a,b)\setminus K$.

Thus $(a,b)\setminus K$ is a union of members of $\mathscr{B}_4$, i.e. $\mathcal{T}_u$-open. Strict: $(0,1]\in\mathcal{T}_u$ but $\notin\mathcal{T}_K$: a $\mathcal{T}_K$ basis element containing 1 cannot be of the form $(a,b)\setminus K$ (as $1\in K$), so it is an interval (a,b) with $b>1$, which is not contained in $(0,1]$.

7. Transitive consequences. $\mathcal{T}_f\subsetneq\mathcal{T}_{\text{std}}\subsetneq\mathcal{T}_K,\mathcal{T}_\ell,\mathcal{T}_u$, so $\mathcal{T}_f$ is strictly coarser than all of $\mathcal{T}_K,\mathcal{T}_\ell,\mathcal{T}_u$; likewise $\mathcal{T}_{\text{left}},\mathcal{T}_{\text{right}}$ are strictly coarser than $\mathcal{T}_K,\mathcal{T}_\ell,\mathcal{T}_u$, and $\mathcal{T}_{\text{std}}\subsetneq\mathcal{T}_u$.

Part 2: Incomparable pairs (with witnesses).

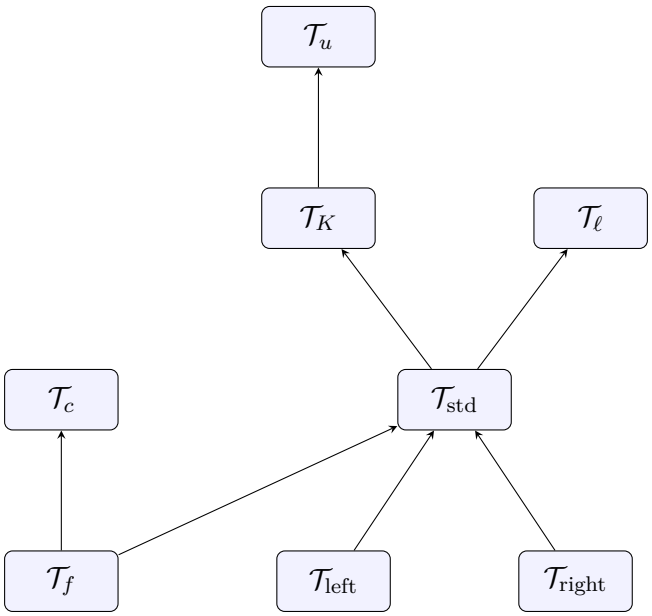
- $\mathcal{T}_K$ **and** $\mathcal{T}_\ell$**.** $(-1,1)\setminus K\in\mathcal{T}_K\setminus\mathcal{T}_\ell$: every $[0,\varepsilon)$ contains some $1/n$. $[0,1)\in\mathcal{T}_\ell\setminus\mathcal{T}_K$: every $\mathcal{T}_K$-basis element containing 0 contains negative numbers.
- $\mathcal{T}_\ell$ **and** $\mathcal{T}_u$**.** $[0,1)\in\mathcal{T}_\ell\setminus\mathcal{T}_u$ (every $(c,0]$ contains negatives); $(0,1]\in\mathcal{T}_u\setminus\mathcal{T}_\ell$ (every $[1,\varepsilon)$ contains numbers >1).
- $\mathcal{T}_{\text{left}}$ **and** $\mathcal{T}_{\text{right}}$**.** $(-\infty,0)\in\mathcal{T}_{\text{left}}\setminus\mathcal{T}_{\text{right}}$ and $(0,\infty)\in\mathcal{T}_{\text{right}}\setminus\mathcal{T}_{\text{left}}$.
- $\mathcal{T}_f$ **and each ray topology.** $(-\infty,0)\in\mathcal{T}_{\text{left}}$ has infinite complement, so $\notin\mathcal{T}_f$; and $\mathbb{R}\setminus\{0\}\in\mathcal{T}_f$ is not a left ray nor $\mathbb{R}$, so $\notin\mathcal{T}_{\text{left}}$. The same witnesses (with $(0,\infty)$) work for $\mathcal{T}_{\text{right}}$.
- $\mathcal{T}_c$ **and each of** $\mathcal{T}_{\text{std}},\mathcal{T}_K,\mathcal{T}_\ell,\mathcal{T}_u,\mathcal{T}_{\text{left}},\mathcal{T}_{\text{right}}$**.**

– $\mathbb{R} \setminus \mathbb{Q} \in \mathcal{T}_c$ is not open in any of the six: every nonempty open set of each contains a nonempty interval minus at most K (or a half-open interval or a ray), hence contains rational numbers.
– Conversely $(0,1) \in \mathcal{T}_{\text{std}} \subseteq \mathcal{T}_K, \mathcal{T}_\ell, \mathcal{T}_u$ is open in them but $\notin \mathcal{T}_c$ (its complement is uncountable); and $(-\infty, 0) \in \mathcal{T}_{\text{left}}$, $(0, \infty) \in \mathcal{T}_{\text{right}}$ are not in $\mathcal{T}_c$ for the same reason.

Comparison table. Entry in row R, column C: $\subsetneq$ means $\mathcal{T}_R$ is strictly coarser than $\mathcal{T}_C$; $\supsetneq$ strictly finer; $\|$ incomparable.

	$\mathcal{T}_f$	$\mathcal{T}_c$	$\mathcal{T}_{\text{std}}$	$\mathcal{T}_K$	$\mathcal{T}_\ell$	$\mathcal{T}_u$	$\mathcal{T}_{\text{left}}$	$\mathcal{T}_{\text{right}}$
$\mathcal{T}_f$	$=$	$\subsetneq$	$\subsetneq$	$\subsetneq$	$\subsetneq$	$\subsetneq$	$\|$	$\|$
$\mathcal{T}_c$	$\supsetneq$	$=$	$\|$	$\|$	$\|$	$\|$	$\|$	$\|$
$\mathcal{T}_{\text{std}}$	$\supsetneq$	$\|$	$=$	$\subsetneq$	$\subsetneq$	$\subsetneq$	$\supsetneq$	$\supsetneq$
$\mathcal{T}_K$	$\supsetneq$	$\|$	$\supsetneq$	$=$	$\|$	$\subsetneq$	$\supsetneq$	$\supsetneq$
$\mathcal{T}_\ell$	$\supsetneq$	$\|$	$\supsetneq$	$\|$	$=$	$\|$	$\supsetneq$	$\supsetneq$
$\mathcal{T}_u$	$\supsetneq$	$\|$	$\supsetneq$	$\supsetneq$	$\|$	$=$	$\supsetneq$	$\supsetneq$
$\mathcal{T}_{\text{left}}$	$\|$	$\|$	$\subsetneq$	$\subsetneq$	$\subsetneq$	$\subsetneq$	$=$	$\|$
$\mathcal{T}_{\text{right}}$	$\|$	$\|$	$\subsetneq$	$\subsetneq$	$\subsetneq$	$\subsetneq$	$\|$	$=$

Hasse diagram. An arrow goes from the coarser topology to the finer one. Only the covering relations are drawn; all other strict inclusions follow by transitivity (e.g. $\mathcal{T}_f \subsetneq \mathcal{T}_K$, $\mathcal{T}_{\text{left}} \subsetneq \mathcal{T}_\ell$), and any two topologies not joined by a directed path are incomparable.



Final summary. Strict chains: $\mathcal{T}_f \subsetneq \mathcal{T}_c$; $\mathcal{T}_f, \mathcal{T}_{\text{left}}, \mathcal{T}_{\text{right}} \subsetneq \mathcal{T}_{\text{std}}$; $\mathcal{T}_{\text{std}} \subsetneq \mathcal{T}_K \subsetneq \mathcal{T}_u$; $\mathcal{T}_{\text{std}} \subsetneq \mathcal{T}_\ell$. Incomparable: $\mathcal{T}_K \parallel \mathcal{T}_\ell$, $\mathcal{T}_\ell \parallel \mathcal{T}_u$, $\mathcal{T}_{\text{left}} \parallel \mathcal{T}_{\text{right}}$, $\mathcal{T}_f \parallel \mathcal{T}_{\text{left}}, \mathcal{T}_{\text{right}}$, and $\mathcal{T}_c$ is comparable only with $\mathcal{T}_f$ (where $\mathcal{T}_f \subsetneq \mathcal{T}_c$).

Problem 1.15

Let $\mathcal{T}_X$ denote the collection of all topologies on a non-empty set X and let $\mathcal{T}_X/\text{hom}$ be the set of homeomorphism classes of topologies on X. Let S_X be the group of all bijective maps on X. Note that each bijection $f\colon X \to X$ induces a bijection $\tilde{f}\colon \mathcal{P}(X) \to \mathcal{P}(X)$ defined by $\tilde{f}(A) = \{f(a) \mid a \in A\}$ for every subset $A \in \mathcal{P}(X)$. Consequently, S_X acts naturally on $\mathcal{T}_X$ via $f.\mathcal{T} = \tilde{f}(\mathcal{T})$ for all $f \in S_X$ and $\mathcal{T} \in \mathcal{T}_X$. Prove that there is a natural bijection between $\mathcal{T}_X/\text{hom}$ and the collection of orbits of the natural action of S_X on $\mathcal{T}_X$.

Solution

submitted by **Manas Singh**

Let $\mathcal{T}_X$ denote the collection of all topologies on X, and let $\mathcal{T}_X/\text{hom}$ denote the set of homeomorphism classes of topologies on X, where $\mathcal{T}_1 \sim \mathcal{T}_2$ if and only if $(X, \mathcal{T}_1)$ and $(X, \mathcal{T}_2)$ are homeomorphic. Let S_X be the group of all bijections $f\colon X \to X$. Each $f \in S_X$ induces the bijection $\tilde{f}\colon \mathcal{P}(X) \to \mathcal{P}(X)$, $\tilde{f}(A) = \{f(a) \mid a \in A\}$, and $\tilde{f}$ preserves arbitrary unions and intersections, with $\tilde{f}(\emptyset) = \emptyset$ and $\tilde{f}(X) = X$.

The action is well-defined. For $\mathcal{T} \in \mathcal{T}_X$, the collection $f \cdot \mathcal{T} = \tilde{f}(\mathcal{T}) = \{f(U) \mid U \in \mathcal{T}\}$ contains $\emptyset$ and X and is closed under arbitrary unions and finite intersections, because $\tilde{f}$ preserves these operations. Hence $f \cdot \mathcal{T} \in \mathcal{T}_X$. Moreover $\text{id} \cdot \mathcal{T} = \mathcal{T}$ and $(g \circ f) \cdot \mathcal{T} = g \cdot (f \cdot \mathcal{T})$, so this is a group action of S_X on $\mathcal{T}_X$.

Key observation. For $f \in S_X$ and $\mathcal{T}_1, \mathcal{T}_2 \in \mathcal{T}_X$,

$$f \cdot \mathcal{T}_1 = \mathcal{T}_2 \iff f\colon (X, \mathcal{T}_1) \to (X, \mathcal{T}_2) \text{ is a homeomorphism.}$$

Indeed, f is a bijection, and $f \cdot \mathcal{T}_1 = \mathcal{T}_2$ means precisely that $U \in \mathcal{T}_1 \iff f(U) \in \mathcal{T}_2$. The forward direction ($U \in \mathcal{T}_1 \Rightarrow f(U) \in \mathcal{T}_2$) says f^{-1} is continuous, and the backward direction says f is continuous. So f is a homeomorphism.

Conversely, every homeomorphism $f\colon (X, \mathcal{T}_1) \to (X, \mathcal{T}_2)$ is in particular a bijection $X \to X$, i.e. an element of S_X, and it satisfies $f \cdot \mathcal{T}_1 = \mathcal{T}_2$.

The bijection. Define $\Phi\colon \mathcal{T}_X/S_X \to \mathcal{T}_X/\text{hom}$ by

$$\Phi(\text{Orb}(\mathcal{T})) = [\mathcal{T}]_{\text{hom}}.$$

- **Well-defined and injective:** By the key observation,

$$\begin{aligned} \text{Orb}(\mathcal{T}_1) = \text{Orb}(\mathcal{T}_2) &\iff \exists f \in S_X :\ f \cdot \mathcal{T}_1 = \mathcal{T}_2 \\ &\iff (X, \mathcal{T}_1) \cong (X, \mathcal{T}_2) \iff [\mathcal{T}_1]_{\text{hom}} = [\mathcal{T}_2]_{\text{hom}}. \end{aligned}$$

  The implication "$\Rightarrow$" shows that Φ does not depend on the choice of representative of the orbit (well-defined), and "$\Leftarrow$" shows that Φ is injective.

- **Surjective:** Given any class $[\mathcal{T}]_{\text{hom}} \in \mathcal{T}_X/\text{hom}$, we have $\Phi(\text{Orb}(\mathcal{T})) = [\mathcal{T}]_{\text{hom}}$.

Therefore Φ is a natural bijection (it requires no choices) between the orbits of the action of S_X on $\mathcal{T}_X$ and $\mathcal{T}_X/\text{hom}$.

Problem 1.16

List down all possible (there are 29) topologies on a three-point set $X = \{a, b, c\}$. Compute the number of orbits under the natural action of S_3 on the set $\mathcal{T}_X$ of all topologies on X.

Solution

submitted by **Manas Singh**

Let $X = \{a, b, c\}$. The group S_3 acts on $\mathcal{T}_X$ by $f \cdot \mathcal{T} = \{f(U) \mid U \in \mathcal{T}\}$. Every topology contains $\emptyset$ and X, so we list all 29 topologies explicitly, grouped by orbit. Within each orbit, the topologies are obtained from one another by permuting a, b, c.

1. **Indiscrete topology** (orbit size 1):
   - $\{\emptyset, X\}$
2. **One non-trivial open set, a singleton** (orbit size 3):
   - $\{\emptyset, \{a\}, X\}$
   - $\{\emptyset, \{b\}, X\}$
   - $\{\emptyset, \{c\}, X\}$
3. **One non-trivial open set, a two-element set** (orbit size 3):
   - $\{\emptyset, \{a, b\}, X\}$
   - $\{\emptyset, \{b, c\}, X\}$
   - $\{\emptyset, \{a, c\}, X\}$
4. **Partition topology** (orbit size 3):
   - $\{\emptyset, \{a\}, \{b, c\}, X\}$
   - $\{\emptyset, \{b\}, \{a, c\}, X\}$
   - $\{\emptyset, \{c\}, \{a, b\}, X\}$
5. **Nested chain** $\{x\} \subset \{x, y\} \subset X$ (orbit size 6, one for each ordered pair (x, y) of distinct points):
   - $\{\emptyset, \{a\}, \{a, b\}, X\}$
   - $\{\emptyset, \{a\}, \{a, c\}, X\}$
   - $\{\emptyset, \{b\}, \{a, b\}, X\}$
   - $\{\emptyset, \{b\}, \{b, c\}, X\}$
   - $\{\emptyset, \{c\}, \{a, c\}, X\}$
   - $\{\emptyset, \{c\}, \{b, c\}, X\}$
6. **Two disjoint singletons and their union** (orbit size 3, determined by the excluded point):
   - $\{\emptyset, \{a\}, \{b\}, \{a, b\}, X\}$
   - $\{\emptyset, \{b\}, \{c\}, \{b, c\}, X\}$
   - $\{\emptyset, \{a\}, \{c\}, \{a, c\}, X\}$
7. **One singleton with both two-element sets containing it** (orbit size 3, determined by the singleton):
   - $\{\emptyset, \{a\}, \{a, b\}, \{a, c\}, X\}$
   - $\{\emptyset, \{b\}, \{a, b\}, \{b, c\}, X\}$

- $\{\emptyset, \{c\}, \{a, c\}, \{b, c\}, X\}$

8. **Two singletons $\{x\}, \{y\}$ with $\{x, y\}$ and $\{x, z\}$** (orbit size 6, one for each ordered pair (x, y) of distinct points, where z is the remaining point):
   - $\{\emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, X\}$
   - $\{\emptyset, \{b\}, \{a\}, \{a, b\}, \{b, c\}, X\}$
   - $\{\emptyset, \{a\}, \{c\}, \{a, c\}, \{a, b\}, X\}$
   - $\{\emptyset, \{c\}, \{a\}, \{a, c\}, \{b, c\}, X\}$
   - $\{\emptyset, \{b\}, \{c\}, \{b, c\}, \{a, b\}, X\}$
   - $\{\emptyset, \{c\}, \{b\}, \{b, c\}, \{a, c\}, X\}$
9. **Discrete topology** (orbit size 1):
   - $\mathcal{P}(X) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{a, c\}, X\}$

Each listed collection contains $\emptyset$ and X and is closed under unions and intersections, so each is a topology. Types 1–9 are pairwise non-homeomorphic: they differ in the number of open sets $(2, 3, 3, 4, 4, 5, 5, 6, 8)$ or, when these agree, in their structure (types 2 and 3 differ in the size of the non-trivial open set; types 4 and 5 differ because the open sets of type 4 are disjoint while those of type 5 are nested; types 6 and 7 differ because type 6 has two singleton open sets and type 7 has only one). Within each type, permuting the points moves any topology to any other, so each type is exactly one orbit.

Summing the orbit sizes gives

$$1 + 3 + 3 + 3 + 6 + 3 + 3 + 6 + 1 = 29,$$

which accounts for all 29 topologies. Hence the number of orbits of S_3 on $\mathcal{T}_X$, i.e. the number of homeomorphism classes of topologies on a three-point set, is exactly **9**.

Solution

submitted by **Sarthak**

All 29 topologies.

Orbit 1: Indiscrete topology

$$\{\emptyset, X\}.$$

Orbit 2: One singleton open

$$\{\emptyset, \{a\}, X\}, \qquad \{\emptyset, \{b\}, X\}, \qquad \{\emptyset, \{c\}, X\}.$$

Orbit 3: One doubleton open

$$\{\emptyset, \{a, b\}, X\}, \qquad \{\emptyset, \{a, c\}, X\}, \qquad \{\emptyset, \{b, c\}, X\}.$$

Orbit 4: One singleton and a doubleton containing it

$$\begin{aligned} &\{\emptyset, \{a\}, \{a, b\}, X\}, \qquad \{\emptyset, \{a\}, \{a, c\}, X\}, \\ &\{\emptyset, \{b\}, \{a, b\}, X\}, \qquad \{\emptyset, \{b\}, \{b, c\}, X\}, \\ &\{\emptyset, \{c\}, \{a, c\}, X\}, \qquad \{\emptyset, \{c\}, \{b, c\}, X\}. \end{aligned}$$

Orbit 5: One singleton and the complementary doubleton

$$\{\emptyset, \{a\}, \{b, c\}, X\}, \qquad \{\emptyset, \{b\}, \{a, c\}, X\}, \qquad \{\emptyset, \{c\}, \{a, b\}, X\}.$$

Orbit 6: Two singletons and their union

$$\{\emptyset, \{a\}, \{b\}, \{a,b\}, X\},$$
$$\{\emptyset, \{a\}, \{c\}, \{a,c\}, X\},$$
$$\{\emptyset, \{b\}, \{c\}, \{b,c\}, X\}.$$

Orbit 7: One singleton and two doubletons containing it

$$\{\emptyset, \{a\}, \{a,b\}, \{a,c\}, X\},$$
$$\{\emptyset, \{b\}, \{a,b\}, \{b,c\}, X\},$$
$$\{\emptyset, \{c\}, \{a,c\}, \{b,c\}, X\}.$$

Orbit 8: Two singletons and three doubletons, omitting one doubleton

$$\{\emptyset, \{a\}, \{b\}, \{a,b\}, \{a,c\}, X\},$$
$$\{\emptyset, \{a\}, \{c\}, \{a,b\}, \{a,c\}, X\},$$
$$\{\emptyset, \{a\}, \{b\}, \{a,b\}, \{b,c\}, X\},$$
$$\{\emptyset, \{b\}, \{c\}, \{a,b\}, \{b,c\}, X\},$$
$$\{\emptyset, \{a\}, \{c\}, \{a,c\}, \{b,c\}, X\},$$
$$\{\emptyset, \{b\}, \{c\}, \{a,c\}, \{b,c\}, X\}.$$

Orbit 9: Discrete topology

$$\{\emptyset, \{a\}, \{b\}, \{c\}, \{a,b\}, \{a,c\}, \{b,c\}, X\}.$$

Thus, the total number of topologies is

$$1 + 3 + 3 + 6 + 3 + 3 + 3 + 6 + 1 = 29.$$

Therefore, the number of orbits of S_3 acting on $\mathcal{T}_X$ is

$$|\mathcal{T}_X / S_3| = 9.$$

Explanation of orbit sizes. The indiscrete topology and the discrete topology are fixed by every permutation of a, b, c. Hence each forms an orbit of size 1.

The remaining seven orbits have sizes 3 or 6, depending on whether the representative topology has nontrivial symmetry under permutations of a, b, c. Since

$$1 + 3 + 3 + 6 + 3 + 3 + 3 + 6 + 1 = 29,$$

the orbit decomposition accounts for all 29 topologies on X.

Problem 1.17

Let X be a set, and let $\mathcal{S}_1$ and $\mathcal{S}_2$ be two subbases on X generating topologies $\mathcal{T}_1$ and $\mathcal{T}_2$, respectively.

(i). If $\mathcal{S}_1 \subseteq \mathcal{S}_2$, prove that $\mathcal{T}_1 \subseteq \mathcal{T}_2$.

(ii). Give an explicit example where $\mathcal{T}_1 = \mathcal{T}_2$ even though $\mathcal{S}_1 \neq \mathcal{S}_2$.

Solution *submitted by* **Manas Singh**

Recall that the basis generated by a subbasis $\mathcal{S}$ is $\mathscr{B}_{\mathcal{S}} = \{U_1 \cap \cdots \cap U_n \mid n \in \mathbb{N},\ U_i \in \mathcal{S}\}$, and the topology generated by $\mathcal{S}$ consists of all unions of elements of $\mathscr{B}_{\mathcal{S}}$.

(i) Suppose $\mathcal{S}_1 \subseteq \mathcal{S}_2$. Let $W \in \mathcal{T}_1$. Then $W = \bigcup_{\alpha \in I} B_\alpha$, where each B_α is a finite intersection of elements of $\mathcal{S}_1$. Since $\mathcal{S}_1 \subseteq \mathcal{S}_2$, each set in that finite intersection belongs to $\mathcal{S}_2$, so $B_\alpha \in \mathscr{B}_{\mathcal{S}_2}$ for all $\alpha \in I$. Thus W is a union of elements of $\mathscr{B}_{\mathcal{S}_2}$, i.e. $W \in \mathcal{T}_2$. Therefore $\mathcal{T}_1 \subseteq \mathcal{T}_2$.

(ii) We give an explicit example where $\mathcal{T}_1 = \mathcal{T}_2$ even though $\mathcal{S}_1 \neq \mathcal{S}_2$.

Let $X = \{1, 2, 3\}$ and define

$$\mathcal{S}_1 = \{\{1\}, \{2\}, X\}, \qquad \mathcal{S}_2 = \{\{1\}, \{2\}, \{1, 2\}, X\}.$$

Both cover X (they contain X), so both are subbases. Clearly $\mathcal{S}_1 \neq \mathcal{S}_2$, since $\{1, 2\} \in \mathcal{S}_2 \setminus \mathcal{S}_1$.

- The finite intersections of elements of $\mathcal{S}_1$ are $\{1\}$, $\{2\}$, X and $\{1\} \cap \{2\} = \emptyset$, so $\mathscr{B}_1 = \{\emptyset, \{1\}, \{2\}, X\}$. Taking all unions (note $\{1\} \cup \{2\} = \{1, 2\}$),

$$\mathcal{T}_1 = \{\emptyset, \{1\}, \{2\}, \{1, 2\}, X\}.$$

- The finite intersections of elements of $\mathcal{S}_2$ are $\{1\}, \{2\}, \{1, 2\}, X$ and $\emptyset$ (for instance $\{1\} \cap \{2\}$), and no new sets arise from intersecting these (e.g. $\{1, 2\} \cap \{1\} = \{1\}$). So $\mathscr{B}_2 = \{\emptyset, \{1\}, \{2\}, \{1, 2\}, X\}$, and since this family is already closed under unions,

$$\mathcal{T}_2 = \{\emptyset, \{1\}, \{2\}, \{1, 2\}, X\}.$$

Thus $\mathcal{T}_1 = \mathcal{T}_2$ although $\mathcal{S}_1 \neq \mathcal{S}_2$.

Solution *submitted by* **Sarthak**

(i) Proof that $\mathcal{T}_1 \subseteq \mathcal{T}_2$. By definition, the topology generated by a subbasis $\mathcal{S}$ is the smallest topology on X containing $\mathcal{S}$. Equivalently, it is the collection of arbitrary unions of finite intersections of elements of $\mathcal{S}$.

Since $\mathcal{S}_1 \subseteq \mathcal{S}_2$, and since $\mathcal{T}_2$ is a topology containing $\mathcal{S}_2$, we have

$$\mathcal{S}_1 \subseteq \mathcal{S}_2 \subseteq \mathcal{T}_2.$$

Thus $\mathcal{T}_2$ is a topology on X that contains $\mathcal{S}_1$. But $\mathcal{T}_1$ is the smallest topology on X containing $\mathcal{S}_1$. Therefore,

$$\mathcal{T}_1 \subseteq \mathcal{T}_2.$$

Alternatively, let $U \in \mathcal{T}_1$. Then U can be written as a union of finite intersections of elements of $\mathcal{S}_1$:

$$U = \bigcup_{\alpha \in A} \Big(\bigcap_{i=1}^{n_\alpha} S_{\alpha i} \Big),$$

where each $S_{\alpha i} \in \mathcal{S}_1$. Since $\mathcal{S}_1 \subseteq \mathcal{S}_2$, each $S_{\alpha i} \in \mathcal{S}_2$. Hence each finite intersection

$$\bigcap_{i=1}^{n_\alpha} S_{\alpha i}$$

belongs to $\mathcal{T}_2$. Since $\mathcal{T}_2$ is closed under arbitrary unions, it follows that

$$U \in \mathcal{T}_2.$$

Therefore,

$$\boxed{\mathcal{T}_1 \subseteq \mathcal{T}_2.}$$

(ii) Example where $\mathcal{T}_1 = \mathcal{T}_2$. Let

$$X = \mathbb{R},$$

and define

$$\mathcal{S}_1 = \{(a, b) : a < b\},$$

the collection of all open intervals in $\mathbb{R}$, and

$$\mathcal{S}_2 = \{(a, b) : a < b\} \cup \{(a, \infty) : a \in \mathbb{R}\}.$$

Clearly,

$$\mathcal{S}_1 \subsetneq \mathcal{S}_2,$$

because each ray (a, ∞) is in $\mathcal{S}_2$ but is not an open interval of the form (c, d).

Let $\mathcal{T}_1$ be the topology generated by $\mathcal{S}_1$, and let $\mathcal{T}_2$ be the topology generated by $\mathcal{S}_2$. The collection $\mathcal{S}_1$ generates the usual topology on $\mathbb{R}$. Also, every ray is a union of open intervals:

$$(a, \infty) = \bigcup_{n=1}^{\infty} (a, a + n).$$

Hence $(a, \infty) \in \mathcal{T}_1$, so

$$\mathcal{S}_2 \subseteq \mathcal{T}_1.$$

Therefore, since $\mathcal{T}_2$ is the smallest topology containing $\mathcal{S}_2$, we get

$$\mathcal{T}_2 \subseteq \mathcal{T}_1.$$

On the other hand, $\mathcal{S}_1 \subseteq \mathcal{S}_2$, so by part (i),

$$\mathcal{T}_1 \subseteq \mathcal{T}_2.$$

Combining both inclusions,

$$\mathcal{T}_1 = \mathcal{T}_2.$$

Thus, even though $\mathcal{S}_1 \subsetneq \mathcal{S}_2$, both subbases generate the same topology, namely the usual topology on $\mathbb{R}$.

Problem Set 2

Problem 2.1

Whether each of the following subsets of $\mathbb{R}^2$ can be written as a Cartesian product $A \times B$, where $A, B \subset \mathbb{R}$?

(i) $\{(x, y) \mid x \text{ is an integer}\}$.

(ii) $\{(x, y) \mid 0 \leq x < 1\}$.

(iii) $\{(x, y) \mid x < y\}$.

(iv) $\{(x, y) \mid x \text{ is an integer, and } y \text{ is not a rational number}\}$.

(v) $\{(x, y) \mid x^2 + y^2 \leq 2026\}$.

Solution *submitted by* **Harisankar B**

We use the following observation repeatedly: if $S = A \times B$ and $(a, b), (a', b') \in S$, then $a, a' \in A$ and $b, b' \in B$, so also $(a, b'), (a', b) \in S$.

(i) **Yes.** $\{(x, y) \mid x \in \mathbb{Z}\} = \mathbb{Z} \times \mathbb{R}$.

(ii) **Yes.** $\{(x, y) \mid 0 \leq x < 1\} = [0, 1) \times \mathbb{R}$.

(iii) **No.** Let $S = \{(x, y) \mid x < y\}$ and suppose $S = A \times B$. Since $(0, 1) \in S$ we get $1 \in B$, and since $(1, 2) \in S$ we get $1 \in A$. Hence $(1, 1) \in A \times B = S$, i.e. $1 < 1$, a contradiction.

(iv) **Yes.** $\{(x, y) \mid x \in \mathbb{Z},\ y \notin \mathbb{Q}\} = \mathbb{Z} \times (\mathbb{R} \setminus \mathbb{Q})$.

(v) **No.** Let $S = \{(x, y) \mid x^2 + y^2 \leq 2026\}$, put $r = \sqrt{2026}$, and suppose $S = A \times B$. Since $(r, 0) \in S$ we get $r \in A$, and since $(0, r) \in S$ we get $r \in B$. Hence $(r, r) \in S$, i.e. $2r^2 = 4052 \leq 2026$, a contradiction.

Problem 2.2

Let $f \colon A \to B$ be a function, with $A_0 \subseteq A$ and $B_0 \subseteq B$.

(i) Show that $A_0 \subseteq f^{-1}(f(A_0))$, and that equality holds for every subset $A_0 \subseteq A$ if and only if f is injective.

(ii) Show that $f(f^{-1}(B_0)) \subseteq B_0$, and that equality holds for every subset $B_0 \subseteq B$ if and only if f is surjective.

Solution *submitted by* **Harisankar B**

(i) Let $a \in A_0$. Then $f(a) \in f(A_0)$, so $a \in f^{-1}(f(A_0))$. Hence $A_0 \subseteq f^{-1}(f(A_0))$.

Suppose f is injective and let $x \in f^{-1}(f(A_0))$. Then $f(x) \in f(A_0)$, so $f(x) = f(a)$ for some $a \in A_0$. By injectivity $x = a \in A_0$. Thus $f^{-1}(f(A_0)) \subseteq A_0$, and equality holds for every $A_0 \subseteq A$.

Conversely, suppose equality holds for every $A_0 \subseteq A$, and let $f(x) = f(y)$. Taking $A_0 = \{x\}$, we have $f(y) = f(x) \in f(\{x\})$, so $y \in f^{-1}(f(\{x\})) = \{x\}$, i.e. $y = x$. Hence f is injective.

(ii) Let $y \in f(f^{-1}(B_0))$. Then $y = f(x)$ for some $x \in f^{-1}(B_0)$, i.e. $f(x) \in B_0$. Hence $y \in B_0$, and $f(f^{-1}(B_0)) \subseteq B_0$.

Suppose f is surjective and let $y \in B_0$. Choose $x \in A$ with $f(x) = y$. Then $f(x) \in B_0$, so $x \in f^{-1}(B_0)$ and $y = f(x) \in f(f^{-1}(B_0))$. Thus $B_0 \subseteq f(f^{-1}(B_0))$, and equality holds for every $B_0 \subseteq B$.

Conversely, suppose equality holds for every $B_0 \subseteq B$. Taking $B_0 = B$ and using $f^{-1}(B) = A$, we get $f(A) = f(f^{-1}(B)) = B$. Hence f is surjective.

Problem 2.3

Let $f\colon A \to B$ be a function, and $B_0, B_1 \subseteq B$. Show that preimages preserve inclusions, unions, intersections, and set differences by proving the following:

(i) If $B_0 \subseteq B_1$, then $f^{-1}(B_0) \subseteq f^{-1}(B_1)$.

(ii) $f^{-1}(B_0 \cup B_1) = f^{-1}(B_0) \cup f^{-1}(B_1)$.

(iii) $f^{-1}(B_0 \cap B_1) = f^{-1}(B_0) \cap f^{-1}(B_1)$.

(iv) $f^{-1}(B_0 \setminus B_1) = f^{-1}(B_0) \setminus f^{-1}(B_1)$.

Solution

submitted by **Harisankar B**

Throughout, recall that $x \in f^{-1}(C)$ if and only if $f(x) \in C$.

(i) Let $B_0 \subseteq B_1$ and $x \in f^{-1}(B_0)$. Then $f(x) \in B_0 \subseteq B_1$, so $x \in f^{-1}(B_1)$. Hence $f^{-1}(B_0) \subseteq f^{-1}(B_1)$.

(ii) For $x \in A$,

$$\begin{aligned} x \in f^{-1}(B_0 \cup B_1) &\iff f(x) \in B_0 \text{ or } f(x) \in B_1 \\ &\iff x \in f^{-1}(B_0) \text{ or } x \in f^{-1}(B_1) \iff x \in f^{-1}(B_0) \cup f^{-1}(B_1). \end{aligned}$$

(iii) For $x \in A$,

$$\begin{aligned} x \in f^{-1}(B_0 \cap B_1) &\iff f(x) \in B_0 \text{ and } f(x) \in B_1 \\ &\iff x \in f^{-1}(B_0) \text{ and } x \in f^{-1}(B_1) \iff x \in f^{-1}(B_0) \cap f^{-1}(B_1). \end{aligned}$$

(iv) For $x \in A$,

$$\begin{aligned} x \in f^{-1}(B_0 \setminus B_1) &\iff f(x) \in B_0 \text{ and } f(x) \notin B_1 \\ &\iff x \in f^{-1}(B_0) \text{ and } x \notin f^{-1}(B_1) \iff x \in f^{-1}(B_0) \setminus f^{-1}(B_1). \end{aligned}$$

In each of (ii)–(iv) the two sets have the same elements, so they are equal.

Problem 2.4

Let $f\colon A \to B$ be a function, and let $A_0, A_1 \subseteq A$.

(i) Show that images preserve inclusions and unions:

(a) If $A_0 \subseteq A_1$, then $f(A_0) \subseteq f(A_1)$.

(b) $f(A_0 \cup A_1) = f(A_0) \cup f(A_1)$.

(ii) Show that images satisfy only one-way inclusions for intersections and set differences in general:

(a) $f(A_0 \cap A_1) \subseteq f(A_0) \cap f(A_1)$.

(b) $f(A_0) \setminus f(A_1) \subseteq f(A_0 \setminus A_1)$.

(iii) Prove that equality holds in both (ii)(a) and (ii)(b) for all subsets $A_0, A_1 \subseteq A$ if and only if f is injective.

Solution *submitted by* **Harisankar B**

(i)(a) Let $A_0 \subseteq A_1$ and $y \in f(A_0)$. Then $y = f(a)$ with $a \in A_0 \subseteq A_1$, so $y \in f(A_1)$.

(i)(b) We have

$$\begin{aligned} y \in f(A_0 \cup A_1) &\iff y = f(a) \text{ for some } a \in A_0 \text{ or some } a \in A_1 \\ &\iff y \in f(A_0) \text{ or } y \in f(A_1) \iff y \in f(A_0) \cup f(A_1). \end{aligned}$$

(ii)(a) Since $A_0 \cap A_1 \subseteq A_0$ and $A_0 \cap A_1 \subseteq A_1$, part (i)(a) gives $f(A_0 \cap A_1) \subseteq f(A_0)$ and $f(A_0 \cap A_1) \subseteq f(A_1)$, hence $f(A_0 \cap A_1) \subseteq f(A_0) \cap f(A_1)$.

(ii)(b) Let $y \in f(A_0) \setminus f(A_1)$. Then $y = f(a)$ for some $a \in A_0$. If $a \in A_1$ we would have $y \in f(A_1)$, which is false; so $a \in A_0 \setminus A_1$ and $y \in f(A_0 \setminus A_1)$.

The reverse inclusions fail in general. Let $f\colon \{1,2\} \to \{0\}$ be the constant map, $A_0 = \{1\}$, $A_1 = \{2\}$. Then $f(A_0 \cap A_1) = f(\emptyset) = \emptyset$, while $f(A_0) \cap f(A_1) = \{0\}$; and $f(A_0) \setminus f(A_1) = \emptyset$, while $f(A_0 \setminus A_1) = f(\{1\}) = \{0\}$.

(iii) Suppose f is injective.

- Let $y \in f(A_0) \cap f(A_1)$, say $y = f(a_0) = f(a_1)$ with $a_0 \in A_0$, $a_1 \in A_1$. Injectivity gives $a_0 = a_1 \in A_0 \cap A_1$, so $y \in f(A_0 \cap A_1)$. With (ii)(a), $f(A_0 \cap A_1) = f(A_0) \cap f(A_1)$.
- Let $y \in f(A_0 \setminus A_1)$, say $y = f(a)$ with $a \in A_0$, $a \notin A_1$. If $y \in f(A_1)$, then $y = f(a')$ for some $a' \in A_1$, and injectivity gives $a = a' \in A_1$, a contradiction. So $y \in f(A_0) \setminus f(A_1)$. With (ii)(b), $f(A_0) \setminus f(A_1) = f(A_0 \setminus A_1)$.

Conversely, suppose equality holds in (ii)(a) and (ii)(b) for all $A_0, A_1 \subseteq A$, and suppose $f(x) = f(x')$ with $x \neq x'$. Take $A_0 = \{x\}$, $A_1 = \{x'\}$. Then $f(A_0 \cap A_1) = f(\emptyset) = \emptyset$, but $f(A_0) \cap f(A_1) = \{f(x)\} \neq \emptyset$, contradicting equality in (ii)(a). Hence f is injective.

Problem 2.5

Let $\sim$ be a relation defined on the Euclidean plane $\mathbb{R}^2$ as follows: two points are related if and only if their distances from the origin are equal.

(i) Show that $\sim$ is an equivalence relation (i.e., reflexivity, symmetry, and transitivity).

(ii) Describe the equivalence class $[P]$ for any point $P \in \mathbb{R}^2$.

(iii) What is the geometric interpretation of the partition of $\mathbb{R}^2$ induced by this relation?

Solution

submitted by **Harisankar B**

For $P = (x, y) \in \mathbb{R}^2$ write $\|P\| = \sqrt{x^2 + y^2}$ for its distance from the origin O. Thus $P \sim Q$ if and only if $\|P\| = \|Q\|$.

(i) Since $\sim$ is defined by equality of the real numbers $\|\cdot\|$, it inherits the properties of equality:

- *Reflexive:* $\|P\| = \|P\|$, so $P \sim P$.
- *Symmetric:* if $\|P\| = \|Q\|$ then $\|Q\| = \|P\|$, so $P \sim Q \Rightarrow Q \sim P$.
- *Transitive:* if $\|P\| = \|Q\|$ and $\|Q\| = \|R\|$ then $\|P\| = \|R\|$, so $P \sim Q$, $Q \sim R \Rightarrow P \sim R$.

Hence $\sim$ is an equivalence relation.

(ii) Let $r = \|P\| \geq 0$. Then

$$[P] = \{Q \in \mathbb{R}^2 \mid \|Q\| = r\} = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = r^2\}.$$

If $P \neq O$ (so $r > 0$), $[P]$ is the circle of radius r centred at the origin passing through P. If $P = O$, then $[P] = \{O\}$, a single point (a "circle of radius 0").

(iii) The equivalence classes partition $\mathbb{R}^2$ into the origin together with the family of concentric circles $\{x^2 + y^2 = r^2\}$, $r > 0$, centred at the origin. Every point of the plane lies on exactly one such circle (the one of radius equal to its distance from O), and distinct radii give disjoint circles. The map $[P] \mapsto \|P\|$ is a bijection from the set of equivalence classes $\mathbb{R}^2/\sim$ onto $[0, \infty)$, so the classes are labelled exactly by the possible radii.

Problem 2.6

Let X be a non-empty set. Suppose that S is the collection of the following two types of open rays: $(a, \infty) = \{x \in X \mid a < x\}$, and $(-\infty, b) = \{x \in X \mid x < b\}$. Prove that S is a subbasis, and it generates the order topology on X.

Solution

submitted by **Kanishk**

Let S be the collection of sets of the form

$$(a, \infty) = \{x \in X : a < x\} \qquad \text{and} \qquad (-\infty, b) = \{x \in X : x < b\}.$$

For two rays of the same type, we have

$$(a, \infty) \cap (c, \infty) = (\max\{a, c\}, \infty) \qquad \text{and} \qquad (-\infty, b) \cap (-\infty, d) = (-\infty, \min\{b, d\}).$$

Also, for two rays of opposite types,

$$(a, \infty) \cap (-\infty, b) = (a, b).$$

Thus, finite intersections of elements of S give open intervals and open rays, which form a basis for the order topology. Hence, S is a subbasis and generates the order topology on X.

Problem 2.7

Let $\{A_n\}_{n=1}^{\infty}$ be a sequence of non-empty countable sets.

(1) Prove that the finite Cartesian product $\prod_{i=1}^{k} A_i = A_1 \times A_2 \times \cdots \times A_k$ is countable for any $k \in \mathbb{N}$.

(2) Determine whether the infinite Cartesian product $\prod_{n=1}^{\infty} A_n$ is necessarily countable. Justify your answer.

Solution

submitted by **Kanishk**

(1) We prove this by induction on k.

For $k = 1$, the result is clear since A_1 is countable.

Suppose that $A_1 \times A_2 \times \cdots \times A_k$ is countable. Since A_{k+1} is also countable and the product of two countable sets is countable, we get that

$$(A_1 \times \cdots \times A_k) \times A_{k+1}$$

is countable. Therefore, by induction, $\prod_{i=1}^{k} A_i$ is countable for every $k \in \mathbb{N}$.

(2) No, the infinite Cartesian product is not necessarily countable. Consider

$$A_n = \{0, 1\}, \qquad n \in \mathbb{N}.$$

Suppose that $\prod_{n=1}^{\infty} A_n$ is countable. Then we can list all its elements as

$$x_1, x_2, x_3, \ldots, \qquad \text{where} \qquad x_i = (x_{i1}, x_{i2}, \ldots).$$

Define a new sequence $y = (y_1, y_2, \ldots)$ by

$$y_n = 1 - x_{nn}.$$

Clearly, y belongs to the infinite product, since every entry is either 0 or 1. But y is different from every x_n, since their n-th entries are different.

This is a contradiction. Hence, $\prod_{n=1}^{\infty} A_n$ is not necessarily countable.

Problem 2.8

Let X be a linearly ordered set. Prove that a singleton set $\{x\}$ is open in the order topology on X if and only if x has an immediate predecessor and an immediate successor, or if x is the minimum/maximum and has an immediate successor/predecessor respectively.

Solution

submitted by **Kanishk**

Suppose first that $\{x\}$ is open in the order topology. Then there exists a basic open set B such that

$$x \in B \subseteq \{x\}.$$

Hence, $B = \{x\}$.

If x is neither the minimum nor the maximum element, then B must be an open interval (a, b), where $a < x < b$. Since $(a, b) = \{x\}$, there is no element between a and x, or between x and b. Thus, a is the immediate predecessor and b is the immediate successor of x.

If x is the minimum element, then

$$(-\infty, b) = \{x\},$$

so b is the immediate successor of x. Similarly, if x is the maximum element, then

$$(a, \infty) = \{x\},$$

so a is the immediate predecessor of x.

Conversely, suppose that x has an immediate predecessor a and an immediate successor b. Then

$$(a, b) = \{x\}.$$

Since (a, b) is open, $\{x\}$ is open.

If x is the minimum element and has an immediate successor b, then

$$(-\infty, b) = \{x\},$$

which is open. Similarly, if x is the maximum element and has an immediate predecessor a, then

$$(a, \infty) = \{x\},$$

which is open.

Therefore, the given conditions are necessary and sufficient for $\{x\}$ to be open.

Problem 2.9

Prove that if V is closed in Y and Y is closed in X, then V is closed in X.

Solution *submitted by* **Karmukilan**

Conventions. A subspace $Y \subseteq X$ carries the topology $\{Y \cap U : U \text{ open in } X\}$. A *neighborhood* of x is an open set containing x. The closure $\overline{A}$ is the intersection of all closed sets containing A; it is the smallest closed set containing A.

Proof. Since V is closed in the subspace Y, the set $Y \setminus V$ is open in Y. By definition of the subspace topology, there is an open set $U \subseteq X$ with

$$Y \setminus V = Y \cap U.$$

Taking complements inside Y gives

$$V = Y \setminus (Y \setminus V) = Y \setminus (Y \cap U) = Y \cap (X \setminus U).$$

Here $X \setminus U$ is closed in X because U is open in X, and Y is closed in X by hypothesis. Hence V is an intersection of two closed subsets of X, so V is closed in X. □

Problem 2.10

Let Y be a subspace of X and let A be a subset of Y. Prove that the topology A inherits as a subspace of Y is identical to the topology it inherits directly as a subspace of X.

Solution *submitted by* **Karmukilan**

Let τ_1 be the topology that A inherits as a subspace of Y, and τ_2 the topology that A inherits as a subspace of X. We show $\tau_1 = \tau_2$.

Proof. Explicitly,

$$\tau_1 = \{A \cap W : W \text{ open in } Y\}, \qquad \tau_2 = \{A \cap U : U \text{ open in } X\}.$$

$\tau_1 \subseteq \tau_2$**:** Let $A \cap W \in \tau_1$ with W open in Y. Then $W = Y \cap U$ for some U open in X. Since $A \subseteq Y$,

$$A \cap W = A \cap (Y \cap U) = (A \cap Y) \cap U = A \cap U \in \tau_2.$$

$\tau_2 \subseteq \tau_1$**:** Let $A \cap U \in \tau_2$ with U open in X. Then $W := Y \cap U$ is open in Y, and since $A \subseteq Y$,

$$A \cap U = (A \cap Y) \cap U = A \cap (Y \cap U) = A \cap W \in \tau_1.$$

Therefore $\tau_1 = \tau_2$. □

Problem 2.11

Let A be a subset of a topological space X and $x \in X$ be an element. Then prove that the following statements are equivalent:

(i). The point $x \in \overline{A}$.

(ii). Every neighborhood of x intersects A.

(iii). Every basis element containing x intersects A.

Solution

submitted by **Karmukilan**

Let $\mathscr{B}$ be a basis for the topology of X.

Proof. **(i) $\Rightarrow$ (ii).** We prove the contrapositive. Suppose some neighborhood U of x satisfies $U \cap A = \emptyset$. Then $A \subseteq X \setminus U$, and $X \setminus U$ is closed since U is open. As $\overline{A}$ is the smallest closed set containing A, we get $\overline{A} \subseteq X \setminus U$. Since $x \in U$, $x \notin X \setminus U$, hence $x \notin \overline{A}$.

(ii) $\Rightarrow$ (iii). Every basis element $B \in \mathscr{B}$ is open, so a basis element containing x is a neighborhood of x. By (ii) it intersects A.

(iii) $\Rightarrow$ (i). Again by contrapositive. Suppose $x \notin \overline{A}$. Then $U := X \setminus \overline{A}$ is open (since $\overline{A}$ is closed) and contains x. Because $\mathscr{B}$ is a basis, there is $B \in \mathscr{B}$ with $x \in B \subseteq U$. Then

$$B \cap A \subseteq U \cap A \subseteq (X \setminus \overline{A}) \cap A = \emptyset,$$

since $A \subseteq \overline{A}$. So B is a basis element containing x that does not intersect A, and (iii) fails.

Thus (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (i), and the three statements are equivalent. □

Problem 2.12

Prove that $\overline{A} = A \cup A'$. Deduce that A is closed in X if and only if $A' \subseteq A$.

Solution

submitted by **Srijit Bhadra**

Solution: Let $x \in \bar{A}$. Now,

$$\begin{aligned} x \in \bar{A} &\iff \text{every neighbourhood of } x \text{ intersects } A \\ &\iff x \in A \text{ or } x \in A' \\ &\iff x \in A \cup A'. \end{aligned}$$

Since x was chosen arbitrarily, we can say $\bar{A} = A \cup A'$.

For the second part, assume A is closed. Then $\bar{A} = A \implies A \cup A' = A \implies A' \subseteq A$. On the other hand, if we assume $A' \subseteq A$, then we get $\bar{A} = A \cup A' = A$. Thus, in this case, we can conclude that A is closed. As both directions of the double implication hold, we have A is closed in X if and only if $A' \subseteq A$. □

Problem 2.13

Prove that $\operatorname{Int} A \cap \operatorname{Bd} A = \emptyset$, and $\overline{A} = \operatorname{Int} A \cup \operatorname{Bd} A$.

Solution

submitted by **Srijit Bhadra**

Solution: We know, $\operatorname{Int}(A) \cap \operatorname{Bd}(A) = \operatorname{Int}(A) \cap \bar{A} \cap \overline{X \setminus A}$. For the sake of contradiction, assume there is a point $x \in \operatorname{Int}(A) \cap \operatorname{Bd}(A) \implies x \in \operatorname{Int}(A) \cap \bar{A} \cap \overline{X \setminus A}$. As $x \in \overline{X \setminus A}$, every neighbourhood of x must intersect $X \setminus A$. Thus, $\operatorname{Int}(A) \cap \overline{X \setminus A} \neq \emptyset$. But this is impossible as $\operatorname{Int}(A) \subseteq A$ ($\Rightarrow\Leftarrow$). Thus, $\operatorname{Int}(A) \cap \operatorname{Bd}(A) = \emptyset$.

For the second part, let $x \in \bar{A}$. Again, this happens if and only if every neighborhood of x intersects A. Now, either there exists a neighborhood U of x such that $x \in U \subseteq A$, or there is no such neighborhood. In the first case, $x \in U \subseteq A \implies x \in \operatorname{Int}(A)$ since $\operatorname{Int}(A)$ is the largest open set contained in A. In the second case, every neighborhood V of x intersects $X \setminus A \implies x \in \overline{X \setminus A} \implies x \in \operatorname{Bd}(A)$. Combining the two cases, $x \in \operatorname{Int}(A) \cup \operatorname{Bd}(A)$. Since $x \in \bar{A}$ was chosen arbitrarily, $\bar{A} \subseteq \operatorname{Int}(A) \cup \operatorname{Bd}(A)$.

For the other way inclusion, notice that $\operatorname{Int}(A) \subseteq A \subseteq \bar{A}$. Also, $\operatorname{Bd}(A) = \bar{A} \cap \overline{X \setminus A} \subseteq \bar{A}$. Therefore, $\operatorname{Int}(A) \cup \operatorname{Bd}(A) \subseteq \bar{A}$. Hence, we conclude that $\bar{A} = \operatorname{Int}(A) \cup \operatorname{Bd}(A)$. □

Problem 2.14

Prove that every order topology is Hausdorff.

Solution

submitted by **Srijit Bhadra**

Solution: Let $X \neq \emptyset$ and $\mathscr{T}$ be the order topology on X. Let $x_1, x_2 \in X$ such that $x_1 \neq x_2$. Consider the basic open set $(x_1, x_2) = \{x \in X : x_1 < x < x_2\} \in \mathscr{T}$. Now, there can be two cases:

Case 1. $(x_1, x_2) = \emptyset$. Observe that

$$(-\infty, x_2) = \bigcup_{a \in X,\ a < x_2} (a, x_2) \in \mathscr{T}.$$

Similarly,

$$(x_1, \infty) = \bigcup_{b \in X,\ b > x_1} (x_1, b) \in \mathscr{T}.$$

Also, $x_1 \in (-\infty, x_2)$ and $x_2 \in (x_1, \infty)$. Crucially, $(-\infty, x_2) \cap (x_1, \infty) = \emptyset$.

Case 2. $(x_1, x_2) \neq \emptyset$. Take any point $c \in (x_1, x_2)$. As we saw earlier, $(-\infty, c)$ and (c, ∞) are open in X. Note that $x_1 \in (-\infty, c)$, $x_2 \in (c, \infty)$, and $(-\infty, c) \cap (c, \infty) = \emptyset$.

Thus, in both cases, we found two disjoint open sets in X with one containing x_1 and the other containing x_2. As $x_1, x_2 \in X$ were chosen arbitrarily, we can say that $(X, \mathscr{T})$ is Hausdorff. Since $(X, \mathscr{T})$ was also chosen arbitrarily, we can say that every order topology is Hausdorff. □

Problem 2.15

Let X be a T_1 space. Then prove that for each pair x_1, x_2 of distinct points, there exist open sets U_1 and U_2 containing x_1 and x_2 respectively such that $x_1 \notin U_2$ and $x_2 \notin U_1$.

Solution

submitted by **Srijit Bhadra**

Solution: Let x_1, x_2 be a pair of distinct points in X. X being a T_1 space, singletons $\{x_1\}$ and $\{x_2\}$ are closed in X. Therefore, $X \setminus \{x_1\}$ and $X \setminus \{x_2\}$ are open in X. Now, $x_2 \in X \setminus \{x_1\}$ (because $x_1 \neq x_2$) but $x_2 \notin X \setminus \{x_2\}$. Similarly, $x_1 \in X \setminus \{x_2\}$ (because $x_1 \neq x_2$) but $x_1 \notin X \setminus \{x_1\}$. As $x_1, x_2 \in X$ were chosen arbitrarily, we can say that for each pair x_1, x_2 of distinct points, there exist open sets U_1 and U_2 containing x_1 and x_2, respectively, such that $x_1 \notin U_2$ and $x_2 \notin U_1$. $\square$

Problem Set 3

Problem 3.1

Give an example of a T_1 space where a sequence converges to more than one point. Conclude that a T_1 space is not necessarily a Hausdorff space.

Solution *submitted by* **Arunodaya**

Consider the set $X = \mathbb{R}$ equipped with the cofinite topology

$$\tau = \{\emptyset\} \cup \{U \subseteq \mathbb{R} : \mathbb{R} \setminus U \text{ is finite}\},$$

that is, the open sets are the empty set or subsets of $\mathbb{R}$ whose complements are finite.

Now, we know that a topological space is T_1 if every singleton set is closed. Let $x \in \mathbb{R}$. Then $\mathbb{R} \setminus \{x\}$ is open because its complement $\{x\}$ is finite. Therefore, $\{x\}$ is closed. Since x is arbitrary, every singleton subset of $\mathbb{R}$ is closed. Hence $(\mathbb{R}, \tau)$ is a T_1 space.

Consider the sequence

$$x_n = n, \quad n \in \mathbb{N}.$$

$x_n \to x$ if for every open neighbourhood U of x, $\exists\, N \in \mathbb{N}$ such that $n \geq N \Rightarrow x_n \in U$.

Let $x \in \mathbb{R}$ be arbitrary. Let U be any open neighbourhood of x. By the definition of the cofinite topology, $\mathbb{R} \setminus U$ is finite. Since $x_n = n$, the sequence consists of distinct natural numbers. Consequently only finitely many terms of the sequence can belong to the finite set $\mathbb{R} \setminus U$. Therefore, $\exists\, N \in \mathbb{N}$ such that

$$n \geq N \Rightarrow n \notin \mathbb{R} \setminus U.$$

Thus, $n \geq N \Rightarrow x_n = n \in U$. So, by the definition of convergence,

$$x_n \to x \quad \forall\, x \in \mathbb{R}.$$

Now, the sequence $x_n = n$ converges to every real number

$\Rightarrow$ its limit is not unique

$\Rightarrow$ the space cannot be Hausdorff.

Thus $(\mathbb{R}, \tau)$ is T_1 but not T_2, i.e. $T_1 \not\Rightarrow T_2$.

Problem 3.2

Prove that the product topology $\mathcal{T}_{\text{prod}}$ on $\mathbb{R}^n$ (where each factor $\mathbb{R}$ is equipped with its standard Euclidean topology) coincides with the Euclidean topology $\mathcal{T}_{\text{Euc}}$ on $\mathbb{R}^n$ induced by the standard Euclidean metric.

Solution

submitted by **Arunodaya**

Let $U = U_1 \times U_2 \times \cdots \times U_n$ be a basic open set in the product topology. Take an arbitrary point $x = (x_1, x_2, x_3, x_4, \ldots, x_n) \in U$. Since each U_i is open in $\mathbb{R}$, $\exists\, \varepsilon_i > 0$ such that

$$(x_i - \varepsilon_i,\, x_i + \varepsilon_i) \subseteq U_i.$$

Choose $\varepsilon = \min\{\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n\}$. Since there are finitely many positive numbers, $\varepsilon > 0$.

Now, consider any point $y = (y_1, y_2, \ldots, y_n) \in B(x, \varepsilon)$. Then $d(x, y) < \varepsilon$. Thus,

$$\sqrt{\sum_{i=1}^{n}(x_i - y_i)^2} < \varepsilon \;\Rightarrow\; \sum_{i=1}^{n}(x_i - y_i)^2 < \varepsilon^2.$$

Since every term in the sum is non-negative, $(x_i - y_i)^2 < \varepsilon^2$ for every $i = 1, 2, \ldots, n$. Consequently $|x_i - y_i| < \varepsilon \leq \varepsilon_i$; thus $y_i \in (x_i - \varepsilon_i,\, x_i + \varepsilon_i) \subseteq U_i$. This holds for every i. Hence

$$y \in U_1 \times U_2 \times \cdots \times U_n \;\Rightarrow\; B(x, \varepsilon) \subseteq U.$$

Since $x \in U$ was arbitrary, every point of U has a Euclidean open ball contained in U. Hence U is Euclidean open.

$$\underline{\mathcal{T}_{\text{prod}} \subseteq \mathcal{T}_{\text{Euc}}}$$

Let V be an arbitrary Euclidean open subset of $\mathbb{R}^n$. Take any point $x = (x_1, x_2, \ldots, x_n) \in V$. Since V is Euclidean open, $\exists\, \delta > 0$ such that $B(x, \delta) \subseteq V$. Now consider the product of open intervals

$$W = \prod_{i=1}^{n}\Big(x_i - \frac{\delta}{\sqrt{n}},\; x_i + \frac{\delta}{\sqrt{n}}\Big).$$

We will show that $W \subseteq B(x, \delta)$. Take any $y = (y_1, y_2, y_3, y_4, \ldots, y_n) \in W$. By definition of W,

$$|x_i - y_i| < \delta/\sqrt{n} \quad \forall\, i = 1, 2, 3, 4, \ldots, n.$$

Therefore

$$d(x, y) = \sqrt{\sum_{i=1}^{n}(x_i - y_i)^2} < \sqrt{\sum_{i=1}^{n}\frac{\delta^2}{n}} = \sqrt{\frac{n\delta^2}{n}} = \delta.$$

Thus $y \in B(x, \delta)$. Consequently $W \subseteq B(x, \delta) \subseteq V$.

Notice that W is a basic open set in the product topology because it is a product of open intervals in $\mathbb{R}$. Hence, every point $x \in V$ has a basic product open neighbourhood contained in V. Therefore, V is open in the product topology. So, $\underline{\mathcal{T}_{\text{Euc}} \subseteq \mathcal{T}_{\text{prod}}}$.

So $\mathcal{T}_{\text{prod}} = \mathcal{T}_{\text{Euc}}$.

Problem 3.3

Let $\{X_\alpha\}_{\alpha\in\Lambda}$ be a family of topological spaces, and let $\phi\colon \Lambda \to \Lambda$ be a bijection. Prove that the product spaces $X = \prod_{\alpha\in\Lambda} X_\alpha$ and $X_\phi = \prod_{\alpha\in\Lambda} X_{\phi(\alpha)}$, both equipped with their respective product topologies, are homeomorphic.

Solution

submitted by **Arunodaya**

We define $f\colon X \to X_\phi$ by

$$f\big((x_\alpha)_{\alpha\in A}\big) = \big(x_{\phi(\alpha)}\big)_{\alpha\in A}.$$

Since ϕ is a bijection, every coordinate in the second product corresponds to exactly one coordinate in the first product. Since ϕ is bijective, it has an inverse

$$\phi^{-1}\colon B \to A.$$

Define $g\colon X_\phi \to X$ by

$$g\big((y_\beta)_{\beta\in B}\big) = \big(y_{\phi^{-1}(\beta)}\big)_{\beta\in B}.$$

Then for every $x = (x_\alpha) \in X$, $g(f(x)) = x$. Similarly for every $y \in X_\phi$, $f(g(y)) = y$. Thus $g = f^{-1}$. Hence f is bijective.

The product topology has a subbasis consisting of sets of the form $\pi_\alpha^{-1}(U_\alpha)$, where U_α is open in X_α and $\pi_\alpha\colon X \to X_\alpha$ is the coordinate projection.

Let U_α be open in $X_{\phi(\alpha)}$. Consider the subbasic open set in X_ϕ

$$V = \{y \in X_\phi : y_\alpha \in U_\alpha\}.$$

$$\begin{aligned} f^{-1}(V) &= \{x \in X : f(x)_\alpha \in U_\alpha\} \\ &= \{x \in X : x_\alpha \in U_\alpha\} \\ &= \pi_\alpha^{-1}(U_\alpha). \end{aligned}$$

Since U_α is open and π_α is continuous, the set is open in $X \Rightarrow$ the inverse image of every subbasic open set is open. Consequently, f is continuous.

By the same argument, for every subbasic open set

$$W = \{x \in X : x_\alpha \in U_\alpha\}$$

we have

$$(f^{-1})^{-1}(W) = \{y \in X_\phi : y_{\phi^{-1}(\alpha)} \in U_\alpha\};$$

this is a subbasic open set in X_ϕ. Hence f^{-1} is continuous.

We proved that f is bijective, continuous and has a continuous inverse; therefore $X \cong X_\phi$.

Problem 3.4

Let $\{X_\alpha\}_{\alpha\in\Lambda}$ be a family of topological spaces, and for each $\alpha \in \Lambda$, let $Y_\alpha \subseteq X_\alpha$ be equipped with the subspace topology. Prove that the product topology on $Y = \prod_{\alpha\in\Lambda} Y_\alpha$ coincides with the subspace topology on Y induced by the product topology on $X = \prod_{\alpha\in\Lambda} X_\alpha$.

Solution

submitted by **Arunodaya**

Let

- τ_1 denote the product topology on Y,
- τ_2 denote the subspace topology on Y inherited from X.

We shall prove $\tau_1 = \tau_2$.

Let B be a basic open set in X. Then

$$B = \bigcap_{\alpha\in F} \pi_\alpha^{-1}(U_\alpha),$$

where each U_α is open in X_α. By the definition of the subspace topology, a basic open set in Y is $B \cap Y$. Now,

$$\begin{aligned} B \cap Y &= Y \cap \bigcap_{\alpha\in F} \pi_\alpha^{-1}(U_\alpha) \\ &= \bigcap_{\alpha\in F} \left(Y \cap \pi_\alpha^{-1}(U_\alpha)\right) \\ &= \bigcap_{\alpha\in F} q_\alpha^{-1}(U_\alpha \cap Y_\alpha). \end{aligned}$$

Since U_α is open in X_α, the set $U_\alpha \cap Y_\alpha$ is open in Y_α by the subspace topology. Therefore, $B \cap Y$ is open in the product topology on Y. Hence $\underline{\tau_2 \subseteq \tau_1}$.

Let C be a basic open set in the product topology on Y. Then

$$C = \bigcap_{\alpha\in F} q_\alpha^{-1}(V_\alpha),$$

where V_α is open in Y_α. Since Y_α has the subspace topology inherited from X_α, $\exists$ an open set U_α in X_α such that $V_\alpha = U_\alpha \cap Y_\alpha$. Therefore,

$$\begin{aligned} C &= \bigcap_{\alpha\in F} q_\alpha^{-1}(U_\alpha \cap Y_\alpha) \\ &= Y \cap \bigcap_{\alpha\in F} \pi_\alpha^{-1}(U_\alpha). \end{aligned}$$

But $\bigcap_{\alpha\in F} \pi_\alpha^{-1}(U_\alpha)$ is open in X because it is a basic open set in the product topology on X. Thus, C is the intersection of Y with an open set in X. Hence C is open in the subspace topology on Y. Therefore $\underline{\tau_1 \subseteq \tau_2}$.

So, $\underline{\tau_1 = \tau_2}$.

Problem 3.5

Let $\{X_\alpha\}_{\alpha\in\Lambda}$ be an indexed collection of topological spaces, and let $X = \prod_{\alpha\in\Lambda} X_\alpha$ be equipped with the product topology $\mathcal{T}_{\text{prod}}$. For any fixed index $\beta \in \Lambda$, consider the β-th coordinate projection map

$$p_\beta \colon \prod_{\alpha\in\Lambda} X_\alpha \longrightarrow X_\beta, \qquad p_\beta\big((x_\alpha)_{\alpha\in\Lambda}\big) = x_\beta.$$

Prove that p_β is an open map. Is p_β necessarily a closed map?

Solution

submitted by **Arunodaya**

Part (i). By the definition of the product topology, every basic open set has the form

$$B = \bigcap_{\alpha\in F} \pi_\alpha^{-1}(U_\alpha),$$

where

- $F \subseteq A$ is finite,
- each U_α is open in X_α.

Case 1: $\beta \in F$. Then $B = \bigcap_{\alpha\in F} \pi_\alpha^{-1}(U_\alpha)$, and the image of B under p_β is

$$p_\beta(B) = U_\beta,$$

provided the other coordinate restrictions are non-empty. Indeed, every element of B has its β-coordinate in U_β. Conversely, any element of U_β can be extended to a point of B if all the other coordinate restrictions are non-empty. Since U_β is open in X_β, the image is open. If another coordinate restriction is empty, then $B = \emptyset$, whose image is also open.

Case 2: $\beta \notin F$. In this case, there is no restriction on the β-coordinate:

$$p_\beta(B) = X_\beta,$$

provided $B \neq \emptyset$. Since X_β is open in itself, the image is open. If $B = \emptyset$, its image is open again.

Now, $U = \bigcup_{\lambda\in\Lambda} B_\lambda$. Using the fact that images preserve arbitrary unions, we get

$$p_\beta(U) = p_\beta\Big(\bigcup_{\lambda\in\Lambda} B_\lambda\Big) = \bigcup_{\lambda\in\Lambda} p_\beta(B_\lambda).$$

From the above step, $p_\beta(B_\lambda)$ is open in X_β. An arbitrary union of open sets is open; thus, $p_\beta(U)$ is open in X_β.

Part (ii). No. Take $X_1 = \mathbb{R}$, $X_2 = \mathbb{R}$ with their usual Euclidean topologies. Then $X = X_1 \times X_2 = \mathbb{R}^2$. Consider the coordinate projection $p_1 \colon \mathbb{R}^2 \to \mathbb{R}$ defined by $p_1(x, y) = x$. Consider

$$F = \{(x, y) \in \mathbb{R}^2 : xy = 1\}.$$

Define $h \colon \mathbb{R}^2 \to \mathbb{R}$ by $h(x, y) = xy$. Since multiplication is continuous, h is continuous. Also $F = h^{-1}(\{1\})$. Since $\{1\}$ is closed in $\mathbb{R}$ and the inverse image of a closed set under a continuous function is closed, we say F is closed in $\mathbb{R}^2$.

We have

$$p_1(F) = \{x \in \mathbb{R} \mid \exists\, y \in \mathbb{R} \text{ such that } xy = 1\} = \mathbb{R} \setminus \{0\}.$$

Indeed, if $x \neq 0$ we can choose $y = 1/x$. But if $x = 0$, no such y exists. Therefore $p_1(F) = \mathbb{R} \setminus \{0\}$.

The set $\mathbb{R} \setminus \{0\} = (-\infty, 0) \cup (0, \infty)$ is not closed in $\mathbb{R}$ because its complement $\{0\}$ is not open in $\mathbb{R}$. Thus, we found a closed set $F \subseteq \mathbb{R}^2$ whose image under p_1 is not closed. Consequently, p_1 is not closed.

So, every coordinate projection is an open map. A coordinate projection need not be a closed map.

Problem 3.6

Let L be a line in the plane $\mathbb{R}^2$, and $\mathbb{R}_\ell$ be the lower limit topology on $\mathbb{R}$. Discuss the topology L inherits as a subspace of $\mathbb{R}_\ell \times \mathbb{R}$ and as a subspace of $\mathbb{R}_\ell \times \mathbb{R}_\ell$.

Solution

submitted by **Pula Karan**

We analyze the subspace topology on L under both product topologies.

Part 1: Subspace Topology inherited from $\mathbb{R}_l \times \mathbb{R}$. A basis for $\mathbb{R}_l \times \mathbb{R}$ consists of rectangles of the form $[a, b) \times (c, d)$ with $a < b$ and $c < d$.

(a) **Vertical Line ($L : x = k$):** For any $(k, y_0) \in L$, a basic open neighborhood in $\mathbb{R}_l \times \mathbb{R}$ is $U = [k, k + \epsilon) \times (y_0 - \epsilon, y_0 + \epsilon)$. Intersecting U with L yields:

$$U \cap L = \{k\} \times (y_0 - \epsilon, y_0 + \epsilon)$$

This traces open intervals along the y-axis. Thus, L inherits the **standard Euclidean topology** ($\mathbb{R}$).

(b) **Horizontal Line ($L : y = k$):** For any $(x_0, k) \in L$, a basic open neighborhood in $\mathbb{R}_l \times \mathbb{R}$ is $U = [x_0, x_0 + \epsilon) \times (k - \epsilon, k + \epsilon)$. Intersecting U with L yields:

$$U \cap L = [x_0, x_0 + \epsilon) \times \{k\}$$

This traces right-half-open intervals along the x-axis. Thus, L inherits the **lower limit topology** ($\mathbb{R}_l$).

(c) **Non-Vertical Line ($L : y = mx + k$, $m \neq 0$):** For any $(x_0, y_0) \in L$, consider basic open sets $U = [x_0, x_0 + \epsilon) \times (y_0 - \delta, y_0 + \delta)$.

- If $m > 0$, as x increases from x_0, y increases from y_0. For sufficiently small $\delta > 0$, $U \cap L$ isolates points corresponding to $x \in [x_0, x_0 + \delta/m)$.
- If $m < 0$, as x increases from x_0, y decreases from y_0. Intersecting $[x_0, x_0 + \epsilon)$ with $(y_0 - \delta, y_0 + \delta)$ again isolates points corresponding to $x \in [x_0, x_0 + \delta/|m|)$.

In both cases, basic open sets induce half-open intervals $[x_0, x_0 + r)$ along L. Thus, L inherits the **lower limit topology** ($\mathbb{R}_l$).

Part 2: Subspace Topology inherited from $\mathbb{R}_l \times \mathbb{R}_l$ (Sorgenfrey Plane). A basis for $\mathbb{R}_l \times \mathbb{R}_l$ consists of half-open rectangles of the form $[a, b) \times [c, d)$.

(a) **Vertical ($x = k$), Horizontal ($y = k$), and Positive Slope ($y = mx + k$, $m > 0$):** For any $(x_0, y_0) \in L$, intersecting $[x_0, x_0 + \epsilon) \times [y_0, y_0 + \epsilon')$ with L isolates points corresponding to $[x_0, x_0 + \delta)$ where $\delta = \min(\epsilon, \epsilon'/m)$ for $m > 0$ (or ϵ if vertical/horizontal). Thus, L inherits the **lower limit topology** ($\mathbb{R}_l$).

(b) **Oblique Line with Negative Slope ($y = mx + k$, $m < 0$):** Let $(x_0, y_0) \in L$ and consider $U = [x_0, x_0 + \epsilon) \times [y_0, y_0 + \epsilon)$. For any $x > x_0$, $y = mx + k < mx_0 + k = y_0$, so $(x, y) \notin U$. For $x < x_0$, $x \notin [x_0, x_0 + \epsilon)$, so $(x, y) \notin U$. Therefore, $U \cap L = \{(x_0, y_0)\}$. Since every singleton is open, L inherits the **discrete topology**.

Problem 3.7

Compare (i) the ordered square I_o^2, (ii) the topology I^2 inherits as a subspace of the product topology $\mathbb{R}^2$, and (iii) the product topology on $I \times I$.

Solution

submitted by **Pula Karan**

Let $I = [0, 1]$. We compare the three topologies on $I^2 = [0, 1] \times [0, 1]$:

1. **Topologies (ii) and (iii) are identical ($\mathcal{T}_{\text{sub}} = \mathcal{T}_{\text{prod}}$):**
   - Topology (ii) is the subspace topology inherited from the Euclidean product topology on $\mathbb{R}^2$. Basis elements are of the form $(U \times V) \cap I^2$, where $U, V \subseteq \mathbb{R}$ are open intervals.
   - Topology (iii) is the product topology on $I \times I$, where each factor I carries the subspace topology from $\mathbb{R}$. Basis elements are of the form $(U \cap I) \times (V \cap I) = (U \times V) \cap I^2$.

   Since their bases coincide, Topologies (ii) and (iii) are identical.
2. **Topology (i) (I_o^2) is strictly finer than Topologies (ii) and (iii):** The ordered square I_o^2 is equipped with the dictionary order topology $\leq_{\text{dic}}$.
   - *Euclidean $\subseteq$ Dictionary Order*: Any standard Euclidean basic open set $U = (a, b) \times (c, d) \subset (0, 1)^2$ can be written as:

     $$U = \bigcup_{x_0 \in (a,b)} \Big(\{x_0\} \times (c, d) \Big) = \bigcup_{x_0 \in (a,b)} \Big((x_0, c), (x_0, d) \Big)_{\text{dic}}$$

     Since each vertical interval $\{x_0\} \times (c, d)$ is open in the dictionary order topology, U is open in I_o^2.
   - *Strict Containment*: Sets of the form $\{x_0\} \times (c, d)$ for $0 < c < d < 1$ are open in I_o^2. However, they contain no 2D Euclidean open ball, so they are not open in the standard topology on I^2.

Conclusion: Topology (ii) = Topology (iii) $\subsetneq$ Topology (i).

Problem 3.8

Let X be a topological space, and $\Delta = \{(x, x) \in X \times X \mid x \in X\}$ be the diagonal of $X \times X$, equipped with the subspace topology inherited from the product topology on $X \times X$.

(i) Prove that X is Hausdorff if and only if the diagonal Δ is a closed subset of $X \times X$.

(ii) Prove that the diagonal map $d\colon X \to \Delta$ defined by $d(x) = (x, x)$ is a homeomorphism.

Solution *submitted by* **Pula Karan**

Part (i).

Proof. ($\implies$) Suppose X is Hausdorff. We show that $(X \times X) \setminus \Delta$ is open. Let $(x, y) \in (X \times X) \setminus \Delta$, which means $x \neq y$. Since X is Hausdorff, there exist disjoint open sets $U, V \subseteq X$ such that $x \in U$, $y \in V$, and $U \cap V = \varnothing$. Then $U \times V$ is an open neighborhood of (x, y) in $X \times X$. If there were some $(z, z) \in (U \times V) \cap \Delta$, then $z \in U \cap V = \varnothing$, a contradiction. Hence $(U \times V) \cap \Delta = \varnothing$, so $U \times V \subseteq (X \times X) \setminus \Delta$. Thus $(X \times X) \setminus \Delta$ is open, so Δ is closed.

($\impliedby$) Suppose Δ is closed in $X \times X$, so $(X \times X) \setminus \Delta$ is open. Let $x, y \in X$ with $x \neq y$. Then $(x, y) \in (X \times X) \setminus \Delta$. By definition of the product topology, there exists a basic open set $U \times V \subseteq X \times X$ such that $(x, y) \in U \times V \subseteq (X \times X) \setminus \Delta$. This implies $x \in U$ and $y \in V$. If there existed $z \in U \cap V$, then $(z, z) \in (U \times V) \cap \Delta$, contradicting $U \times V \subseteq (X \times X) \setminus \Delta$. Therefore $U \cap V = \varnothing$, so X is Hausdorff. □

Part (ii).

Proof. We check the three properties of a homeomorphism for $d\colon X \to \Delta$:

1. **Bijectivity:**
   - *Injective*: $d(x_1) = d(x_2) \implies (x_1, x_1) = (x_2, x_2) \implies x_1 = x_2$.
   - *Surjective*: Any element in Δ is of the form $(x, x) = d(x)$ for some $x \in X$.
2. **Continuity of d:** The component functions of d are $d_1 = \mathrm{id}_X$ and $d_2 = \mathrm{id}_X$. Since the identity map $\mathrm{id}_X\colon X \to X$ is continuous, $d\colon X \to X \times X$ is continuous by the characteristic property of product spaces. Restricting the codomain to the subspace Δ preserves continuity.
3. **Continuity of d^{-1}:** The inverse map $d^{-1}\colon \Delta \to X$ is given by $d^{-1}(x, x) = x$. This is the restriction of the first coordinate projection $p_1\colon X \times X \to X$ to Δ. Since projection maps are continuous, its restriction $d^{-1} = p_1|_\Delta$ is continuous.

Thus, $d\colon X \to \Delta$ is a homeomorphism. □

Problem 3.9

Let X and Y be topological spaces, and let $f, g\colon X \to Y$ be continuous maps. Assume that Y is a Hausdorff space. Prove the following statements:

(i) The coincidence set $E = \{x \in X \mid f(x) = g(x)\}$ is closed in X.

(ii) If D is a dense subset of X and $f|_D = g|_D$, then $f = g$ on all of X.

(iii) The graph of f, defined as $\Gamma_f = \{(x, f(x)) \in X \times Y \mid x \in X\}$, is closed in the product space $X \times Y$.

(iv) Assume that $f\colon X \to Y$ is a continuous, open, and surjective map. Then the fiber relation set $R = \{(x_1, x_2) \in X \times X \mid f(x_1) = f(x_2)\}$ is closed in $X \times X$ if and only if Y is Hausdorff.

Solution *submitted by* **Pula Karan**

Part (i).

Proof. Define $h\colon X \to Y \times Y$ by $h(x) = (f(x), g(x))$. Since f and g are continuous, h is continuous. Because Y is Hausdorff, the diagonal $\Delta_Y = \{(y, y) \in Y \times Y\}$ is closed in $Y \times Y$ by Problem 3.8(i). Notice that:

$$x \in E \iff f(x) = g(x) \iff (f(x), g(x)) \in \Delta_Y \iff x \in h^{-1}(\Delta_Y)$$

Thus $E = h^{-1}(\Delta_Y)$. Since Δ_Y is closed and h is continuous, E is closed in X. □

Part (ii).

Proof. Consider the coincidence set $E = \{x \in X \mid f(x) = g(x)\}$. By part (i), E is closed in X. Since $f|_D = g|_D$, we have $D \subseteq E$. Taking closures gives $\overline{D} \subseteq \overline{E} = E$. Since D is dense in X, $\overline{D} = X$. Therefore $X \subseteq E$, which means $E = X$, so $f(x) = g(x)$ for all $x \in X$. □

Part (iii).

Proof. Define $F\colon X \times Y \to Y \times Y$ by $F(x, y) = (f(x), y)$. F is continuous because its component maps $f \circ p_1$ and p_2 are continuous. Since Y is Hausdorff, Δ_Y is closed in $Y \times Y$. Notice that:

$$(x, y) \in \Gamma_f \iff y = f(x) \iff (f(x), y) \in \Delta_Y \iff F(x, y) \in \Delta_Y$$

Thus $\Gamma_f = F^{-1}(\Delta_Y)$. Since F is continuous and Δ_Y is closed, Γ_f is closed in $X \times Y$. □

Part (iv).

Proof. Consider the product map $f \times f\colon X \times X \to Y \times Y$ defined by $(f \times f)(x_1, x_2) = (f(x_1), f(x_2))$. Since f is continuous, open, and surjective, $f \times f$ is also continuous, open, and surjective. Notice that $R = (f \times f)^{-1}(\Delta_Y)$.

($\Longleftarrow$) If Y is Hausdorff, Δ_Y is closed in $Y \times Y$. By continuity of $f \times f$, $R = (f \times f)^{-1}(\Delta_Y)$ is closed in $X \times X$.

($\implies$) Suppose R is closed in $X \times X$. Since $f \times f$ is surjective, $(f \times f)(R) = \Delta_Y$. Taking complements yields:

$$(Y \times Y) \setminus \Delta_Y = (f \times f)((X \times X) \setminus R)$$

Since R is closed, $(X \times X) \setminus R$ is open. Because $f \times f$ is an open map, $(Y \times Y) \setminus \Delta_Y$ is open in $Y \times Y$, which means Δ_Y is closed in $Y \times Y$. By Problem 3.8(i), Y is Hausdorff. □

Problem 3.10

Let X be a topological space and $A \subset X$ be dense in X. Let Y be a Hausdorff space and $f\colon A \to Y$ be a continuous map. Show that if f may be extended to a continuous map $g\colon X \to Y$, then g is uniquely determined by f.

Solution

submitted by **Pula Karan**

Proof. Suppose $g_1, g_2\colon X \to Y$ are two continuous extensions of $f\colon A \to Y$, so $g_1|_A = f$ and $g_2|_A = f$.

1. Since $g_1|_A = g_2|_A = f$, the maps g_1 and g_2 agree on the subset $A \subseteq X$.
2. A is dense in X, and Y is a Hausdorff space.
3. By **Problem 3.9(ii)**, if two continuous functions from X into a Hausdorff space Y agree on a dense subset $A \subseteq X$, they must agree on all of X.

Therefore, $g_1(x) = g_2(x)$ for all $x \in X$, proving that the continuous extension g is uniquely determined by f. □

Problem Set 4

Problem 4.1

Let X and Y be two topological spaces; and $a \in X$, $b \in Y$ be two given points. Prove that the maps $f\colon X \to X \times Y$ and $g\colon Y \to X \times Y$ given by

$$f(x) = (x, b) \quad \text{and} \quad g(y) = (a, y)$$

are embeddings.

Solution

submitted by **Dhrubojoti**

Proof. Recall that a map $h\colon Z \to W$ is an embedding if it is a homeomorphism from Z onto $h(Z)$, equipped with the subspace topology.

1. The map $f\colon X \to X \times Y$, $f(x) = (x, b)$. *Step 1: f is continuous.* Let $U \times V$ be an arbitrary basic open set in $X \times Y$, where $U \subseteq X$ and $V \subseteq Y$ are open. Then

$$\begin{aligned} f^{-1}(U \times V) &= \{x \in X : f(x) \in U \times V\} \\ &= \{x \in X : (x, b) \in U \times V\} \\ &= \begin{cases} U, & b \in V, \\ \emptyset, & b \notin V. \end{cases} \end{aligned}$$

In either case, $f^{-1}(U \times V)$ is open in X. Hence f is continuous.

Step 2: f is injective. Suppose that $f(x_1) = f(x_2)$. Then

$$(x_1, b) = (x_2, b).$$

Therefore, $x_1 = x_2$, so f is injective.

Step 3: The inverse map $f^{-1}\colon f(X) \to X$ is continuous. Observe that

$$f(X) = X \times \{b\}.$$

Consider the first projection map

$$\pi_X\colon X \times Y \to X, \qquad \pi_X(x, y) = x.$$

Since projection maps are continuous, its restriction

$$\pi_X|_{f(X)}\colon f(X) \to X$$

is continuous. But

$$\pi_X(f(x)) = \pi_X(x, b) = x.$$

Thus

$$f^{-1} = \pi_X|_{f(X)}.$$

Hence f^{-1} is continuous.

Therefore, f is a homeomorphism from X onto $f(X)$, and consequently f is an embedding.

2. The map $g\colon Y \to X \times Y$, $g(y) = (a, y)$. *Step 1: g is continuous.* Let $U \times V$ be an arbitrary basic open set in $X \times Y$. Then

$$\begin{aligned} g^{-1}(U \times V) &= \{y \in Y : g(y) \in U \times V\} \\ &= \{y \in Y : (a, y) \in U \times V\} \\ &= \begin{cases} V, & a \in U, \\ \emptyset, & a \notin U. \end{cases} \end{aligned}$$

In either case, $g^{-1}(U \times V)$ is open in Y. Hence g is continuous.

Step 2: g is injective. Suppose that $g(y_1) = g(y_2)$. Then

$$(a, y_1) = (a, y_2).$$

Therefore, $y_1 = y_2$, so g is injective.

Step 3: The inverse map $g^{-1}\colon g(Y) \to Y$ is continuous. Observe that

$$g(Y) = \{a\} \times Y.$$

Consider the second projection map

$$\pi_Y\colon X \times Y \to Y, \qquad \pi_Y(x, y) = y.$$

Since projection maps are continuous, its restriction

$$\pi_Y|_{g(Y)}\colon g(Y) \to Y$$

is continuous. But

$$\pi_Y(g(y)) = \pi_Y(a, y) = y.$$

Thus

$$g^{-1} = \pi_Y|_{g(Y)}.$$

Hence g^{-1} is continuous.

Therefore, g is a homeomorphism from Y onto $g(Y)$, and consequently g is an embedding.

Hence both f and g are embeddings.

Problem 4.2

Suppose that $f, g\colon X \to \mathbb{R}$ are two continuous maps.

(i). Prove that the set $\{x \in X \mid f(x) \leq g(x)\}$ is closed in X.

(ii). Prove that the map $h\colon X \to \mathbb{R}$ given by $h(x) = \min\{f(x), g(x)\}$ is continuous.

Solution *submitted by* **Dhrubojoti**

Proof.

(i) The set $\{x \in X \mid f(x) \leq g(x)\}$ is closed in X. Let

$$A = \{x \in X \mid f(x) \leq g(x)\}.$$

We know that f and g are continuous. Since the subtraction map

$$\varphi\colon \mathbb{R} \times \mathbb{R} \to \mathbb{R}, \qquad \varphi(u, v) = u - v$$

is continuous, the map

$$F\colon X \to \mathbb{R}, \qquad F(x) = f(x) - g(x)$$

is continuous. Now observe that

$$\begin{aligned} A &= \{x \in X \mid f(x) \leq g(x)\} \\ &= \{x \in X \mid f(x) - g(x) \leq 0\} \\ &= F^{-1}((-\infty, 0]). \end{aligned}$$

Since $(-\infty, 0]$ is closed in $\mathbb{R}$ and F is continuous, its inverse image is closed in X. Therefore,

$$\boxed{\{x \in X \mid f(x) \leq g(x)\} \text{ is closed in } X.}$$

(ii) The map $h\colon X \to \mathbb{R}$ given by $h(x) = \min\{f(x), g(x)\}$ is continuous. We use the identity

$$\min\{a, b\} = \frac{a + b - |a - b|}{2}, \qquad a, b \in \mathbb{R}.$$

Therefore, for every $x \in X$,

$$h(x) = \frac{f(x) + g(x) - |f(x) - g(x)|}{2}.$$

Since f and g are continuous, the maps

$$f + g\colon X \to \mathbb{R} \quad \text{and} \quad f - g\colon X \to \mathbb{R}$$

are continuous. Also, the absolute value map

$$|\cdot|\colon \mathbb{R} \to \mathbb{R}, \qquad t \mapsto |t|$$

is continuous. Hence the composition

$$x \mapsto |f(x) - g(x)|$$

is continuous. Consequently,

$$h(x) = \frac{f(x) + g(x) - |f(x) - g(x)|}{2}$$

is continuous, since sums, differences and scalar multiples of continuous real-valued functions are continuous. Thus,

$$\boxed{h\colon X \to \mathbb{R} \text{ is continuous.}}$$

Problem 4.3

Let $\{A_\alpha\}$ be a collection of subsets of X such that $X = \bigcup_\alpha A_\alpha$. Suppose that $f\colon X \to Y$ is a map such that the restriction map $f|_{A_\alpha}\colon A_\alpha \to Y$ is continuous for each α.

(i). Prove that if the collection $\{A_\alpha\}$ is finite and each set A_α is closed, then f is continuous.

(ii). Find an example where the collection $\{A_\alpha\}$ is countable and each set A_α is closed, but f is not continuous.

(iii). An indexed family of sets $\{A_\alpha\}$ is said to be *locally finite* if each point x has a neighborhood that intersects A_α for only finitely many values of α. Prove that if the family $\{A_\alpha\}$ is locally finite and each set A_α is closed, then f is continuous.

Solution

submitted by **Dhrubojoti**

Recall. A map $f\colon X \to Y$ is continuous if and only if the preimage of every closed subset of Y is closed in X. Also, if A is a closed subset of X, then every relatively closed subset of A is closed in X.

(i) The collection $\{A_\alpha\}$ is finite and each A_α is closed.

Proof. Suppose

$$X = A_1 \cup A_2 \cup \cdots \cup A_n,$$

where each A_i is closed in X. We are given that the restriction

$$f|_{A_i}\colon A_i \to Y$$

is continuous for every $i = 1, 2, \ldots, n$. We need to prove that $f\colon X \to Y$ is continuous.

Let C be an arbitrary closed subset of Y. Consider

$$f^{-1}(C) = \{x \in X : f(x) \in C\}.$$

Since $X = \bigcup_{i=1}^n A_i$, we can write

$$f^{-1}(C) = \bigcup_{i=1}^{n} (f^{-1}(C) \cap A_i).$$

Now observe that

$$f^{-1}(C) \cap A_i = (f|_{A_i})^{-1}(C).$$

Since $f|_{A_i}$ is continuous and C is closed in Y, $(f|_{A_i})^{-1}(C)$ is closed in A_i. But A_i itself is closed in X. Therefore, a relatively closed subset of A_i is also closed in X. Thus, $f^{-1}(C) \cap A_i$ is closed in X for every i. Consequently,

$$f^{-1}(C) = \bigcup_{i=1}^{n} (f^{-1}(C) \cap A_i)$$

is a finite union of closed subsets of X. Hence, $f^{-1}(C)$ is closed in X. Since C was an arbitrary closed subset of Y, we conclude that

$$\boxed{f\colon X \to Y \text{ is continuous.}}$$

□

(ii) Find a countable collection of closed sets covering X for which f is not continuous. Example. Take

$$X = \mathbb{R}, \qquad Y = \mathbb{R}.$$

Define

$$A_0 = \{0\}$$

and, for $n \in \mathbb{N}$,

$$A_n = \left[-n, -\frac{1}{n}\right] \cup \left[\frac{1}{n}, n\right].$$

Step 1. Show that each A_n is closed. Clearly, $A_0 = \{0\}$ is closed in $\mathbb{R}$. Also, each A_n is a finite union of closed intervals. Hence, every A_n is closed in $\mathbb{R}$.

Step 2. Show that the collection is countable and covers $\mathbb{R}$. The collection

$$\{A_0, A_1, A_2, \dots\}$$

is countable. Moreover, every nonzero real number belongs to some A_n. Indeed, given $x \neq 0$, choose n sufficiently large such that

$$n \geq |x| \quad \text{and} \quad \frac{1}{n} \leq |x|.$$

Then $x \in A_n$. Since $0 \in A_0$, we obtain

$$\boxed{\mathbb{R} = \bigcup_{n=0}^{\infty} A_n.}$$

Step 3. Define the map f. Define $f\colon \mathbb{R} \to \mathbb{R}$ by

$$f(x) = \begin{cases} 1, & x = 0, \\ 0, & x \neq 0. \end{cases}$$

We now show that the restriction of f to every A_n is continuous.

- On $A_0 = \{0\}$, the restriction is the constant function $f(0) = 1$, which is continuous.
- On every A_n, $n \geq 1$, we have $0 \notin A_n$. Therefore, $f|_{A_n}$ is the constant zero function, which is continuous.

Thus, every restriction $f|_{A_n}\colon A_n \to \mathbb{R}$ is continuous.

Step 4. Show that f is not continuous. Consider the open set

$$U = \left(\frac{1}{2}, \frac{3}{2}\right) \subseteq \mathbb{R}.$$

We have

$$f^{-1}(U) = \{0\}.$$

But $\{0\}$ is not open in $\mathbb{R}$. Therefore, f is not continuous.

Hence, we have constructed a countable closed cover of $\mathbb{R}$ such that every restriction is continuous, but the original map is not continuous. Thus,

$$\boxed{\text{The conclusion of (i) need not hold for a countable cover.}}$$

□

(iii) Prove the result when $\{A_\alpha\}$ is locally finite and each A_α is closed.

Proof. We are given that:

1. $X = \bigcup_\alpha A_\alpha$.
2. Every A_α is closed in X.
3. The family $\{A_\alpha\}$ is locally finite.
4. Every restriction $f|_{A_\alpha} : A_\alpha \to Y$ is continuous.

We need to prove that $f : X \to Y$ is continuous. We will use the following important result.

Lemma. *The union of a locally finite family of closed subsets of a topological space is closed.*

Proof of the lemma. Let $\{F_\alpha\}$ be a locally finite family of closed subsets of X. We want to show that

$$F = \bigcup_\alpha F_\alpha$$

is closed in X. Take an arbitrary point $x \in X \setminus F$. Since the family is locally finite, there exists an open neighbourhood U of x that intersects only finitely many members of the family. Suppose these members are

$$F_{\alpha_1}, F_{\alpha_2}, \ldots, F_{\alpha_n}.$$

Now, $x \notin F$, so in particular,

$$x \notin F_{\alpha_i}, \qquad i = 1, \ldots, n.$$

Since each F_{α_i} is closed, its complement is open. Thus, $X \setminus F_{\alpha_i}$ is an open neighbourhood of x. Consider

$$V = U \cap \bigcap_{i=1}^{n} (X \setminus F_{\alpha_i}).$$

This is an open neighbourhood of x, being a finite intersection of open sets. We claim that

$$V \cap F = \emptyset.$$

Indeed:

- For the finitely many sets $F_{\alpha_1}, \ldots, F_{\alpha_n}$, the set V does not intersect any of them.
- For every other index α, U, and hence V, does not intersect F_α, by the choice of U.

Therefore, V does not intersect their union F. So every point in $X \setminus F$ has an open neighbourhood contained in $X \setminus F$. Hence, $X \setminus F$ is open, which means

$$\bigcup_\alpha F_\alpha \text{ is closed in } X.$$

This proves the lemma.

Now we prove the required result. Let C be an arbitrary closed subset of Y. Consider $f^{-1}(C)$. Since $X = \bigcup_\alpha A_\alpha$, we have

$$\begin{aligned} f^{-1}(C) &= \bigcup_\alpha (f^{-1}(C) \cap A_\alpha) \\ &= \bigcup_\alpha (f|_{A_\alpha})^{-1}(C). \end{aligned}$$

Now, since every restriction $f|_{A_\alpha}$ is continuous and C is closed in Y, $(f|_{A_\alpha})^{-1}(C)$ is closed in A_α. Since A_α is closed in X, it follows that

$$F_\alpha := (f|_{A_\alpha})^{-1}(C)$$

is closed in X. Furthermore, since $F_\alpha \subseteq A_\alpha$ and the family $\{A_\alpha\}$ is locally finite, the family $\{F_\alpha\}$ is also locally finite. Therefore, by our lemma, the union

$$f^{-1}(C) = \bigcup_\alpha F_\alpha$$

is closed in X. Since C was an arbitrary closed subset of Y, the preimage of every closed subset of Y is closed in X. Consequently,

$$\boxed{f\colon X \to Y \text{ is continuous.}}$$ □

Problem 4.4

Let $f\colon X \to A$ and $g\colon Y \to B$ be two continuous maps. Define $f \times g\colon X \times Y \to A \times B$ by $(f \times g)(x, y) = (f(x), g(y))$. Prove that $f \times g$ is a continuous map.

Solution

submitted by **Dhrubojoti**

Proof. We are given that

$$f\colon X \to A \quad \text{and} \quad g\colon Y \to B$$

are continuous maps. We need to prove that

$$f \times g\colon X \times Y \to A \times B$$

is continuous. Recall that the product topology on $A \times B$ has a basis consisting of sets of the form $U \times V$, where U is open in A and V is open in B. We will use the characterization of continuity in terms of the preimages of basic open sets.

Step 1. Consider an arbitrary basic open set in $A \times B$. Let $U \times V$ be an arbitrary basic open set in $A \times B$, where $U \subseteq A$ and $V \subseteq B$ are open. We have

$$\begin{aligned}(f \times g)^{-1}(U \times V) &= \{(x, y) \in X \times Y : (f \times g)(x, y) \in U \times V\} \\ &= \{(x, y) \in X \times Y : (f(x), g(y)) \in U \times V\} \\ &= \{(x, y) \in X \times Y : f(x) \in U,\ g(y) \in V\} \\ &= f^{-1}(U) \times g^{-1}(V).\end{aligned}$$

Therefore,

$$(f \times g)^{-1}(U \times V) = f^{-1}(U) \times g^{-1}(V).$$

Step 2. Show that the preimage is open in $X \times Y$. Since $f\colon X \to A$ is continuous and U is open in A, we know that $f^{-1}(U)$ is open in X. Similarly, since $g\colon Y \to B$ is continuous and V is open in B, $g^{-1}(V)$ is open in Y. Hence, $f^{-1}(U) \times g^{-1}(V)$ is open in $X \times Y$, by the definition of the product topology. Consequently, $(f \times g)^{-1}(U \times V)$ is open in $X \times Y$.

Step 3. Conclude continuity. We have shown that the preimage under $f \times g$ of every basic open set in $A \times B$ is open in $X \times Y$. Since every open subset of $A \times B$ is a union of basic open sets, the preimage of every open subset of $A \times B$ is open in $X \times Y$. Therefore,

$$\boxed{f \times g\colon X \times Y \to A \times B \text{ is continuous.}}$$ □

Problem 4.5

Let (X, d) be a metric space and let $S \subseteq X$ be a non-empty subset. For any $x \in X$, define the distance from x to S by $d(x, S) = \inf_{s \in S} d(x, s)$.

(1) Show that for any $x, y \in X$, $|d(x, S) - d(y, S)| \leq d(x, y)$.

(2) Conclude that $f_S \colon X \to \mathbb{R}$ defined by $f_S(x) = d(x, S)$ is continuous on X.

Solution

submitted by **Aaditya**

Proof.

(1) By the triangle inequality, for every $s \in S$,

$$d(x, s) \leq d(x, y) + d(y, s).$$

Taking the infimum over $s \in S$, we get

$$d(x, S) \leq d(x, y) + d(y, S).$$

Thus,

$$d(x, S) - d(y, S) \leq d(x, y).$$

Interchanging x and y gives

$$d(y, S) - d(x, S) \leq d(x, y).$$

Therefore,

$$\boxed{|d(x, S) - d(y, S)| \leq d(x, y).}$$

(2) Let $x_0 \in X$ and $\varepsilon > 0$. Choose $\delta = \varepsilon$.

If $d(x, x_0) < \delta$, then by (1),

$$\begin{aligned} |f_S(x) - f_S(x_0)| &= |d(x, S) - d(x_0, S)| \\ &\leq d(x, x_0) \\ &< \varepsilon. \end{aligned}$$

Hence, f_S is continuous on X. □

Problem 4.6

Let $\{A_n\}_{n\in\mathbb{N}}$ be a sequence of connected subspaces of X such that $A_n \cap A_{n+1} \neq \emptyset$ for all $n \in \mathbb{N}$. Prove that $\bigcup_{n=1}^{\infty} A_n$ is connected.

Solution

submitted by **Aaditya**

Proof.

Let

$$A = \bigcup_{n=1}^{\infty} A_n.$$

Suppose, for contradiction, that A is disconnected. Then

$$A = U \cup V,$$

where U, V are non-empty, disjoint and open in A.

Since each A_n is connected and

$$A_n = (A_n \cap U) \cup (A_n \cap V),$$

we must have

$$A_n \subseteq U \quad \text{or} \quad A_n \subseteq V.$$

Without loss of generality, let $A_1 \subseteq U$.

Since $A_1 \cap A_2 \neq \emptyset$, there exists $p \in A_1 \cap A_2$. Thus $p \in U$, so $A_2 \not\subseteq V$. Consequently, $A_2 \subseteq U$.

Repeating this argument inductively, we obtain

$$A_n \subseteq U \quad \text{for every } n \in \mathbb{N}.$$

Therefore,

$$A = \bigcup_{n=1}^{\infty} A_n \subseteq U,$$

contradicting $V \neq \emptyset$.

Hence,

$$\boxed{\bigcup_{n=1}^{\infty} A_n \text{ is connected.}}$$

□

Problem 4.7

Determine whether or not $\mathbb{R}^\omega$ is connected under the uniform topology.

Solution

submitted by **Aaditya**

Proof.

The uniform metric is

$$\bar{\rho}(x, y) = \sup_n \min\{|x_n - y_n|, 1\}.$$

Define

$$\mathcal{B} = \left\{ x \in \mathbb{R}^\omega : \sup_n |x_n| < \infty \right\}$$

and

$$\mathcal{U} = \mathbb{R}^\omega \setminus \mathcal{B}.$$

We show that both sets are open.

First, observe that

$$\bar{\rho}(x, y) < 1 \implies |x_n - y_n| < 1 \quad \forall n.$$

Claim 1: $\mathcal{B}$ is open.

Let $x \in \mathcal{B}$. Then $|x_n| \leq M$ for some $M > 0$ and all n.

If $\bar{\rho}(x, y) < 1$, then

$$|y_n| \leq |x_n| + |y_n - x_n| < M + 1.$$

Hence $y \in \mathcal{B}$, proving that $\mathcal{B}$ is open.

Claim 2: $\mathcal{U}$ is open.

Let $x \in \mathcal{U}$ and suppose $\bar{\rho}(x, y) < 1$.

If y were bounded, say $|y_n| \leq M$, then

$$|x_n| \leq |y_n| + |x_n - y_n| < M + 1.$$

This contradicts the fact that x is unbounded.

Thus $y \in \mathcal{U}$, proving that $\mathcal{U}$ is open.

Finally, $\mathcal{B}$ and $\mathcal{U}$ are disjoint, non-empty, and

$$\mathcal{B} \cup \mathcal{U} = \mathbb{R}^\omega.$$

Therefore, they form a separation of $\mathbb{R}^\omega$. Hence,

$$\boxed{\mathbb{R}^\omega \text{ is not connected.}}$$

□

Problem 4.8

Determine whether or not the lower limit topology $\mathbb{R}_\ell$ on $\mathbb{R}$ is connected.

Solution

submitted by **Aaditya**

Proof.

The lower limit topology has the basis

$$\mathcal{B} = \{[a, b) : a < b\}.$$

Consider

$$U = (-\infty, 0), \qquad V = [0, \infty).$$

We show that both sets are open.

If $x \in U$, then

$$x \in [x, 0) \subseteq U.$$

Since $[x, 0)$ is a basis element, U is open.

Similarly, if $x \in V$, then

$$x \in [x, x + 1) \subseteq V.$$

Thus, V is also open.

Moreover,

$$U \cap V = \emptyset, \qquad U \cup V = \mathbb{R}.$$

Both sets are non-empty. Therefore, (U, V) forms a separation of $\mathbb{R}_\ell$.

Hence,

$$\boxed{\mathbb{R}_\ell \text{ is not connected.}}$$

□

Problem 4.9

Let $C \subset X$ be a connected subspace of X and $A \subset X$. Prove that if C intersects both A and $X \setminus A$, then C intersects $\operatorname{Bd} A$.

Solution

submitted by **Harisankar B**

Recall that $\operatorname{Bd} A = \overline{A} \cap \overline{X \setminus A}$, and that $X \setminus \overline{X \setminus A} = \operatorname{Int} A$ and $X \setminus \overline{A} = \operatorname{Int}(X \setminus A)$.

Proof. Suppose, for contradiction, that $C \cap \operatorname{Bd} A = \emptyset$. Consider the two sets

$$P = C \cap \operatorname{Int} A, \qquad Q = C \cap (X \setminus \overline{A}).$$

Both $\operatorname{Int} A$ and $X \setminus \overline{A}$ are open in X, so P and Q are open in the subspace C. They are disjoint, since $\operatorname{Int} A \subseteq A \subseteq \overline{A}$.

$P \cup Q = C$. Let $c \in C$. If $c \in \overline{A}$, then $c \notin \operatorname{Bd} A$ forces $c \notin \overline{X \setminus A}$, i.e. $c \in \operatorname{Int} A$, so $c \in P$. Otherwise $c \in X \setminus \overline{A}$, so $c \in Q$.

$P \neq \emptyset$. Pick $a \in C \cap A$. Then $a \in \overline{A}$, and as above $a \notin \operatorname{Bd} A$ gives $a \in \operatorname{Int} A$; hence $a \in P$.

$Q \neq \emptyset$. Pick $b \in C \cap (X \setminus A)$. Then $b \in \overline{X \setminus A}$, and $b \notin \operatorname{Bd} A$ forces $b \notin \overline{A}$; hence $b \in Q$.

Thus P, Q form a separation of C, contradicting the connectedness of C. Therefore $C \cap \operatorname{Bd} A \neq \emptyset$. □

Problem 4.10

Let X and Y be two connected subspaces and $A \subsetneq X$ and $B \subsetneq Y$. Prove that $(X \times Y) \setminus (A \times B)$ is connected.

Solution

submitted by **Harisankar B**

Let $Z = (X \times Y) \setminus (A \times B)$. We use two standard facts:

(a) For fixed $x \in X$ the map $y \mapsto (x, y)$ is an embedding of Y onto $\{x\} \times Y$ (Problem 4.1); hence $\{x\} \times Y$ is connected. Likewise $X \times \{y\}$ is connected for every $y \in Y$.

(b) A union of connected subspaces having a point in common is connected; in particular, the union of two connected subspaces that intersect is connected.

Proof. Since $A \subsetneq X$ and $B \subsetneq Y$, we can fix $x_0 \in X \setminus A$ and $y_0 \in Y \setminus B$. Note that a point (x, y) lies in $A \times B$ only if $x \in A$ *and* $y \in B$; hence

$$Z = \{(x, y) \in X \times Y : x \notin A \text{ or } y \notin B\}.$$

For $x \in X \setminus A$ and $y \in Y \setminus B$ define

$$T_x = (\{x\} \times Y) \cup (X \times \{y_0\}), \qquad S_y = (X \times \{y\}) \cup (\{x_0\} \times Y).$$

Each T_x and S_y lies in Z. Points of $\{x\} \times Y$ have first coordinate $x \notin A$; points of $X \times \{y_0\}$ have second coordinate $y_0 \notin B$. Similarly for S_y, using $y \notin B$ and $x_0 \notin A$.

Each T_x and S_y is connected. T_x is the union of the connected sets $\{x\} \times Y$ and $X \times \{y_0\}$ (fact (a)), which meet at (x, y_0); so T_x is connected by fact (b). Likewise S_y is the union of the connected sets $X \times \{y\}$ and $\{x_0\} \times Y$, which meet at (x_0, y).

They share a common point. Every T_x contains $X \times \{y_0\} \ni (x_0, y_0)$, and every S_y contains $\{x_0\} \times Y \ni (x_0, y_0)$.

They cover Z. If $(x, y) \in Z$ with $x \notin A$, then $(x, y) \in \{x\} \times Y \subseteq T_x$; if $y \notin B$, then $(x, y) \in X \times \{y\} \subseteq S_y$.

Therefore

$$Z = \bigcup_{x \in X \setminus A} T_x \cup \bigcup_{y \in Y \setminus B} S_y$$

is a union of connected subspaces all containing the point (x_0, y_0), and hence Z is connected by fact (b). □

Problem 4.11

Let $Y \subset X$; and let X and Y be connected. Prove that if A and B form a separation of $X \setminus Y$, then both $Y \cup A$ and $Y \cup B$ are connected.

Solution

submitted by **Harisankar B**

We use the following characterisation of separations (Munkres, Lemma 23.1): *if W is a subspace of X, then a separation of W is a pair of disjoint non-empty sets whose union is W, neither of which contains a limit point (in X) of the other.* We also use that if Y is connected and $W = C \cup D$ is a separation with $Y \subseteq W$, then Y lies entirely in C or entirely in D.

Proof. We show that $Y \cup A$ is connected; the argument for $Y \cup B$ is identical with A and B interchanged.

Suppose, for contradiction, that C, D form a separation of $Y \cup A$. Since Y is connected, Y lies entirely in one of them; say $Y \subseteq C$. Then $D \subseteq (Y \cup A) \setminus Y \subseteq A$. We claim that

$$X = (C \cup B) \cup D$$

is a separation of X.

Union and disjointness. $C \cup D = Y \cup A$ and $A \cup B = X \setminus Y$, so $(C \cup B) \cup D = Y \cup A \cup B = X$. Moreover $D \cap C = \emptyset$, and $D \cap B \subseteq A \cap B = \emptyset$.

Non-emptiness. $D \neq \emptyset$ and $C \neq \emptyset$, as C, D is a separation; hence $C \cup B \neq \emptyset$.

No limit points across. By Lemma 23.1 applied to the separations C, D of $Y \cup A$ and A, B of $X \setminus Y$:

- D contains no limit point of C, and C contains no limit point of D;
- A contains no limit point of B, and B contains no limit point of A.

Since $D \subseteq A$, the set D contains no limit point of B; so D contains no limit point of $C \cup B$ (a limit point of a union is a limit point of one of the sets). Also every limit point of D is a limit point of A, so B contains no limit point of D; together with the first bullet, $C \cup B$ contains no limit point of D.

By Lemma 23.1, $C \cup B$ and D form a separation of X, contradicting the connectedness of X. Hence $Y \cup A$ is connected, and similarly $Y \cup B$ is connected. □

Problem Set 5

Problem 5.1

Prove that the punctured Euclidean space $X = \mathbb{R}^n \setminus \{0\}$ is path connected if and only if $n > 1$.

Solution

submitted by **Mayuri**

We show that $X = \mathbb{R}^n \setminus \{0\}$ is path connected if and only if $n > 1$ (here $n \geq 1$).

Case $n = 1$. Then $X = (-\infty, 0) \cup (0, \infty)$, a union of two disjoint nonempty open sets, so X is not connected, hence not path connected. Directly: if $\gamma\colon [0,1] \to X$ were a path from -1 to 1, the intermediate value theorem would give t with $\gamma(t) = 0 \notin X$, a contradiction.

Case $n \geq 2$. Let $x, y \in X$. Recall the segment $[x, y] = \{(1-t)x + ty : t \in [0,1]\}$ is the image of a path (a straight line path).

If $0 \notin [x, y]$, the straight line path from x to y lies in X.

Otherwise $0 = (1-t)x + ty$ for some $t \in [0,1]$. Since $x, y \neq 0$ we have $t \in (0,1)$, so x and y are linearly dependent: $y = -cx$ with $c = \frac{1-t}{t} > 0$. Since $n \geq 2$, choose $z \in \mathbb{R}^n$ linearly independent from x (then also from y, as y is a nonzero multiple of x). In particular $z \neq 0$.

Claim. $0 \notin [x, z]$.

If $0 = (1-s)x + sz$ with $s \in [0,1]$, then $s \neq 0$ (as $x \neq 0$) and $s \neq 1$ (as $z \neq 0$), so $z = -(\frac{1-s}{s})x$, contradicting linear independence. The same argument shows $0 \notin [z, y]$.

Hence the segment from x to z followed by the segment from z to y is a path in X from x to y. So X is path connected. □

Problem 5.2

Prove that $\mathbb{R}^2 \setminus \mathbb{Q}^2$ with the subspace topology of $\mathbb{R}^2$ is connected, but $\mathbb{R} \setminus \mathbb{Q}$ with the subspace topology of $\mathbb{R}$ is not connected.

Solution

submitted by **Mayuri**

$\mathbb{R}^2 \setminus \mathbb{Q}^2$ *is connected.* We show it is path connected, which implies connected. Let $x, y \in \mathbb{R}^2 \setminus \mathbb{Q}^2$ with $x \neq y$.

For a point $p \in \mathbb{R}^2$, consider the lines through p; they are parametrized by their direction angle $\theta \in [0, \pi)$, so there are uncountably many. For each $q \in \mathbb{Q}^2$ there is exactly one line through x and q (note $q \neq x$ since $x \notin \mathbb{Q}^2$). Since $\mathbb{Q}^2$ is countable, only countably many lines through x meet $\mathbb{Q}^2$. So all but countably many lines through x are contained entirely in $\mathbb{R}^2 \setminus \mathbb{Q}^2$. The same holds for y.

Choose a line L_x through x missing $\mathbb{Q}^2$. Then choose a line L_y through y missing $\mathbb{Q}^2$ with a direction different from that of L_x (possible, since uncountably many directions are available and only countably many are bad, and we exclude just one). Then L_x and L_y are not parallel, so they meet at a single point p. Go from x to p along L_x, then from p to y along L_y. This is a path in $\mathbb{R}^2 \setminus \mathbb{Q}^2$ from x to y. Hence $\mathbb{R}^2 \setminus \mathbb{Q}^2$ is path connected, and therefore connected.

$\mathbb{R} \setminus \mathbb{Q}$ *is not connected.* Let $U = (-\infty, 0) \cap (\mathbb{R} \setminus \mathbb{Q})$ and $V = (0, \infty) \cap (\mathbb{R} \setminus \mathbb{Q})$. Both are open in the subspace $\mathbb{R} \setminus \mathbb{Q}$ and they are disjoint. Since $0 \in \mathbb{Q}$, we have $0 \notin \mathbb{R} \setminus \mathbb{Q}$, so $U \cup V = \mathbb{R} \setminus \mathbb{Q}$. Both are nonempty ($-\sqrt{2} \in U$, $\sqrt{2} \in V$). So $\mathbb{R} \setminus \mathbb{Q}$ is the union of two disjoint nonempty open sets, hence disconnected. □

Problem 5.3

Let X and Y be topological spaces, and suppose there exist embeddings $f\colon X \to Y$ and $g\colon Y \to X$. Is it necessary that X and Y are homeomorphic? Prove or provide a counterexample.

Solution

submitted by **Mayuri**

No, it is not necessary.

Counterexample. Let $X = [0, 1]$ and $Y = (0, 1)$, both with the subspace topology from $\mathbb{R}$.

- $f\colon X \to Y$, $f(x) = \frac{1}{4} + \frac{x}{2}$. This is continuous and injective, with image $[\frac{1}{4}, \frac{3}{4}]$, and the inverse $[\frac{1}{4}, \frac{3}{4}] \to [0, 1]$, $y \mapsto 2y - \frac{1}{2}$, is continuous. So f is a homeomorphism onto its image, i.e. an embedding.
- $g\colon Y \to X$, the inclusion $g(y) = y$. It is an embedding, since the topology on Y is the subspace topology inherited from $\mathbb{R}$, which is also the subspace topology inherited from X.

But X and Y are not homeomorphic: $X = [0, 1]$ is compact (Heine–Borel), whereas $Y = (0, 1)$ is not, since the open cover $\{(\frac{1}{n}, 1) : n \geq 2\}$ has no finite subcover (a finite subfamily has union $(\frac{1}{N}, 1)$ for the largest $n = N$ used, which misses $\frac{1}{2N}$, a point of Y). Compactness is preserved by homeomorphisms, so $X \ncong Y$. □

Problem 5.4

Let X be a non-empty topological space and let $\mathscr{B}$ be a base for the topology on X. Prove that X is compact if and only if every open cover of X consisting of elements of $\mathscr{B}$ has a finite subcover.

Solution

submitted by **Mayuri**

Let $\mathscr{B}$ be a base for the topology on X.

($\Longrightarrow$) If X is compact, then every open cover of X has a finite subcover. A cover consisting of elements of $\mathscr{B}$ is in particular an open cover (basis elements are open), so it has a finite subcover.

($\Longleftarrow$) Suppose every open cover of X consisting of elements of $\mathscr{B}$ has a finite subcover. Let $\mathcal{U}$ be an arbitrary open cover of X. For each $x \in X$ choose $U_x \in \mathcal{U}$ with $x \in U_x$, and since U_x is open and $\mathscr{B}$ is a base, choose $B_x \in \mathscr{B}$ with $x \in B_x \subseteq U_x$.

Then $\{B_x : x \in X\}$ is a cover of X (each $x \in B_x$) by elements of $\mathscr{B}$. By hypothesis, there are $x_1, \ldots, x_n \in X$ with $X = B_{x_1} \cup \cdots \cup B_{x_n}$. Since $B_{x_i} \subseteq U_{x_i}$, we get

$$X = U_{x_1} \cup \cdots \cup U_{x_n},$$

so $\{U_{x_1}, \ldots, U_{x_n}\}$ is a finite subcover of $\mathcal{U}$. Hence X is compact. □

Problem 5.5

Let $\mathcal{T}_1$ and $\mathcal{T}_2$ be two topologies on X such that both $(X, \mathcal{T}_1)$ and $(X, \mathcal{T}_2)$ are compact Hausdorff spaces. Prove that either $\mathcal{T}_1 = \mathcal{T}_2$ or they are not comparable.

Solution

submitted by **Prabhu**

Proof. Suppose $\mathcal{T}_1$ and $\mathcal{T}_2$ are comparable. Without loss of generality, let

$$\mathcal{T}_1 \subseteq \mathcal{T}_2.$$

Consider the identity map

$$I \colon (X, \mathcal{T}_2) \to (X, \mathcal{T}_1), \qquad I(x) := x.$$

Clearly, I is *bijective*.

For all $U \in \mathcal{T}_1$, we have

$$I^{-1}(U) = U \in \mathcal{T}_2.$$

Hence, I is *continuous*.

Since I is a bijective continuous map from a compact space to a Hausdorff space, I is a *homeomorphism*. So, I^{-1} is also continuous.

Thus, for all $U \in \mathcal{T}_2$, we have

$$(I^{-1})^{-1}(U) = I(U) = U \in \mathcal{T}_1.$$

Hence,

$$\mathcal{T}_2 \subseteq \mathcal{T}_1.$$

Therefore,

$$\boxed{\mathcal{T}_1 = \mathcal{T}_2.}$$

Hence, either $\mathcal{T}_1 = \mathcal{T}_2$ or they are not comparable. □

Problem 5.6

Prove that every compact subspace of a metric space is closed and bounded in that metric topology. Find an example of a metric space in which a closed and bounded subspace is not compact.

Solution

submitted by **Prabhu**

Proof. Let (X, d) be a metric space and let $K \subseteq X$ be compact.

We know that every metric space is Hausdorff and a compact subspace of a Hausdorff space is closed. Since X is a metric space it is Hausdorff. And K is a compact subspace of X, thus it is *closed* in X.

Let $a \in X$. Consider the collection

$$\mathcal{U} := \{B(a, n) : n \in \mathbb{N}\},$$

where

$$B(a,n) := \{x \in X : d(x,a) < n\}.$$

For all $x \in K$, $d(x,a)$ is finite; so, $\exists\, n \in \mathbb{N}$ such that $d(x,a) < n$; i.e., $x \in B(a,n)$. Thus, $\mathcal{U}$ is an open cover of A.

Since K is compact, there exists a finite subcover $\{B(a,n_1),\ldots,B(a,n_m)\}$. Therefore,

$$K \subseteq \bigcup_{i=1}^{m} B(a,n_i).$$

Let $N := \max\{n_1,n_2,\ldots,n_m\}$. Since $B(a,n_i) \subseteq B(a,N)$ for every i, we get

$$K \subseteq \bigcup_{i=1}^{m} B(a,n_i) \subseteq B(a,N).$$

Therefore, K is *bounded*. Hence,

$$\boxed{K \text{ is closed and bounded.}}$$

□

Example: Consider the metric space $X := (0,1)$ with the usual metric ($\because d(x,y) := |x-y|$), and let $A := X$. Clearly, A is closed ($\because$ it is the entire space) and bounded ($\because \forall x \in A,\ 0 < x < 1$) in X.

However, consider the collection

$$\mathcal{U} = \left\{ \left(\frac{1}{n}, 1\right) : n \in \mathbb{N},\ n \geq 2 \right\}.$$

Each member of $\mathcal{U}$ is open in the subspace $(0,1)$. And,

$$(0,1) = \bigcup_{n=2}^{\infty} \left(\frac{1}{n}, 1\right).$$

Thus, $\mathcal{U}$ is an open cover of A.

For every finite subcollection $\left\{\left(\frac{1}{n_1},1\right), \left(\frac{1}{n_2},1\right), \ldots, \left(\frac{1}{n_k},1\right)\right\}$, let $N = \max\{n_1,n_2,\ldots,n_k\}$. Since $n_i \leq N$ for every $i = 1,\ldots,k$, we have

$$\left(\frac{1}{n_i},1\right) \subseteq \left(\frac{1}{N},1\right).$$

Therefore,

$$\bigcup_{i=1}^{k} \left(\frac{1}{n_i},1\right) \subseteq \left(\frac{1}{N},1\right) \subsetneq (0,1).$$

It cannot cover $(0,1)$, since points in $(0,1/N]$ remain uncovered. Hence, $\mathcal{U}$ has no finite subcover. Therefore,

$$\boxed{(0,1) \text{ is closed and bounded but not compact in } (0,1).}$$

Problem 5.7

Let X be a compact metric space. Prove that if $f\colon X \to X$ is an isometry (distance preserving map) of X, then f is a homeomorphism.

Solution

submitted by **Jai**

Proof. Let d denote the metric on X, so that $d(f(x), f(y)) = d(x, y)$ for all $x, y \in X$.

Step 1: f is continuous and injective. Given $\varepsilon > 0$, take $\delta = \varepsilon$; then $d(x, y) < \delta$ implies $d(f(x), f(y)) = d(x, y) < \varepsilon$, so f is continuous. If $f(x) = f(y)$, then $d(x, y) = d(f(x), f(y)) = 0$, so $x = y$; hence f is injective.

Step 2: f is surjective. Suppose, for contradiction, that there is $y \in X \setminus f(X)$. Since X is compact and f is continuous, $f(X)$ is compact, hence closed in the metric space X. Therefore there exists $\varepsilon > 0$ such that $B(y, \varepsilon) \cap f(X) = \emptyset$, i.e.

$$d(y, f(x)) \geq \varepsilon \qquad \text{for all } x \in X.$$

Define a sequence $(x_n)_{n\geq 0}$ by $x_n = f^n(y)$ (with $f^0 = \mathrm{id}_X$). For $n > m \geq 0$, since f^m is an isometry,

$$d(x_n, x_m) = d\big(f^m(f^{\,n-m}(y)),\ f^m(y)\big) = d\big(f^{\,n-m}(y),\ y\big) = d\big(y, f(f^{\,n-m-1}(y))\big) \geq \varepsilon,$$

because $n - m \geq 1$. Thus any two distinct terms of the sequence are at distance at least ε, so no subsequence of (x_n) is Cauchy and hence none converges. This contradicts the fact that a compact metric space is sequentially compact. Hence f is surjective.

Step 3: f^{-1} is continuous. By Steps 1 and 2, f is a bijection. For $a, b \in X$,

$$d(f^{-1}(a), f^{-1}(b)) = d\big(f(f^{-1}(a)), f(f^{-1}(b))\big) = d(a, b),$$

so f^{-1} is also an isometry, hence continuous by Step 1.

Therefore f is a continuous bijection with continuous inverse, i.e. a homeomorphism. □

Problem 5.8

Let $\mathbb{R}$ be given the cocountable topology. Is the closed interval $[0, 1]$ a compact subspace of $\mathbb{R}$ under this topology?

Solution

submitted by **Jai**

Answer. No, $[0, 1]$ is not compact.

Proof. Recall that in the cocountable topology, $U \subseteq \mathbb{R}$ is open if and only if $U = \emptyset$ or $\mathbb{R} \setminus U$ is countable. Let

$$C = \{c_n = \tfrac{1}{n} : n \in \mathbb{N}\} \subseteq [0, 1],$$

a countably infinite set. Define

$$U_0 = \mathbb{R} \setminus C, \qquad U_n = \mathbb{R} \setminus \{c_k : k \neq n\} \quad (n \in \mathbb{N}).$$

Each of these sets has countable complement, so each is open in $\mathbb{R}$. Note that

$$U_0 \cap C = \emptyset, \qquad U_n \cap C = \{c_n\}.$$

The family covers $[0,1]$**.** If $x \in [0,1] \setminus C$, then $x \in U_0$; if $x = c_n$ for some n, then $x \in U_n$. Hence $\{U_n \cap [0,1]\}_{n\geq 0}$ is an open cover of $[0,1]$ in the subspace topology.

No finite subcover. Let $F \subseteq \{0\} \cup \mathbb{N}$ be finite. Choose $m \in \mathbb{N}$ with $m \notin F$. Then

$$c_m \notin \bigcup_{n\in F} U_n,$$

since U_0 contains no point of C, and for $n \in F \setminus \{0\}$ the only point of C in U_n is $c_n \neq c_m$. So no finite subfamily covers $[0,1]$.

Thus $[0,1]$ is not compact in the cocountable topology. □

Remark. In fact, the same argument shows that a subset of $\mathbb{R}$ is compact in the cocountable topology if and only if it is finite.

Problem 5.9

Let $f\colon X \to Y$ be a continuous map between topological spaces. Show that if X is compact and Y is Hausdorff, then f is a closed map.

Solution

submitted by **Jai**

We use three standard facts, proved here for completeness.

Lemma (A)**.** *A closed subset of a compact space is compact.*

Proof. Let $C \subseteq X$ be closed, X compact, and let $\{U_\alpha\}$ be an open cover of C by open sets of X. Then $\{U_\alpha\} \cup \{X \setminus C\}$ is an open cover of X, so it has a finite subcover. Discarding $X \setminus C$ (which contains no point of C) leaves a finite subcover of C. □

Lemma (B)**.** *A continuous image of a compact space is compact.*

Proof. Let K be compact, $f\colon K \to Y$ continuous, and $\{V_\alpha\}$ an open cover of $f(K)$. Then $\{f^{-1}(V_\alpha)\}$ is an open cover of K; pick a finite subcover $f^{-1}(V_{\alpha_1}), \dots, f^{-1}(V_{\alpha_n})$. Then $V_{\alpha_1}, \dots, V_{\alpha_n}$ cover $f(K)$. □

Lemma (C)**.** *A compact subset K of a Hausdorff space Y is closed.*

Proof. Let $y \in Y \setminus K$. For each $k \in K$, by the Hausdorff property choose disjoint open sets $U_k \ni k$ and $V_k \ni y$. The sets $\{U_k\}_{k\in K}$ cover K, so there are $k_1, \dots, k_n$ with $K \subseteq U_{k_1} \cup \cdots \cup U_{k_n}$. Then $V = V_{k_1} \cap \cdots \cap V_{k_n}$ is an open neighbourhood of y disjoint from $U_{k_1} \cup \cdots \cup U_{k_n} \supseteq K$. Hence $Y \setminus K$ is open. □

Solution to Problem 5.9. Let $C \subseteq X$ be closed. By Lemma A, C is compact. Since f is continuous, $f(C)$ is compact by Lemma B. Since Y is Hausdorff, $f(C)$ is closed in Y by Lemma C. Hence f maps closed sets to closed sets, i.e. f is a closed map. □

Problem 5.10

Prove that if Y is compact, then the projection $\pi_X \colon X \times Y \to X$ is a closed map.

Solution

submitted by **Jai**

Proof. Let $C \subseteq X \times Y$ be closed. We show that $X \setminus \pi_X(C)$ is open.

Let $x_0 \in X \setminus \pi_X(C)$. Then $(x_0, y) \notin C$ for every $y \in Y$. Since $X \times Y \setminus C$ is open, for each $y \in Y$ there is a basic open set $U_y \times V_y$ with

$$(x_0, y) \in U_y \times V_y \subseteq (X \times Y) \setminus C,$$

where $U_y \subseteq X$ and $V_y \subseteq Y$ are open. The sets $\{V_y\}_{y \in Y}$ form an open cover of the compact space Y, so there exist $y_1, \ldots, y_n \in Y$ with

$$Y = V_{y_1} \cup \cdots \cup V_{y_n}.$$

Put $U = U_{y_1} \cap \cdots \cap U_{y_n}$ (with $U = X$ if $Y = \emptyset$). Then U is an open neighbourhood of x_0.

Claim. $U \cap \pi_X(C) = \emptyset$.

Suppose instead that $x \in U \cap \pi_X(C)$. Then $(x, y) \in C$ for some $y \in Y$. Choose i with $y \in V_{y_i}$. Since $x \in U \subseteq U_{y_i}$, we get

$$(x, y) \in U_{y_i} \times V_{y_i} \subseteq (X \times Y) \setminus C,$$

contradicting $(x, y) \in C$. This proves the claim.

Thus every point x_0 of $X \setminus \pi_X(C)$ has an open neighbourhood $U \subseteq X \setminus \pi_X(C)$, so $X \setminus \pi_X(C)$ is open and $\pi_X(C)$ is closed. Hence π_X is a closed map. □

Problem Set 6

Problem 6.1

Let $p\colon X \twoheadrightarrow Y$ be a closed continuous surjective map such that $p^{-1}(y)$ is compact for each $y \in Y$. Prove that if Y is compact, then so is X.

Solution *submitted by* **Yashpal Prasad**

Proof. Let $\mathcal{U} = \{U_\alpha\}_{\alpha \in A}$ be an arbitrary open covering of the space X. We wish to extract a finite subcovering of X from $\mathcal{U}$ to establish that X is compact. For each point $y \in Y$, the fiber $p^{-1}(y) \subseteq X$ is a compact subspace by hypothesis. Since $\mathcal{U}$ covers X, it covers $p^{-1}(y)$ in particular; hence, we can extract a finite subcollection of open sets from $\mathcal{U}$, say $U_{\alpha_{1,y}}, U_{\alpha_{2,y}}, \ldots, U_{\alpha_{k_y,y}}$, which covers $p^{-1}(y)$.

Define the union of these selected open sets as $S_y = \bigcup_{i=1}^{k_y} U_{\alpha_{i,y}}$, so that $p^{-1}(y) \subseteq S_y$. The complement $V_y = X \setminus S_y$ is a closed subset of X because S_y is a finite union of open sets. Because $p\colon X \to Y$ is a closed map, the image $p(V_y)$ is a closed subset of Y, which implies that its complement $W_y = Y \setminus p(V_y)$ is an open subset of Y.

We claim that $y \in W_y$. Suppose to the contrary that $y \notin W_y$. Then $y \in p(V_y)$, meaning there exists some $x \in V_y$ such that $p(x) = y$, or $x \in p^{-1}(y)$. But $V_y = X \setminus S_y$ is disjoint from S_y, whereas $p^{-1}(y) \subseteq S_y$, yielding a contradiction. Thus, $y \in W_y$ for every $y \in Y$.

The collection of open sets $\{W_y\}_{y \in Y}$ forms an open covering of Y. Since Y is compact, this covering contains a finite subcovering, say $W_{y_1}, W_{y_2}, \ldots, W_{y_m}$. For each j, we have $p^{-1}(W_{y_j}) \subseteq X \setminus V_{y_j} = S_{y_j} = \bigcup_{i=1}^{k_{y_j}} U_{\alpha_{i,y_j}}$. Taking the inverse images over the finite subcovering yields:

$$X = p^{-1}(Y) = \bigcup_{j=1}^{m} p^{-1}(W_{y_j}) \subseteq \bigcup_{j=1}^{m} \left(\bigcup_{i=1}^{k_{y_j}} U_{\alpha_{i,y_j}} \right).$$

Conclusion: We have expressed X as a finite union of sets belonging to $\mathcal{U}$. Therefore, X is compact. □

Problem 6.2

Let X be a compact metric space. Prove that if $f\colon X \to X$ is an isometry (distance preserving map) of X, then f is a homeomorphism.

Solution

submitted by **Yashpal Prasad**

Proof. Let $f\colon X \to X$ satisfy $d(f(x), f(y)) = d(x, y)$ for all $x, y \in X$. A homeomorphism is a continuous bijection with continuous inverse. We check each part in turn.

Step 1: f is continuous. Given $\epsilon > 0$, take $\delta = \epsilon$. If $d(x, y) < \delta$, then $d(f(x), f(y)) = d(x, y) < \epsilon$. So f is (uniformly) continuous.

Step 2: f is injective. If $f(x) = f(y)$, then $d(x, y) = d(f(x), f(y)) = 0$. Hence $x = y$.

Step 3: f is surjective (this is where compactness is used). Suppose, for contradiction, that $f(X) \neq X$, and pick $y \in X \setminus f(X)$.

(a) *$f(X)$ is closed.* By Step 1, f is continuous, and X is compact. The continuous image of a compact set is compact (standard theorem), so $f(X)$ is compact. A compact subset of a metric space is closed, so $f(X)$ is closed.

(b) *y is at positive distance from $f(X)$.* Since $y \notin f(X)$ and $f(X)$ is closed, some open ball $B(y, \epsilon)$ misses $f(X)$. That is, $d(y, z) \geq \epsilon$ for every $z \in f(X)$.

(c) *Build a sequence with no convergent subsequence.* Define $x_n = f^n(y)$ for $n \geq 0$, where $f^0 = \mathrm{id}$. Take $m > n \geq 0$. Since f is an isometry, so is f^n, and therefore

$$d(x_n, x_m) = d\big(f^n(y), f^n(f^{m-n}(y))\big) = d\big(y, f^{m-n}(y)\big).$$

Because $m - n \geq 1$, the point $f^{m-n}(y) = f(f^{m-n-1}(y))$ lies in $f(X)$. By (b), $d(x_n, x_m) \geq \epsilon$.

(d) *Contradiction.* Every pair of distinct terms of (x_n) is at distance at least ϵ. So no subsequence is Cauchy, and hence none converges. But X is a compact metric space, so it is sequentially compact (standard theorem), and (x_n) must have a convergent subsequence. This is a contradiction.

Therefore $f(X) = X$, so f is surjective.

Step 4: f^{-1} is continuous. By Steps 2 and 3, f is a bijection, so $f^{-1}\colon X \to X$ exists. For $a, b \in X$, write $a = f(x)$ and $b = f(y)$. Then

$$d\big(f^{-1}(a), f^{-1}(b)\big) = d(x, y) = d(f(x), f(y)) = d(a, b).$$

So f^{-1} is also an isometry, and by the same argument as in Step 1 it is continuous.

Conclusion: f is a continuous bijection with continuous inverse, so f is a homeomorphism. □

Problem 6.3

Prove that the composition of two quotient maps is a quotient map.

Solution *submitted by* **Yashpal Prasad**

Proof. Let $p\colon X \to Y$ and $q\colon Y \to Z$ be quotient maps. By definition, both p and q are continuous and surjective maps. Consider the composite map $qp\colon X \to Z$ defined by $(qp)(x) = q(p(x))$. The composition of continuous functions is continuous, and the composition of surjective functions is surjective; hence qp is continuous and surjective.

To prove that qp is a quotient map, we must show that a subset $U \subseteq Z$ is open in Z if and only if its inverse image $(qp)^{-1}(U)$ is open in X. By set-theoretic properties of inverse images under compositions, $(qp)^{-1}(U) = p^{-1}(q^{-1}(U))$.

Suppose $U \subseteq Z$ is such that $(qp)^{-1}(U)$ is open in X. Since p is a quotient map, a subset $V \subseteq Y$ is open in Y if and only if $p^{-1}(V)$ is open in X. Setting $V = q^{-1}(U)$, we have $p^{-1}(V) = (qp)^{-1}(U)$ open in X, which implies $q^{-1}(U)$ is open in Y. Now, since q is a quotient map and $q^{-1}(U)$ is open in Y, it follows that U is open in Z. Conversely, if U is open in Z, then $q^{-1}(U)$ is open in Y by continuity of q, and $p^{-1}(q^{-1}(U))$ is open in X by continuity of p.

Conclusion: U is open in Z if and only if $(qp)^{-1}(U)$ is open in X. Thus, qp is a quotient map. □

Problem 6.4

Prove that a surjective continuous map $f\colon X \twoheadrightarrow Y$ that is either an open map or a closed map is a quotient map.

Solution *submitted by* **Indrajeet Kumawat**

Proof. A surjective continuous map $f\colon X \to Y$ is a quotient map if for any subset $U \subseteq Y$, U is open in Y if and only if $f^{-1}(U)$ is open in X (equivalently with closed sets). We analyze the two cases separately:

Case 1 (f is an open map): Let $U \subseteq Y$ be any subset. If U is open in Y, then $f^{-1}(U)$ is open in X by continuity. Conversely, suppose $f^{-1}(U)$ is open in X. Since f is an open map, the image $f(f^{-1}(U))$ is open in Y. Because f is surjective, $f(f^{-1}(U)) = U$, so U is open in Y.

Case 2 (f is a closed map): Let $C \subseteq Y$ be any subset. If C is closed in Y, then $f^{-1}(C)$ is closed in X by continuity. Conversely, suppose $f^{-1}(C)$ is closed in X. Since f is a closed map, $f(f^{-1}(C))$ is closed in Y. Since f is surjective, $f(f^{-1}(C)) = C$, so C is closed in Y.

Conclusion: In both cases, the quotient map condition holds. Therefore, f is a quotient map. □

Problem 6.5

Prove that a bijective quotient map is a homeomorphism.

Solution *submitted by* **Indrajeet Kumawat**

Proof. Let $f\colon X \to Y$ be a quotient map that is also bijective. Because f is a quotient map, it is continuous and surjective. Because f is bijective, it possesses an inverse function $f^{-1}\colon Y \to X$. To prove that f is a homeomorphism, we must show that f^{-1} is continuous (since f is already continuous).

Showing f^{-1} is continuous is equivalent to showing that for any open set $U \subseteq X$, its inverse image under f^{-1}, which is $(f^{-1})^{-1}(U) = f(U)$, is open in Y. Let U be an arbitrary open subset of X. Because f is injective, $f^{-1}(f(U)) = U$. By definition, f is a quotient map, meaning a subset $V \subseteq Y$ is open in Y if and only if $f^{-1}(V)$ is open in X. Choosing $V = f(U)$, we have $f^{-1}(V) = U$, which is given to be open in X. Therefore, $V = f(U)$ is open in Y.

Conclusion: Since $f(U)$ is open in Y for every open set $U \subseteq X$, the inverse map f^{-1} is continuous. Thus, f is a homeomorphism. □

Problem 6.6

Let $f_1\colon X_1 \to Y_1$ and $f_2\colon X_2 \to Y_2$ be maps between topological spaces, and let $f_1 \times f_2\colon X_1 \times X_2 \to Y_1 \times Y_2$ denote the product map $(x_1, x_2) \mapsto (f_1(x_1), f_2(x_2))$.

(a) Prove that if f_1 and f_2 are continuous, then $f_1 \times f_2$ is continuous with respect to the product topologies.

(b) Give a counterexample to show that if $p_1\colon X_1 \to Y_1$ and $p_2\colon X_2 \to Y_2$ are quotient maps, $p_1 \times p_2$ need not be a quotient map.

Solution *submitted by* **Indrajeet Kumawat**

(a) A sub-basis for the product topology on $Y_1 \times Y_2$ consists of sets of the form $U_1 \times U_2$, where $U_1 \subseteq Y_1$ and $U_2 \subseteq Y_2$ are open. To prove $f_1 \times f_2$ is continuous, we show that the inverse image of any basic open set $U_1 \times U_2$ is open in $X_1 \times X_2$. Let $(x_1, x_2) \in (f_1 \times f_2)^{-1}(U_1 \times U_2)$. Then $(f_1(x_1), f_2(x_2)) \in U_1 \times U_2$, which implies $f_1(x_1) \in U_1$ and $f_2(x_2) \in U_2$, or equivalently $x_1 \in f_1^{-1}(U_1)$ and $x_2 \in f_2^{-1}(U_2)$. Thus, $(f_1 \times f_2)^{-1}(U_1 \times U_2) = f_1^{-1}(U_1) \times f_2^{-1}(U_2)$. Since f_1 and f_2 are continuous, $f_1^{-1}(U_1)$ and $f_2^{-1}(U_2)$ are open in X_1 and X_2 respectively, making their Cartesian product open in $X_1 \times X_2$.

Conclusion (a): $f_1 \times f_2$ is continuous. □

(b) Consider the quotient map $p_1 = \pi\colon \mathbb{R} \to \mathbb{R}/\mathbb{N}$ obtained by collapsing the closed set of integers $\mathbb{N} = \{1, 2, 3, \dots\}$ to a single point $*$, and let $p_2 = \mathrm{id}_{\mathbb{Q}}\colon \mathbb{Q} \to \mathbb{Q}$ be the identity map on the rational numbers, which is a homeomorphism and hence a quotient map. We examine their product map $F = \pi \times \mathrm{id}_{\mathbb{Q}}\colon \mathbb{R} \times \mathbb{Q} \to (\mathbb{R}/\mathbb{N}) \times \mathbb{Q}$. By part (a), F is continuous, and it is clearly surjective. However, we claim that F is not a quotient map.

To see why, note that the fibers of F over points in the codomain are either single points or sets of the form $\mathbb{N} \times \{s\}$ for $s \in \mathbb{Q}$. A subset $U \subseteq \mathbb{R} \times \mathbb{Q}$ is saturated with respect to F if for each rational s, either the entire vertical slice $\mathbb{N} \times \{s\}$ is contained in U or none of it is. We can construct a saturated open set U by taking an enumeration of irrational numbers $r_n = \sqrt{2}/n$ for $n \geq 1$ and defining continuous functions $h_n(s) = \min\{1/4, |s - r_n|\}$ for $s \in \mathbb{Q}$. Because each r_n is irrational, $h_n(s) > 0$ everywhere on $\mathbb{Q}$. We define the open set

$$U = \bigcup_{n=1}^{\infty} \{(x, s) \in \mathbb{R} \times \mathbb{Q} : |x - n| < h_n(s)\}.$$

Since $(n, s) \in U$ for every integer n and rational s, the set contains $\mathbb{N} \times \mathbb{Q}$, ensuring it is saturated.

Let $W = F(U)$ be the image of U. Because U is saturated, the inverse image $F^{-1}(W) = U$ is open in $\mathbb{R} \times \mathbb{Q}$. If F were a quotient map, its image W would necessarily be open in $(\mathbb{R}/\mathbb{N}) \times \mathbb{Q}$. Notice that the point $(*, 0)$ belongs to W. If W were open, there would exist an open neighborhood V of $*$ in $\mathbb{R}/\mathbb{N}$ and a positive number $\delta > 0$ such that $V \times ((-\delta, \delta) \cap \mathbb{Q}) \subseteq W$. The inverse image $\pi^{-1}(V)$ is an open neighborhood in $\mathbb{R}$ containing all integers $n \geq 1$, meaning for each n there exists some $0 < \eta_n < 1/4$ such that $(n - \eta_n, n + \eta_n) \subseteq \pi^{-1}(V)$. By choosing n sufficiently large such that $|r_n| < \delta/2$ and picking a rational s close to r_n via the density of $\mathbb{Q}$, we can construct a point $x = n + (|s - r_n| + \eta_n)/2$ that satisfies $|s - r_n| < |x - n| < \eta_n$ while simultaneously contradicting the bound established by $h_n(s)$. This contradiction shows that the image W cannot be open in the product topology.

Conclusion (b): The product of quotient maps $p_1 \times p_2$ is not necessarily a quotient map. □

Problem 6.7

Let X and Y be topological spaces, and $f\colon X \to Y$ be a bijective continuous map. Show that if Y is Hausdorff, then X is also Hausdorff.

Solution

submitted by **Himanshu Sharma**

Proof. To show that X is Hausdorff, we must demonstrate that for any two distinct points $x_1, x_2 \in X$ with $x_1 \neq x_2$, there exist disjoint open neighborhoods $U_1, U_2 \subseteq X$ containing x_1 and x_2 respectively. Since $f\colon X \to Y$ is injective, $f(x_1) \neq f(x_2)$. Because Y is a Hausdorff space, there exist disjoint open sets $V_1, V_2 \subseteq Y$ such that $f(x_1) \in V_1$, $f(x_2) \in V_2$, and $V_1 \cap V_2 = \emptyset$.

Since f is continuous, the inverse images $U_1 = f^{-1}(V_1)$ and $U_2 = f^{-1}(V_2)$ are open subsets of X containing x_1 and x_2 respectively. Furthermore, their intersection is:

$$U_1 \cap U_2 = f^{-1}(V_1) \cap f^{-1}(V_2) = f^{-1}(V_1 \cap V_2) = f^{-1}(\emptyset) = \emptyset.$$

Conclusion: We have found disjoint open neighborhoods separating x_1 and x_2. Therefore, X is Hausdorff. □

Problem 6.8

Let X be a compact topological space and $p\colon X \twoheadrightarrow Y$ be a quotient map.

(a) Prove that Y is compact.

(b) Show that if Y is Hausdorff, then p is a closed map.

Solution *submitted by* **Himanshu Sharma**

(a) A quotient map $p\colon X \to Y$ is continuous and surjective. It is a fundamental topological theorem that the continuous image of a compact space is compact. Since X is compact and $Y = p(X)$, Y is compact.

Conclusion (a): Y is compact. □

(b) Let $C \subseteq X$ be an arbitrary closed set in X. We must prove that $p(C)$ is closed in Y. Because X is compact and C is closed in X, C is compact. Consider the restriction map $p|_C\colon C \to p(C)$, which is continuous. Thus, $p(C) = p|_C(C)$ is the continuous image of a compact space, so $p(C)$ is compact in Y. Since Y is Hausdorff, any compact subspace of a Hausdorff space is closed. Consequently, $p(C)$ is closed in Y.

Conclusion (b): p is a closed map. □

Problem 6.9

Let $p\colon X \to Y$ be a continuous surjection. Prove that p is a closed map if and only if for every subset $S \subseteq Y$ and every open set $U \subseteq X$ containing $p^{-1}(S)$, there exists an open set $V \subseteq Y$ containing S such that $p^{-1}(V) \subseteq U$.

Solution *submitted by* **Himanshu Sharma**

Proof. **Proof of (⇒):** Suppose p is a closed map. Let $S \subseteq Y$ and let $U \subseteq X$ be an open set containing $p^{-1}(S)$. Consider the closed set $C = X \setminus U$. Since p is a closed map, $p(C)$ is closed in Y, so $V = Y \setminus p(C)$ is open in Y. For any $s \in S$, $p^{-1}(\{s\}) \subseteq p^{-1}(S) \subseteq U$, meaning no point in the fiber belongs to C. Thus $s \notin p(C)$, so $s \in V$, implying $S \subseteq V$. Furthermore, $p^{-1}(V) = X \setminus p^{-1}(p(C)) \subseteq X \setminus C = U$.

Proof of (⇐): Conversely, suppose the condition holds. Let $C \subseteq X$ be closed. We must show $p(C)$ is closed in Y, or $Y \setminus p(C)$ is open. Let $y \in Y \setminus p(C)$ and set $S = \{y\}$. The fiber $p^{-1}(\{y\})$ is disjoint from C, so $p^{-1}(\{y\}) \subseteq X \setminus C$. Let $U = X \setminus C$, which is open. By hypothesis, there exists an open set $V_y \subseteq Y$ containing y such that $p^{-1}(V_y) \subseteq U = X \setminus C$. This implies $V_y \cap p(C) = \emptyset$, so $V_y \subseteq Y \setminus p(C)$. Thus, $Y \setminus p(C)$ is open.

Conclusion: p is a closed map if and only if the stated condition holds. □

Problem Set 7 (Munkres §22)

Problem 7.1

Check the details of Example 3.[a]

[a] Example 3 of §22 (The Quotient Topology) in J. R. Munkres, *Topology*, 2nd ed.

Solution *submitted by* **Karan Gangadharan**

The quotient topology on A induced by p is given by

$$\tau_A = \{U \subseteq A : p^{-1}(U) \text{ is open in } \mathbb{R}\}.$$

Since

$$p^{-1}(\{a\}) = (-\infty, 0), \qquad p^{-1}(\{b\}) = (0, \infty), \qquad p^{-1}(\{c\}) = \{0\},$$

and

$$p^{-1}(\{a, b\}) = \mathbb{R} \setminus \{0\},$$

we observe that the open subsets of A are

$$\varnothing, \quad \{a\}, \quad \{b\}, \quad \{a, b\}, \quad \{a, b, c\}.$$

Hence, the quotient topology on A is

$$\boxed{\tau_A = \{\varnothing, \{a\}, \{b\}, \{a, b\}, \{a, b, c\}\}.}$$

Problem 7.2

(a) Let $p\colon X \to Y$ be a continuous map. Show that if there is a continuous map $f\colon Y \to X$ such that $p \circ f$ equals the identity map of Y, then p is a quotient map.

(b) If $A \subset X$, a *retraction* of X onto A is a continuous map $r\colon X \to A$ such that $r(a) = a$ for each $a \in A$. Show that a retraction is a quotient map.

Solution *submitted by* **Karan Gangadharan**

Proof. **(a)** Let $p\colon X \to Y$ be a continuous map. Suppose there exists a continuous map $f\colon Y \to X$ such that

$$p \circ f = I_Y,$$

where I_Y is the identity map on Y.

We know that p is continuous. Also, since

$$p(f(y)) = y, \qquad \forall y \in Y,$$

the map p is surjective.

Now, let $U \subseteq Y$ be such that $p^{-1}(U)$ is open in X. Since f is continuous, we have $f^{-1}\big(p^{-1}(U)\big)$ open in Y. But

$$f^{-1}\big(p^{-1}(U)\big) = (p \circ f)^{-1}(U) = I_Y^{-1}(U) = U.$$

Therefore, U is open in Y.

Thus, p is a continuous surjective map satisfying

$$U \text{ is open in } Y \iff p^{-1}(U) \text{ is open in } X.$$

Hence, p is a quotient map.

$\boxed{p \text{ is a quotient map.}}$

(b) Let $A \subseteq X$ and let $r\colon X \to A$ be a retraction such that

$$r(a) = a, \qquad \forall a \in A.$$

We know that r is continuous by definition. Also, since $r(a) = a$ for every $a \in A$, the map r is surjective.

Now, let $U \subseteq A$ be such that $r^{-1}(U)$ is open in X. Since A has the subspace topology, we have $r^{-1}(U) \cap A$ open in A. But, using the property of retraction,

$$\begin{aligned} r^{-1}(U) \cap A &= \{x \in A : r(x) \in U\} \\ &= \{x \in A : x \in U\} \\ &= U. \end{aligned}$$

Therefore, U is open in A.

Thus, r is a continuous surjective map such that

$$U \text{ is open in } A \iff r^{-1}(U) \text{ is open in } X.$$

Hence, r is a quotient map.

$\boxed{\text{Every retraction is a quotient map.}}$ □

Problem 7.3

Let $\pi_1\colon \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ be projection on the first coordinate. Let A be the subspace of $\mathbb{R} \times \mathbb{R}$ consisting of all points $x \times y$ for which either $x \geq 0$ or $y = 0$ (or both); let $q\colon A \to \mathbb{R}$ be obtained by restricting π_1. Show that q is a quotient map that is neither open nor closed.

Solution

submitted by **Karan Gangadharan**

Proof. Let

$$\pi_1\colon \mathbb{R} \times \mathbb{R} \to \mathbb{R}$$

be the projection onto the first coordinate, defined by $\pi_1(x, y) = x$. Let

$$A = \{(x, y) \in \mathbb{R} \times \mathbb{R} : x \geq 0 \text{ or } y = 0\}.$$

Define $g\colon A \to \mathbb{R}$ by restricting π_1 to A. Thus, $g(x, y) = x$.

We shall show that g is a quotient map but is neither open nor closed.

(i) g is a quotient map: Since g is the restriction of the continuous projection map π_1, it is continuous. Also, g is surjective, because for every $x \in \mathbb{R}$,

$$(x, 0) \in A \quad \text{and} \quad g(x, 0) = x.$$

Now, consider the map $f\colon \mathbb{R} \to A$ defined by $f(x) = (x, 0)$. Clearly, f is continuous and

$$(g \circ f)(x) = g(x, 0) = x.$$

Therefore, $g \circ f = I_{\mathbb{R}}$. By Problem 7.2(a), since g and f are continuous and $g \circ f = I_{\mathbb{R}}$, it follows that

$$\boxed{g \text{ is a quotient map.}}$$

(ii) g is not an open map: Consider the set

$$U = A \cap ((-1, 1) \times (1/2, 3/2)).$$

Since $(-1, 1) \times (1/2, 3/2)$ is open in $\mathbb{R}^2$, U is open in A. Now, by the definition of A, if $(x, y) \in U$, then $x \geq 0$, since $y \neq 0$. Consequently,

$$g(U) = [0, 1).$$

But $[0, 1)$ is not open in $\mathbb{R}$. Thus, g does not map every open subset of A onto an open subset of $\mathbb{R}$. Hence,

$$\boxed{g \text{ is not an open map.}}$$

(iii) g is not a closed map: Consider the subset

$$F = \left\{ \left(\frac{1}{n}, n \right) : n \in \mathbb{N} \right\}.$$

Clearly, $F \subseteq A$, since $1/n > 0$ for every $n \in \mathbb{N}$. We claim that F is closed in A. Indeed, F is closed in $\mathbb{R}^2$, because the second coordinates of its points tend to infinity, so it has no accumulation point in $\mathbb{R}^2$. Therefore, F is also closed in the subspace A. Now,

$$g(F) = g\left(\left\{ \left(\frac{1}{n}, n \right) : n \in \mathbb{N} \right\} \right) = \left\{ \frac{1}{n} : n \in \mathbb{N} \right\}.$$

But

$$\frac{1}{n} \longrightarrow 0 \quad \text{as } n \longrightarrow \infty,$$

and $0 \notin g(F)$. Hence, $g(F)$ is not closed in $\mathbb{R}$. Therefore, g does not map every closed subset of A onto a closed subset of $\mathbb{R}$. Thus,

$$\boxed{g \text{ is not a closed map.}}$$

Combining (i), (ii) and (iii), we conclude that

$$\boxed{g\colon A \to \mathbb{R} \text{ is a quotient map which is neither open nor closed.}} \qquad \square$$

Problem 7.4

(a) Define an equivalence relation on the plane $X = \mathbb{R}^2$ as follows:

$$x_0 \times y_0 \sim x_1 \times y_1 \quad \text{if} \quad x_0 + y_0^2 = x_1 + y_1^2.$$

Let X^* be the corresponding quotient space. It is homeomorphic to a familiar space; what is it? [*Hint:* Set $g(x \times y) = x + y^2$.]

(b) Repeat (a) for the equivalence relation

$$x_0 \times y_0 \sim x_1 \times y_1 \quad \text{if} \quad x_0^2 + y_0^2 = x_1^2 + y_1^2.$$

Solution *submitted by* **Karan Gangadharan**

Proof. **(a)** Define an equivalence relation on $X = \mathbb{R}^2$ by

$$(x_0, y_0) \sim (x_1, y_1) \iff x_0 + y_0^2 = x_1 + y_1^2.$$

Let $X^* = X/\!\sim$ be the corresponding quotient space. We shall show that X^* is homeomorphic to $\mathbb{R}$.

Define a map $g\colon X \to \mathbb{R}$ by

$$g(x, y) = x + y^2.$$

Since g is a polynomial function, it is continuous. Also, g is surjective, since for every $t \in \mathbb{R}$, $g(t, 0) = t$.

Now, define a map $f\colon \mathbb{R} \to X$ by $f(t) = (t, 0)$. Clearly, f is continuous and

$$(g \circ f)(t) = g(t, 0) = t.$$

Thus, $g \circ f = I_{\mathbb{R}}$. Therefore, by Problem 7.2(a), g is a quotient map.

Let $q\colon X \to X^*$ be the natural quotient map defined by $q(x, y) = [(x, y)]$. Define $\overline{g}\colon X^* \to \mathbb{R}$ by

$$\overline{g}([(x, y)]) = x + y^2.$$

We show that $\overline{g}$ is a homeomorphism.

(i) $\overline{g}$ is well-defined: Suppose $[(x_0, y_0)] = [(x_1, y_1)]$. Then, by the definition of the equivalence relation, $x_0 + y_0^2 = x_1 + y_1^2$. Hence, $\overline{g}([(x_0, y_0)]) = \overline{g}([(x_1, y_1)])$. Therefore, $\overline{g}$ is well-defined.

(ii) $\overline{g}$ is bijective: Suppose $\overline{g}([(x_0, y_0)]) = \overline{g}([(x_1, y_1)])$. Then $x_0 + y_0^2 = x_1 + y_1^2$. Thus, $(x_0, y_0) \sim (x_1, y_1)$, which implies $[(x_0, y_0)] = [(x_1, y_1)]$. Therefore, $\overline{g}$ is injective. Further, since g is surjective, for every $t \in \mathbb{R}$, there exists $(x, y) \in X$ such that $\overline{g}([(x, y)]) = t$. Hence, $\overline{g}$ is surjective. Therefore, $\overline{g}$ is bijective.

(iii) $\overline{g}$ is a homeomorphism: We have

$$g = \overline{g} \circ q.$$

Since g is continuous and q is a quotient map, the quotient property of q implies that $\overline{g}$ is continuous.

Also, let U be any open subset of X^*. Since $\overline{g}$ is injective, we have

$$g^{-1}(\overline{g}(U)) = q^{-1}(U).$$

Since q is a quotient map and U is open in X^*, $q^{-1}(U)$ is open in X. Thus, $g^{-1}(\overline{g}(U))$ is open in X. Since g is a quotient map, $\overline{g}(U)$ is open in $\mathbb{R}$. Therefore, $\overline{g}$ is an open map.

Hence, $\overline{g}$ is a continuous bijective open map, and consequently it is a homeomorphism. Thus,

$$\boxed{X^* \cong \mathbb{R}.}$$

(b) Define an equivalence relation on $X = \mathbb{R}^2$ by

$$(x_0, y_0) \sim (x_1, y_1) \iff x_0^2 + y_0^2 = x_1^2 + y_1^2.$$

Let $X^* = X/\sim$ be the corresponding quotient space. We shall show that X^* is homeomorphic to $[0, \infty)$.

Define a map $g\colon X \to [0, \infty)$ by

$$g(x, y) = x^2 + y^2.$$

Since g is a polynomial function, it is continuous. Also, g is surjective, since for every $t \in [0, \infty)$, $g(\sqrt{t}, 0) = t$.

Now, define a map $f\colon [0, \infty) \to X$ by $f(t) = (\sqrt{t}, 0)$. Clearly, f is continuous and

$$(g \circ f)(t) = g(\sqrt{t}, 0) = t.$$

Thus, $g \circ f = I_{[0,\infty)}$. Therefore, by Problem 7.2(a), g is a quotient map.

Let $q\colon X \to X^*$ be the natural quotient map defined by $q(x, y) = [(x, y)]$. Define $\overline{g}\colon X^* \to [0, \infty)$ by

$$\overline{g}([(x, y)]) = x^2 + y^2.$$

We show that $\overline{g}$ is a homeomorphism.

(i) $\overline{g}$ is well-defined: Suppose $[(x_0, y_0)] = [(x_1, y_1)]$. Then, by the definition of the equivalence relation, $x_0^2 + y_0^2 = x_1^2 + y_1^2$. Hence, $\overline{g}([(x_0, y_0)]) = \overline{g}([(x_1, y_1)])$. Therefore, $\overline{g}$ is well-defined.

(ii) $\overline{g}$ is bijective: Suppose $\overline{g}([(x_0, y_0)]) = \overline{g}([(x_1, y_1)])$. Then $x_0^2 + y_0^2 = x_1^2 + y_1^2$. Thus, $(x_0, y_0) \sim (x_1, y_1)$, which implies $[(x_0, y_0)] = [(x_1, y_1)]$. Therefore, $\overline{g}$ is injective. Further, since g is surjective, for every $t \in [0, \infty)$, there exists $(x, y) \in X$ such that $\overline{g}([(x, y)]) = t$. Hence, $\overline{g}$ is surjective. Therefore, $\overline{g}$ is bijective.

(iii) $\overline{g}$ is a homeomorphism: We have

$$g = \overline{g} \circ q.$$

Since g is continuous and q is a quotient map, the quotient property of q implies that $\overline{g}$ is continuous.

Also, let U be any open subset of X^*. Since $\overline{g}$ is injective, we have

$$g^{-1}(\overline{g}(U)) = q^{-1}(U).$$

Since q is a quotient map and U is open in X^*, $q^{-1}(U)$ is open in X. Thus, $g^{-1}(\overline{g}(U))$ is open in X. Since g is a quotient map, $\overline{g}(U)$ is open in $[0, \infty)$. Therefore, $\overline{g}$ is an open map.

Hence, $\overline{g}$ is a continuous bijective open map, and consequently it is a homeomorphism. Thus,

$$\boxed{X^* \cong [0, \infty).}$$ □

Problem 7.5

Let $p\colon X \to Y$ be an open map. Show that if A is open in X, then the map $q\colon A \to p(A)$ obtained by restricting p is an open map.

Solution *submitted by* **Alokjeet Sahoo**

Proof. We are given that $p\colon X \to Y$ is an open map and that A is an open subset of X. Consider the restriction of p to A, defined by

$$q\colon A \longrightarrow p(A), \qquad q(x) = p(x), \quad x \in A.$$

We need to show that q is open.

Note that a map is open if the image of every open subset of its domain is open in its codomain.

Let U be an arbitrary open subset of A. Since U is open in the subspace topology on A, there exists an open subset V of X such that

$$U = A \cap V.$$

Now, both A and V are open in X. Therefore, their intersection is also open in X. Hence, U is open in X.

Since $p\colon X \to Y$ is an open map, the image of every open subset of X is open in Y. Consequently, $p(U)$ is open in Y.

On the other hand, since $U \subseteq A$, we have

$$p(U) \subseteq p(A).$$

Also, by the definition of the restriction q, we obtain

$$q(U) = p(U).$$

Thus, $q(U)$ is an open subset of Y contained in $p(A)$. By the definition of the subspace topology, $q(U)$ is therefore open in $p(A)$.

Since U was an arbitrary open subset of A, the image of every open subset of A under q is open in $p(A)$. Therefore,

$$q\colon A \to p(A) \text{ is an open map.}$$ □

Problem 7.6

Recall that $\mathbb{R}_K$ denotes the real line in the K-topology (see §13). Let Y be the quotient space obtained from $\mathbb{R}_K$ by collapsing the set K to a point; let $p\colon \mathbb{R}_K \to Y$ be the quotient map.

(a) Show that Y satisfies the T_1 axiom, but is not Hausdorff.

(b) Show that $p \times p\colon \mathbb{R}_K \times \mathbb{R}_K \to Y \times Y$ is not a quotient map. [*Hint:* The diagonal is not closed in $Y \times Y$, but its inverse image is closed in $\mathbb{R}_K \times \mathbb{R}_K$.]

Solution *submitted by* **Alokjeet Sahoo**

Recall that a basis for the K-topology consists of sets of the form

$$(a, b) \qquad \text{and} \qquad (a, b) \setminus K,$$

where $a, b \in \mathbb{R}$ and $a < b$.

(a)

Proof. We shall prove the two assertions separately.

Step 1: Prove that Y satisfies the T_1 axiom. Recall that a topological space is said to satisfy the T_1 axiom if every singleton subset is closed.

Under the quotient map p, all the points of K are identified to a single point. Let this point be denoted by $y_0 = p(K)$. Thus, the equivalence classes are

$$p^{-1}(\{y_0\}) = K$$

and

$$p^{-1}(\{p(x)\}) = \{x\}, \qquad x \in \mathbb{R} \setminus K.$$

We first observe that K is closed in $\mathbb{R}_K$. Indeed, $\mathbb{R} \setminus K$ is open in the K-topology. To see this, note that every point outside K has a basic neighbourhood contained in $\mathbb{R} \setminus K$. For the point 0, we can take a neighbourhood of the form

$$(-\varepsilon, \varepsilon) \setminus K.$$

For any other point outside K, we can choose a sufficiently small ordinary open interval avoiding K. Hence, K is closed in $\mathbb{R}_K$.

Also, every singleton $\{x\}$ is closed in $\mathbb{R}_K$, since the K-topology is finer than the usual topology on $\mathbb{R}$, which is a T_1 space.

Now, consider an arbitrary point $y \in Y$. There are two possibilities:

Case 1: If $y = y_0 = p(K)$, then

$$p^{-1}(\{y\}) = K,$$

which is closed in $\mathbb{R}_K$.

Case 2: If $y = p(x)$ for some $x \in \mathbb{R} \setminus K$, then

$$p^{-1}(\{y\}) = \{x\},$$

which is also closed in $\mathbb{R}_K$.

Therefore, in both cases, $p^{-1}(\{y\})$ is closed in $\mathbb{R}_K$. By the definition of the quotient topology, a subset F of Y is closed if and only if $p^{-1}(F)$ is closed in $\mathbb{R}_K$. Consequently, $\{y\}$ is closed in Y. Since y was arbitrary, every singleton subset of Y is closed. Thus,

$$Y \text{ satisfies the } T_1 \text{ axiom.}$$

Step 2: Prove that Y is not Hausdorff. Recall that a topological space is Hausdorff if every pair of distinct points can be separated by disjoint open neighbourhoods. We shall show that the two distinct points

$$y_0 = p(K) \qquad \text{and} \qquad y_1 = p(0)$$

cannot be separated by disjoint open neighbourhoods. Notice that $0 \notin K$, so $y_0 \neq y_1$.

Suppose, on the contrary, that Y is Hausdorff. Then there exist disjoint open neighbourhoods U and V of y_0 and y_1, respectively. Thus,

$$y_0 \in U, \qquad y_1 \in V, \qquad U \cap V = \varnothing.$$

Since p is a quotient map, the sets $p^{-1}(U)$ and $p^{-1}(V)$ are open in $\mathbb{R}_K$. Furthermore, since $y_0 \in U$ and $p^{-1}(\{y_0\}) = K$, we have

$$K \subseteq p^{-1}(U).$$

Similarly, since $y_1 = p(0) \in V$, we have

$$0 \in p^{-1}(V).$$

Now, $p^{-1}(V)$ is an open neighbourhood of 0 in the K-topology. By the definition of the K-topology, every open neighbourhood of 0 contains a set of the form

$$(-\varepsilon, \varepsilon) \setminus K$$

for some $\varepsilon > 0$. Therefore,

$$(-\varepsilon, \varepsilon) \setminus K \subseteq p^{-1}(V). \tag{1}$$

Since

$$\frac{1}{n} \longrightarrow 0 \qquad \text{as } n \longrightarrow \infty,$$

we can choose n sufficiently large so that

$$0 < \frac{1}{n} < \frac{\varepsilon}{2}.$$

Put $k = \frac{1}{n}$. Then $k \in K$ and

$$k \in p^{-1}(U).$$

Since $p^{-1}(U)$ is open in $\mathbb{R}_K$ and contains k, there exists $\delta > 0$ such that

$$(k - \delta, k + \delta) \setminus K \subseteq p^{-1}(U). \tag{2}$$

By choosing δ sufficiently small, we may also assume that

$$\delta < \varepsilon - k.$$

Consequently,

$$(k - \delta, k + \delta) \subseteq (-\varepsilon, \varepsilon).$$

Since every nonempty open interval contains points outside K, we can choose

$$z \in (k - \delta, k + \delta) \setminus K.$$

Using (2), we obtain

$$z \in p^{-1}(U).$$

Also, since $z \in (-\varepsilon, \varepsilon) \setminus K$, relation (1) gives

$$z \in p^{-1}(V).$$

Therefore,

$$p^{-1}(U) \cap p^{-1}(V) \neq \varnothing.$$

But this is impossible, because inverse images preserve intersections and

$$p^{-1}(U) \cap p^{-1}(V) = p^{-1}(U \cap V) = p^{-1}(\varnothing) = \varnothing.$$

We have obtained a contradiction. Hence, the two distinct points $p(K)$ and $p(0)$ cannot be separated by disjoint open neighbourhoods. Therefore,

$$Y \text{ is not Hausdorff.}$$

Combining the two steps, we conclude that

$$Y \text{ is } T_1 \text{ but is not Hausdorff.} \qquad \square$$

(b)

Proof. We shall use the fact that the diagonal of a topological space is closed in its product with itself if and only if the space is Hausdorff. Since, by part (a), Y is not Hausdorff, its diagonal

$$\Delta_Y = \{(y, y) : y \in Y\}$$

is not closed in $Y \times Y$.

Now, consider the inverse image of Δ_Y under the map $p \times p$. By definition,

$$\begin{aligned}(p \times p)^{-1}(\Delta_Y) &= \{(x_1, x_2) \in \mathbb{R}_K \times \mathbb{R}_K : (p(x_1), p(x_2)) \in \Delta_Y\} \\ &= \{(x_1, x_2) \in \mathbb{R}_K \times \mathbb{R}_K : p(x_1) = p(x_2)\}.\end{aligned}$$

Recall that two points of $\mathbb{R}_K$ have the same image under p precisely when either they are equal or both belong to K. Therefore,

$$(p \times p)^{-1}(\Delta_Y) = \Delta_{\mathbb{R}_K} \cup (K \times K),$$

where

$$\Delta_{\mathbb{R}_K} = \{(x, x) : x \in \mathbb{R}\}$$

is the diagonal of $\mathbb{R}_K \times \mathbb{R}_K$.

We now show that this inverse image is closed in $\mathbb{R}_K \times \mathbb{R}_K$.

First, observe that $\mathbb{R}_K$ is Hausdorff because its topology is finer than the usual topology on $\mathbb{R}$, which is Hausdorff. It is a standard result that the diagonal of a Hausdorff space is closed in its product with itself. Hence,

$$\Delta_{\mathbb{R}_K}$$

is closed in $\mathbb{R}_K \times \mathbb{R}_K$.

Secondly, from part (a), we know that K is closed in $\mathbb{R}_K$. Therefore, by the product topology,

$$K \times K$$

is closed in $\mathbb{R}_K \times \mathbb{R}_K$.

Since the union of finitely many closed sets is closed, we conclude that

$$\Delta_{\mathbb{R}_K} \cup (K \times K)$$

is closed in $\mathbb{R}_K \times \mathbb{R}_K$. Thus,

$$(p \times p)^{-1}(\Delta_Y) \text{ is closed in } \mathbb{R}_K \times \mathbb{R}_K.$$

Finally, recall the defining property of a quotient map. A surjective map $f\colon X \to Z$ is a quotient map if and only if, for every subset $B \subseteq Z$,

$$B \text{ is closed in } Z \iff f^{-1}(B) \text{ is closed in } X.$$

In particular, if $p \times p$ were a quotient map, the closedness of $(p \times p)^{-1}(\Delta_Y)$ would imply that Δ_Y is closed in $Y \times Y$. However, we have already established that Δ_Y is not closed, since Y is not Hausdorff. This is a contradiction. Therefore,

$$p \times p\colon \mathbb{R}_K \times \mathbb{R}_K \longrightarrow Y \times Y \text{ is not a quotient map.}$$ □